

Analysis of Aerosol Deposition in a Simplified, 2D Tracheobronchial Tree Model

Kaan Borekci, Audra Rosenzweig, and Shuyi Jin

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1. Introduction

1.1 Background and Motivation:

Understanding the transport and deposition of aerosols within the human respiratory system is critical for evaluating the health impacts of inhaled pollutants, viruses, and particulate matter, as well as optimizing targeted drug delivery. The respiratory tract is a complex, branching geometric network where air undergoes various transitional flow regimes. Particles suspended in this inhaled air deposit along the airway walls primarily through inertial impaction, sedimentation, and diffusion.

1.2 Problem Statement:

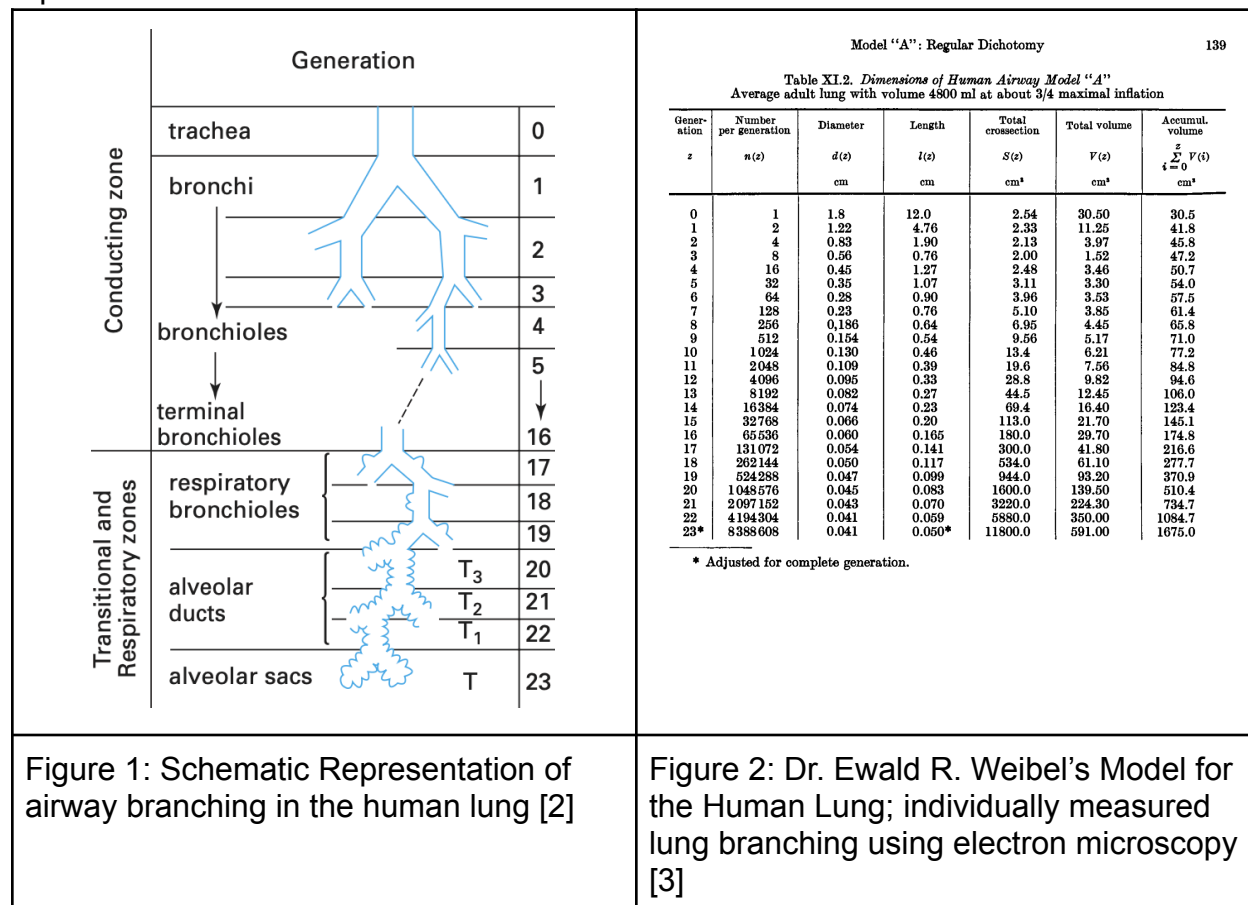
To quantify and compare the airflow dynamics and local deposition efficiency of microparticles (1micron and 10microns and 100microns) in a simplified, 2D, symmetric airway model (generations 0-6). The project will evaluate how unsteadiness (transient sinusoidal breathing) alters boundary layer separation and inertial impaction compared to standard steady state approximations.

2. Methods

2.1 Geometry

The 2D, symmetric geometry designed in ANSYS DesignModeler represents a simplified model of the upper regions of the human lung, starting with the trachea Generation 0 (G0) through G6. The airway dimensions were obtained from Weibel's Model, collected empirically, as seen in Figure 1. A symmetric model was considered, with bifurcation half angles of 35° , consistent with literature [2]. A fillet of radius 0.5 mm was used for every sharp bifurcation and sharp transition in exterior walls; this was an arbitrary value used without any scientific evidence, mainly since it was the smallest value that could be set without disturbing the overall geometry, a persistent issue that occurred within DesignModeler. In a real 3D lung, airways branch outward in all directions without colliding. However, when projecting a constant 70° bifurcation angle onto a flat 2D plane, the inner branches inevitably overlap and intersect as the generations progress. Therefore, the design choice of creating outlets at otherwise intersecting branches was made. To prevent outflow out of the first available outlet, boundary conditions were set to "outflow" instead of the default "pressure outlet"; where each generation was assigned a weight based on the inlet having 100% of the flow (indicated by 1). The full set of boundary conditions are presented in section 2.6.2. Figure 3 shows our geometry. Note that the domain area was determined to be 0.00664 m^2 through Fluent's surface area calculation, and verified by hand calculating each

rectangle with respect to the Weibel Model values as an approximation not considering smoothed, 0.5 mm fillet regions. This value was used for determining the representative mesh sizes and refinement ratios referred to in section 2.7.



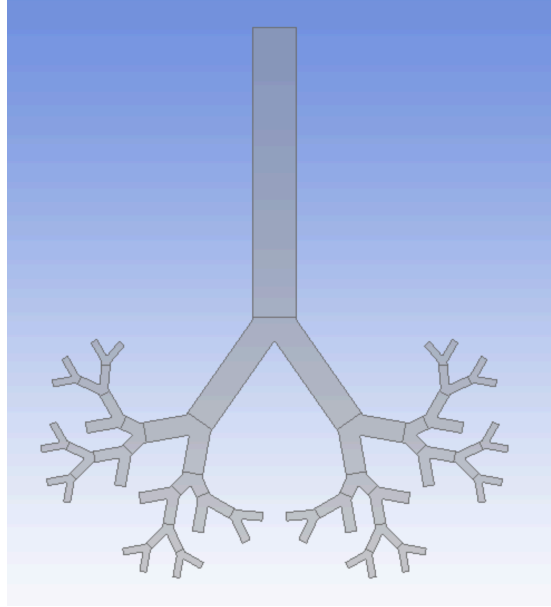


Figure 3: Simplified (Gen 0-6), 2D, Symmetric Airway Model designed in ANSYS DesignModeler using Weibel's Lung Model measurements

2.2 Approach

In this project, we considered four cases: two preliminary and two final. The two preliminaries are steady and transient flow without particles. The goal of these was to establish the expected baseline flow profiles. The steady state is to simulate peak inhalation to identify recirculation dead zones, and thus expected particle deposition. The goal of the transient case is to implement a sinusoidal inlet velocity profile to evaluate how the flow and wake regions develop and shift during a breathing cycle.

The third and fourth cases are those of interest for the project goals. They take the same flow profiles as in Cases 1 and 2 and inject particles of $1\mu\text{m}$, $10\mu\text{m}$, and $100\mu\text{m}$. The goal is to calculate baseline deposition via inertial impact at peak flow (steady state, Case 3) and to observe if that deposition pattern changes when exposed to a transient, sinusoidal flow profile (Case 4).

2.3 Assumptions

2.3.1 General Assumptions:

1. **Isothermal flow:** air at 37°C , 1atm.
2. **Incompressible flow:** $M = 0.007 \ll 0.3$.
3. **Rigid lung:** appropriate until G6 as earlier generations are mostly made of rigid cartilage tissue structure.

4. **Symmetric lung assumption:** left and right lung identical, 70° angle between all branches.
5. **2D geometry:** sufficient for identifying deposition due to inertia, which is dominant in G0-6.
6. **Smooth walls:** neglecting surface roughness from mucosal layer and cartilage rings.
7. **One way DPM coupling:** fluid affects particles, but particles do not alter flow. Valid because the particles are very dilute.
8. **No particle-particle interactions:** valid because of dilute flow.
9. **Spherical particles:** particles are perfect spheres. Valid because small particle size (surface tension dominates).
10. **Neglect Brownian Diffusion for Particles of sizes 1µm-10µm:** in upper regions of the lung (G3-G6), where inertia dominates, Brownian diffusion for small particles is negligible [7]

2.3.2 Steady State Assumptions:

1. **Peak inhalation with 1 second duration:** overestimation of inertia influenced/dominated particles
2. **10 µg/s Total Inlet Particle Concentration:** Assumes 1 actuation (puff) = 1 second, 100 µg drug aerosol enters mouth, 10% of drug reaches trachea G0, so total mass flow rate entering is = 1e-8 kg/s. For 1000 parcels, parcel mass flow rate would thus be: 1e-11. This assumption does not affect any of the particle dynamics (since one-way-coupling is enabled), it is only to have a physical sense of the results.[8]

2.3.3 Transient Assumptions:

1. **Particles only enter body during first inhalation** (e.g. an inhaler without a spacer, 1 actuation)
2. **Perfect sinusoidal flow:** period of 2s

2.4 Flow Characteristics: Regime and Key Features

2.4.1 Reynolds Number Range and Flow Regime:

- Many studies in literature use values between 1.5-3.5 m/s for normal breathing (up to = 40 m/s per when coughing, running, sneezing) [9, 10]

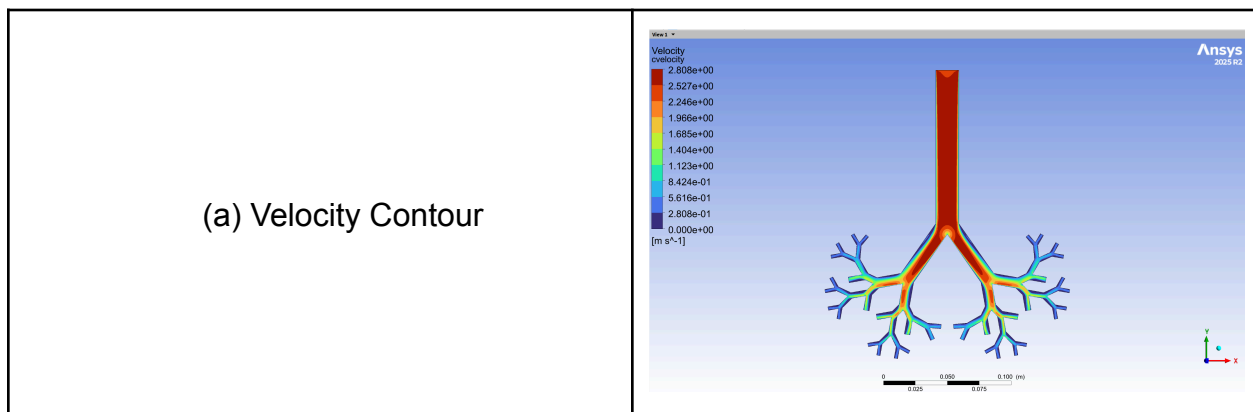
For our purposes we will start off by considering ~2.5 m/s under 1 atm and 37 Celsius (body temperature) at the inlet, G0:

G0 Inlet Velocity - U [m/s]	2.5
Dynamic Viscosity - μ [kg/(ms)]	1.89E-05
Density - ρ [kg/m³]	1.138
G0 Diameter - d [m]	1.80E-02
$Re = \frac{\rho U d}{\mu}$	2704 - Transitional Regime

Table 1: Flow Characteristics

2.4.2 Steady Flow Characteristics

Prior to any particle simulations, the steady state flow-only solution was performed on the old coarse mesh (more on mesh in section 2.7). This allowed us to observe the characteristic flow features and profiles within the domain, informing us of expected particle dynamics. Similar to Assignment 4's bifurcating 2D artery, we expected similar flow behavior in our multi-branch domain. As seen from the Velocity and Pressure contours in Figures 4a, 4b, stagnation points emerge in bifurcations, where static pressure is high and velocity is low. After the bifurcations, boundary layer separations occur within each generation, where recirculation zones emerge, as seen in the velocity streamline plot Figure 5b. We expect that smaller particles will be caught up in these recirculation zones, however, a more detailed set of hypotheses will be considered within the Discrete Phase Model section of this report.



(b) Pressure Contour

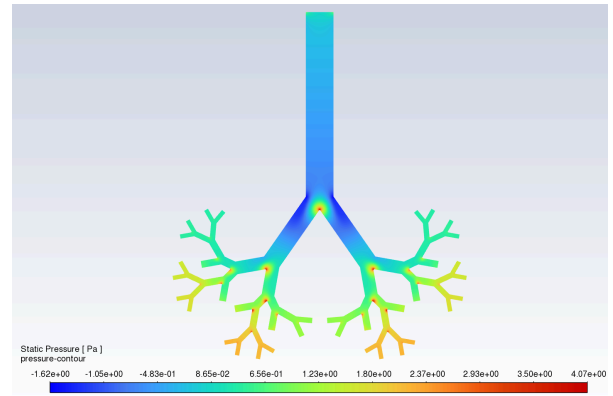
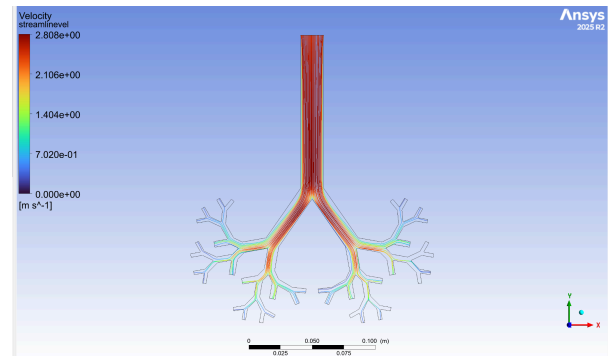


Figure 4a, 4b: Contour Plots of Flow within 2D Lung Domain

(a) Full Domain Velocity Streamline



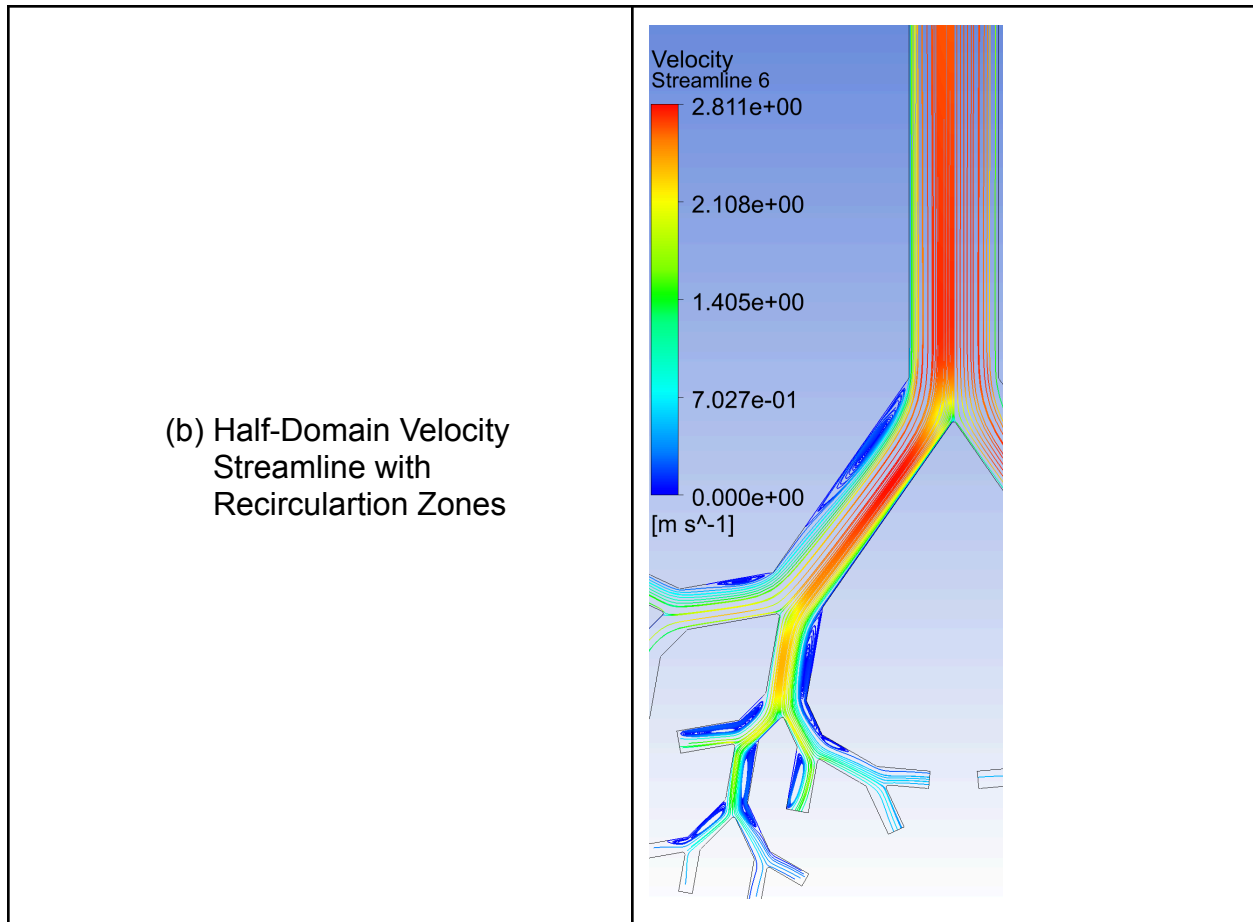
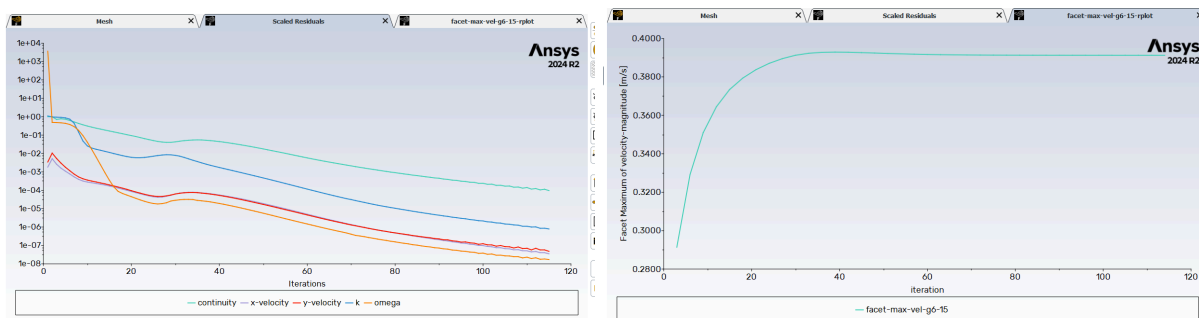


Figure 5a, 5b: Velocity Streamlines

For every steady-state trial presented in this report, 2 plots were considered as evidence of convergence: scaled residuals with tolerances set to $1e-4$, and the facet maximum y-component velocity magnitude for a random outlet in generation 6, both which can be observed in Figures 6a, 6b below. As will be discussed further within the Mesh section, some additional scalars values were considered to verify convergence for the medium and fine meshes, due to continuity converging at $1e-3$, above the set threshold, likely due to increased resolution of recirculating flow features within the domain.



Figures 6a, 6b: Evidence of Solution Convergence

2.4.3 Transient Flow Characteristics

The simulation will start on an inhalation.

2.4.3.1 Inhalation

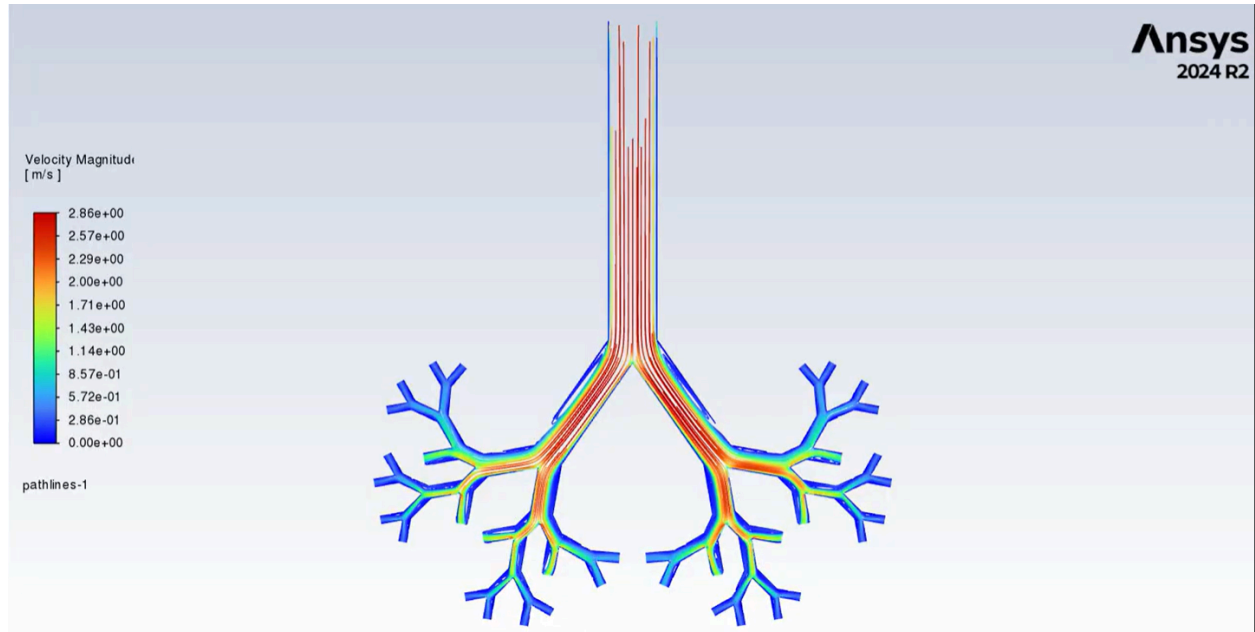


Figure 7: Maximum inhalation velocity profile and streamlines. Recirculation regions at every bifurcation.

2.4.3.2 Exhalation

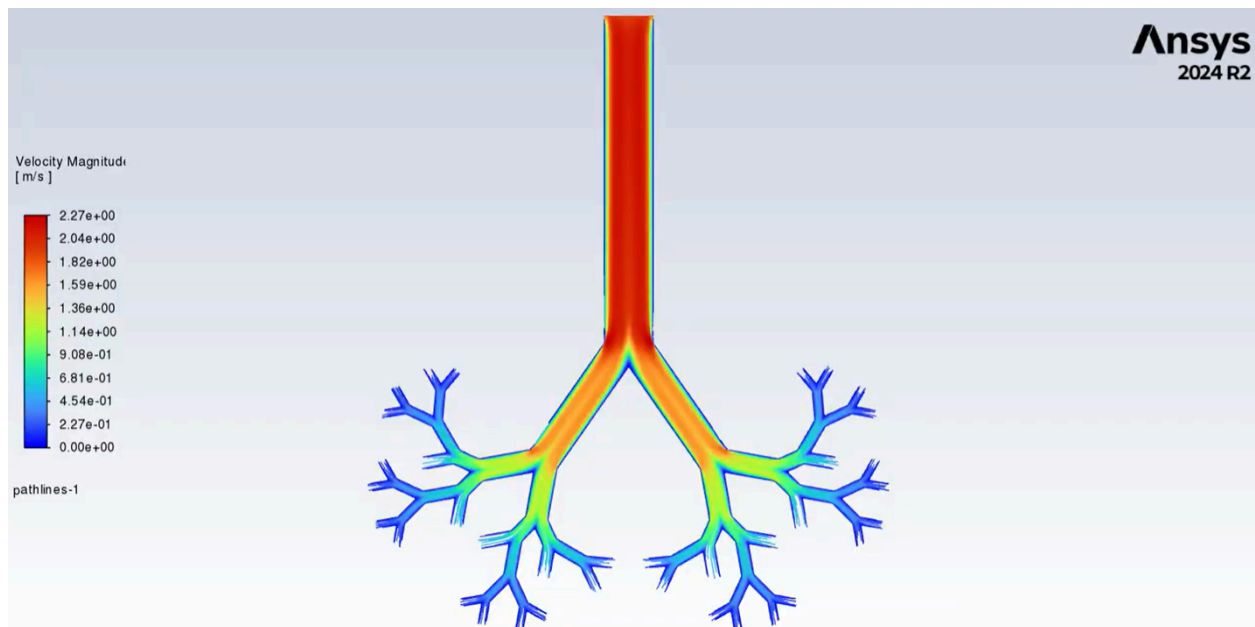
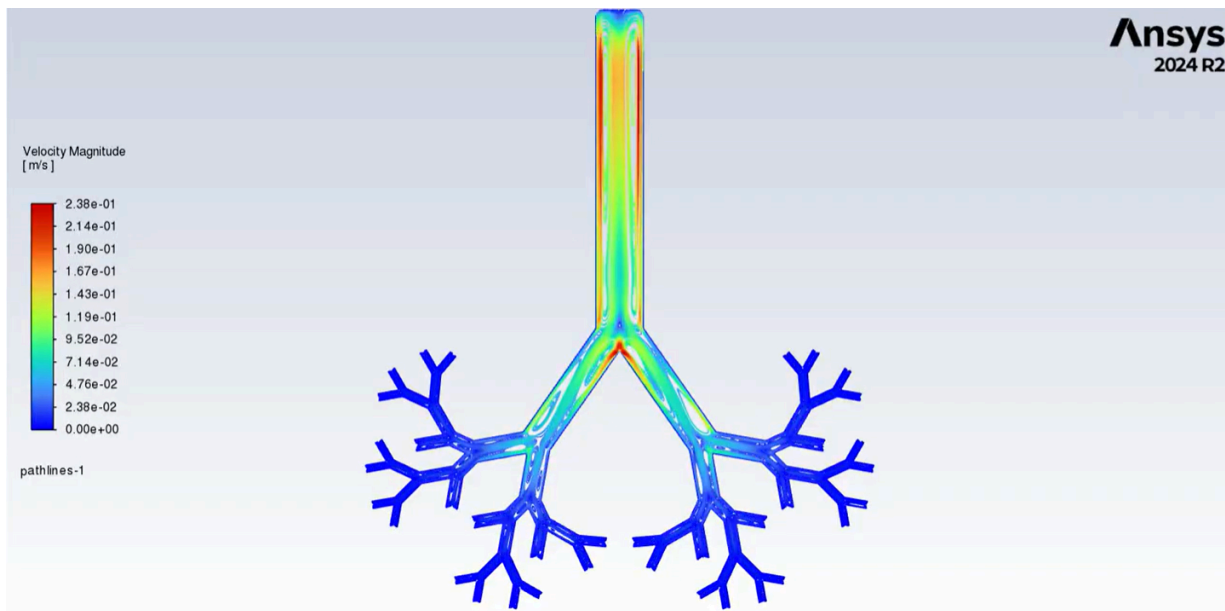
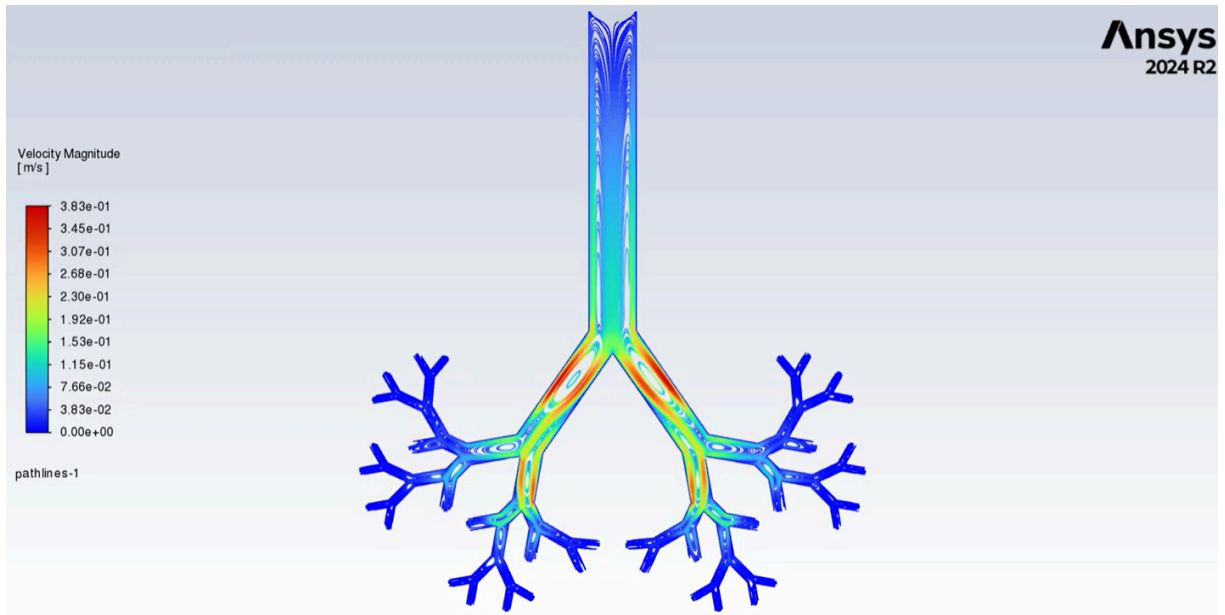


Figure 8: Maximum exhalation velocity profile and streamlines. Very small recirculation regions and dead zones

2.4.3.3 Transition



Figures 9a and 9b: Transition from inhalation to exhalation and from exhalation to inhalation, respectively

2.5 Discrete Phase Model

2.5.1 Equations of Motion

12.2.1.1. Particle Force Balance

Ansys Fluent predicts the trajectory of a discrete phase particle (or droplet or bubble) by integrating the force balance on the particle, which is written in a Lagrangian reference frame. This force balance equates the particle inertia with the forces acting on the particle, and can be written as

$$m_p \frac{d\vec{u}_p}{dt} = m_p \frac{\vec{u} - \vec{u}_p}{\tau_r} + m_p \frac{\vec{g}(\rho_p - \rho)}{\rho_p} + \vec{F} \quad (12.1)$$

Particle Acceleration ← $m_p \frac{d\vec{u}_p}{dt}$ = $m_p \frac{\vec{u} - \vec{u}_p}{\tau_r}$ + $m_p \frac{\vec{g}(\rho_p - \rho)}{\rho_p}$ + $\vec{F}$ → Gravity/Buoyancy
Drag Term ←

where m_p is the particle mass, $\vec{u}$ is the fluid phase velocity, $\vec{u}_p$ is the particle velocity, ρ is the fluid density, ρ_p is the density of the particle, $\vec{F}$ is an additional force, $m_p \frac{\vec{u} - \vec{u}_p}{\tau_r}$ is the drag force, and τ_r is the droplet or particle relaxation time [215] (p. 1069) calculated by:

$$\tau_r = \frac{\rho_p d_p^2}{18\mu} \frac{24}{C_d Re} \quad (12.2)$$

Stokes Number Relationship (next slide) ← $\frac{\rho_p d_p^2}{18\mu}$ $\frac{24}{C_d Re}$ → Pure Stokes-Drag Correction

Here, μ is the molecular viscosity of the fluid, d_p is the particle diameter, and Re is the relative Reynolds number, which is defined as

$$Re \equiv \frac{\rho d_p |\vec{u}_p - \vec{u}|}{\mu} \quad (12.3)$$

Image 1: Discrete Particle Model Equation of Motion from Fluent Theory Guide [4]

The Discrete Phase Model (DPM) in ANSYS Fluent computed particle trajectories by integrating a force balance written in the Lagrangian reference frame. Instead of applying the force balance on each individual particle, Fluent downsamples the total number of particles into parcels, also referred to as streams. Each parcel represents a certain number of the total particle count. Equation 12.1 in Image 1 from the Fluent Theory Guide, equates the particle acceleration on the left hand side to the sum of forces acting on each parcel on the right hand side. The forces acting on the parcel are divided in 3:

1. Drag Term
2. Gravity & Buoyancy
3. External Forces like electromagnetic, lift, intra-particle collisions and interactions (4 way coupling), ... = 0 in our case

The dominant term for aerosol transport is the drag term, which drives the particle toward the local fluid velocity at a rate governed by the particle relaxation time, τ_r . Particle relaxation time, as defined by Equation 12.2, is the characteristic time scale over which a particle adjusts its velocity to the local flow velocity. The leading factor is the Stokes drag result for creeping/Stokes flow around a sphere, and the second Stokes-Drag correction extends the formula beyond the Stokes Regime ($Re < 1$). The

relative Reynolds number given in Equation 12.3 determines which drag law regime applies based on the particle diameter and slip velocity.

The ratio of τ_r relative to a characteristic flow timescale defines the Stokes number, which is discussed in the next section.

2.5.2 Stokes Number and Particle Dynamics Hypotheses

Particle transport behavior is heavily dictated by the Stokes Number, which represents the ratio of particle inertia to fluid viscous forces:

$$Stk = \frac{\rho_p d_p^2 U_f}{18 \mu_f L_c} = \frac{\text{particle inertia}}{\text{fluid viscous forces}}$$

ρ_p = particle density

d_p = particle diameter

U_f = characteristic fluid velocity

μ_f = dynamic viscosity of the fluid

L_c = characteristic length (often parent airway diameter)

Particle Size (Diameter - μm)	Stokes Number	Reasoning
1 micron	4.07E-04	Particles follow streamlines and diffuse to alveoli at far branches
10 micron	4.07E-02	Particles will be filtered in higher regions due to inertia
100 micron	4.07	Inertia completely dominates

Hypotheses of Particle Behavior based on Stokes Number Calculations:

- *Particles with $Stk < 1$* , such as the 1 μm particles, will follow the fluid streamlines, and will be taken up by the further branches and alveoli through diffusion.
- *Particles with $Stk \approx 1$* , such as the 10 μm particles, will be influenced by inertia and will be mechanically filtered by the branching in the upper regions of the lung. However, a significant portion is expected to escape further into the lung.
- *Particles with $Stk > 1$* , such as the 100 μm particles, will be severely impacted by inertia, and almost all will be trapped in the upper regions of the lung.

2.5.3 Discrete Random Walk

To account for the effect of turbulent velocity fluctuations on particle trajectories, the Discrete Random Walk (DRW) model within the Discrete Phase Model was enabled. Rather than tracking parcels through the mean velocity field alone, the DRW model superimposes a stochastic fluctuating velocity component:

$$u' = \zeta \sqrt{u'^2} \quad \text{Equation 12.27 in Fluent Theory Guide [4]}$$

$$\sqrt{u'^2} = \sqrt{v'^2} = \sqrt{w'^2} = \sqrt{2k/3} \quad \text{Equation 12.28 in Fluent Theory Guide [4]}$$

, where ζ represents the Gaussian random number and k is the local turbulent kinetic energy from the SST k - ω model (held constant over each eddy's characteristic lifetime before being resampled). The **number of tries** parameter in Fluent, which we set to a 100 tries for steady-state simulations, specifies how many independent random realizations of the turbulent field each parcel experiences: repeating the trajectory with different random walk seeds and averaging the results captures the statistical effect of turbulent dispersion on deposition. Note that for unsteady tracking, this value is set to 1 and not adjustable [11]. Intuitively, it makes sense that the higher the number of tries, the better, as the randomness due to turbulence will be minimized. An introductory workshop document provided by ANSYS for starting with the Discrete Phase Model suggests a value of 10 tries [12]. However, we set this value to 100 tries consistent with Sturm's findings for turbulent dispersion in the lungs [13]. A potential future study might consider evaluating different numbers of tries similar to our parcel independence analysis discussed in section 3.1.2.1, especially if considering a 3D geometry with more complex turbulent flow features. However, it was a design consideration on our end to not pursue this further, due to limited time and the simplified complexity of the 2D domain.

2.6 Model Setup and Equations to be Solved

2.6.1 Turbulence Model

The turbulence model used is SST k - ω with low-Re corrections. This was chosen because it has good near-wall resolution (which is important for particle deposition) and has good wall separation and recirculation modeling. The low-Re correction was chosen because we are operating in the transitional and laminar regimes.

2.6.2 Boundary Conditions

Table 2: Particle and fluid boundary conditions for steady state and transient flow

Boundary Condition:		Fluid	Particle
Inlet	Steady State	Velocity Inlet: 2.5 m/s	Group Injection (1μm, 10μm, and 100μm Particles)
	Transient	Velocity Inlet: $v = -2.5 \sin(\pi(t - 1))$,	Group Injection (1μm, 10μm, and 100μm Particles), 1000 streams per timestep
Outlets		Outflow - Weighted by Generation	Escape
Walls		No Slip	Trap

Note: Density of the particles were defined to be 1000 kg/m³.

In section 2.3.2 (Steady-State Assumptions), we had mentioned that the mass flow rate of the system is arbitrarily set, since one-way coupling is active and particle-particle or particle-fluid interactions are not considered. Furthermore, since Fluent performs calculations on parcels, the specific number of particles is irrelevant. However, we believe that it is still beneficial to calculate the number of particles based on the material and geometric properties of the particles to get a physical intuition. Assuming 10μg/s (1e-8 kg/s) of a drug reaches the trachea inlet G0, along with the particle density of 1000 kg/m³ and mass of a single spherical particle $m = \rho(\frac{\pi}{6})d^3$:

Table 3: Single Particle Mass and Total Particle Count

Size	Single Particle Mass	Particles per Second
1 μm	5.24E-16 kg	~ 19.1 million
10 μm	5.24E-13 kg	~ 19,100
100 μm	5.24E-10 kg	~ 19

Based on these calculations, alongside the fact that the steady-state results in section 3.1.1 use 500 streams x 100 DRW tries = 50,000 parcels, the total particles entering the

domain per second are significantly less than the 50,000 parcels tracked in the simulation. This is not a contradiction: as mentioned, under one-way coupling, parcels are statistical trajectory samples rather than representations of individual particles, and the mass flow rate of 10 $\mu\text{g/s}$ set is nominal. In practice, aerosolized drug dosages have particles with diameters on the order of magnitude of $1\text{e-}0\ \mu\text{m}$, since the particles of this size are small enough to pass the upper airway and deposit into the deeper alveolar regions of the lung, where drug uptake occurs through diffusion [8].

2.6.3 Transient Specific Conditions

Table 4: Transient simulation timestep setup

Time Step Size [s]	No. of Time Steps	Total Time [s]	No. of Breaths
0.005	600	3	1.5

2.7 Mesh

Three structured meshes were generated using ANSYS Meshing, with the intention of performing a Grid Independence Study, further discussed in section 3.1.2.2. The meshing strategy targeted approximately 12 elements across the width of each airway branch on the **coarsest mesh**, with successive refinements by a factor of 2 in the face sizing to produce the medium and fine meshes (note that this was misconveyed during the final presentation). This was due to the initial attempt of trying to have a refinement ratio of 2 between meshes, and simplify the grid independence process. However, an initial coarse mesh was constructed, but yielded a refinement ratio below the minimum threshold of 1.3 required by Çelik et al. GCI procedure [14]; consequently, the original medium mesh was redesignated as the current coarse mesh, and the new medium and fine meshes were regenerated at higher resolution, producing refinement ratios of $r_{32}=1.45$, and $r_{21}=1.73$ both satisfying the criterion.

Face sizing was applied with a soft behavior setting rather than a hard constraint in order to enable the Capture Curvature option with a curvature normal angle of 5° as well as adding an inflation layer. This was a deliberate choice: a hard face sizing enforces uniform element size across the entire face, but the bifurcations and surrounding walls require locally increased resolution to accurately capture the flow separation and particle impact behaviors at those critical sites. 10 inflation layers were applied on all wall boundaries with a growth layer of 1.2 and a maximum total thickness of 0.4 mm, which ultimately provided adequate near-wall resolution for the Low-Re SST $k-\omega$ turbulence model, further discussed next. These specifications were determined through trial and error, starting with the mesh settings used for Assignment 5 where grid

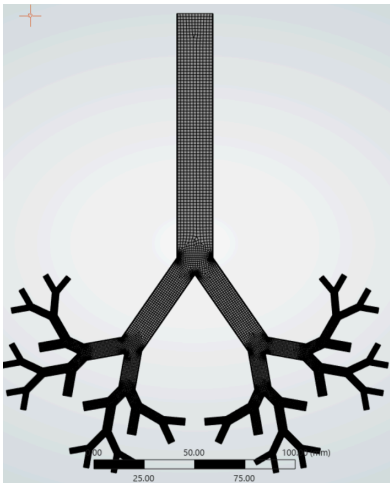
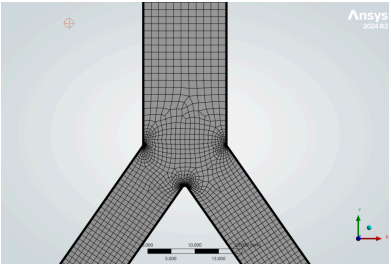
independence was first introduced in the course, and also considering the settings in Assignments 4 where a similar geometry was considered with the bifurcating artery.

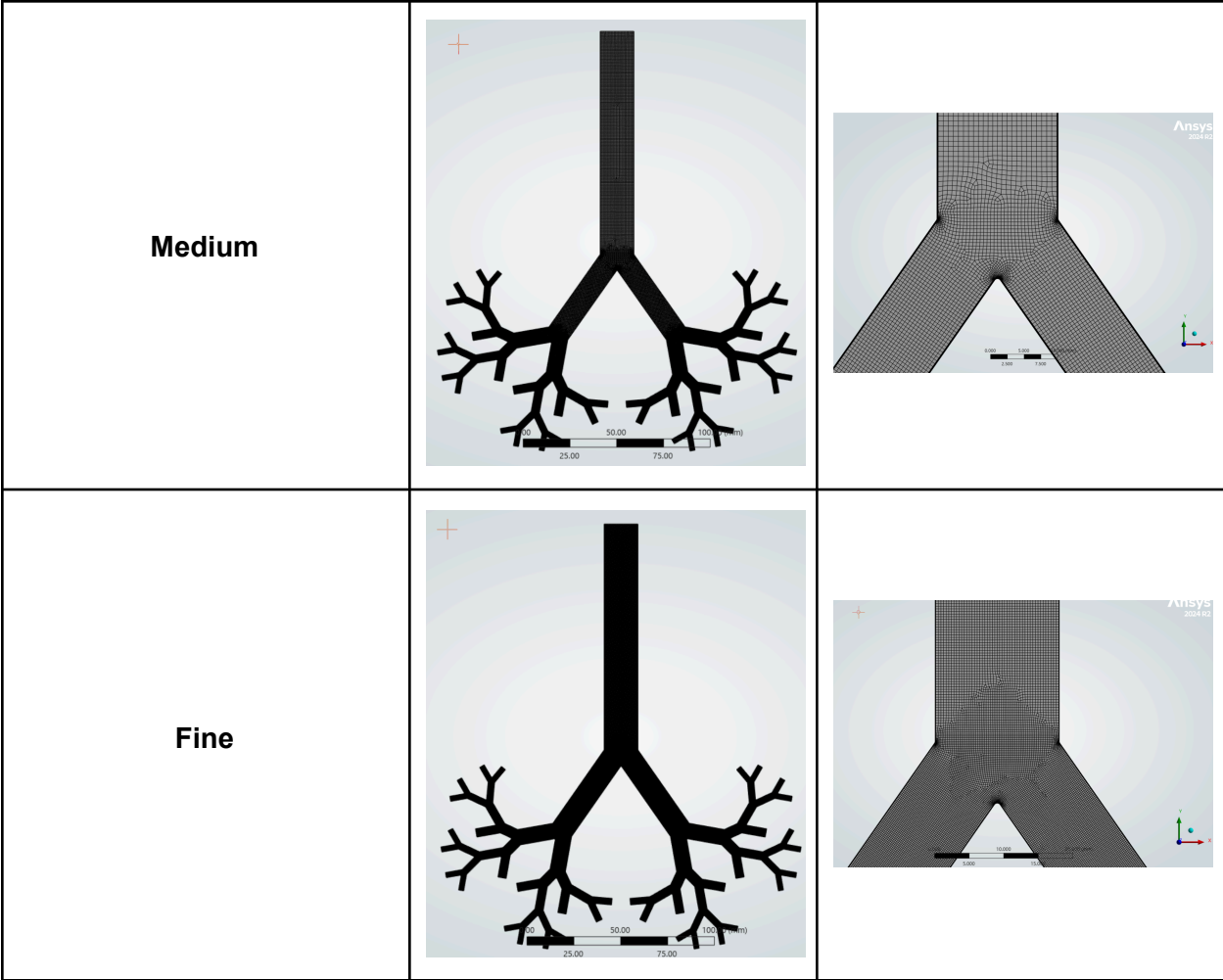
The resulting wall y^+ values confirmed that the inflation later sizing was appropriate for the chosen turbulence model. Even on the coarsest mesh, the maximum y^+ was 0.39, well below the $y^+ < 1$ threshold required (introduced in turbulence modeling lectures) for the SST $k-\omega$ model to resolve the viscous sublayer directly rather than relying on wall functions. These y^+ values, seen in Table 5, verify the near-wall velocity gradients that govern the wall shear stress and by extension, the near-wall particle deposition behavior, are fully resolved across all 3 meshes.

Table 5: Mesh Element Statistics and Wall y^+ Metrics

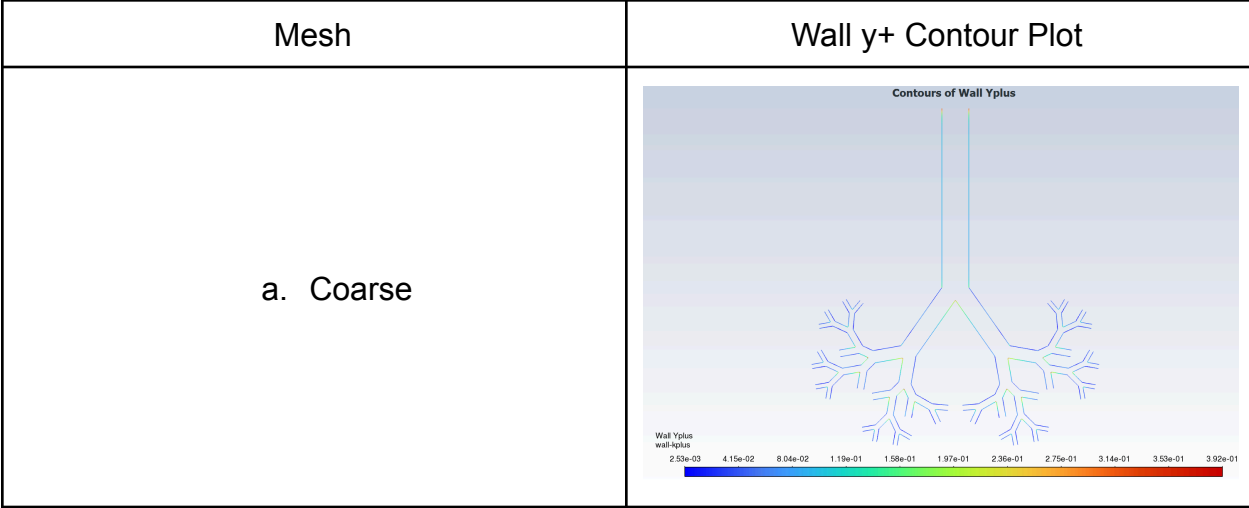
Mesh:	Coarse	Medium	Fine
Nodes:	113105	234519	698788
Elements:	109206	228640	688209
Wall y^+ min	0.002531	0.001258	0.000584
Wall y^+ max	0.391986	0.212602	0.113257

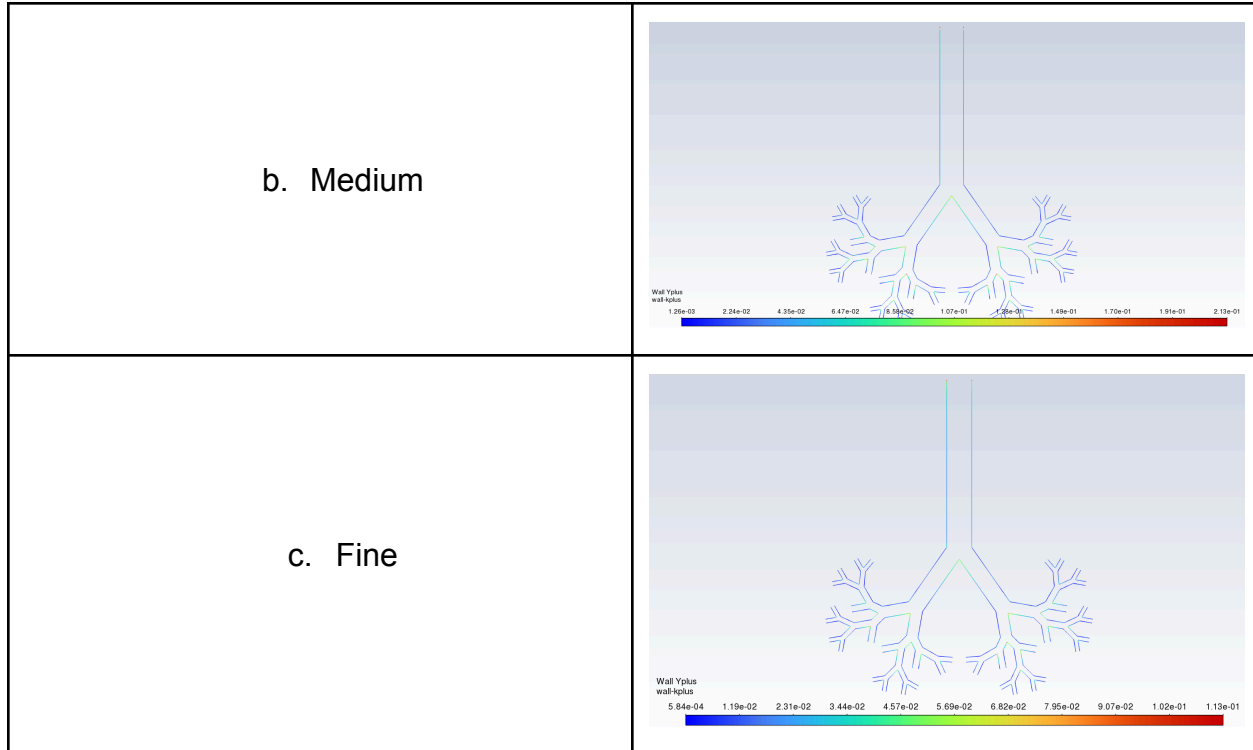
Table 6: Full Airway Domain, and Carina Bifurcation Mesh Visualizations

Mesh	Full Image	Carina (G0 to G1 Transition)
Coarse		



Figures 10a , 10b, 10c: Wall y^+ Contours for the 3 Meshes. All $y^+_{max} < 1$:





3. Results

3.1 Steady State Flow with Discrete Phase

3.1.1 $1\mu\text{m}$, $10\mu\text{m}$, and $100\mu\text{m}$ Cases

Figure 11 shows the overall deposition fraction for each particle size on the fine mesh simulations with 500 streams and 100 DRW tries. The results span three orders of magnitude in deposition efficiency, out of 50,000 parcels: 16, 10,345, and all 50,000 parcels were trapped respectively. This progression is consistent with our hypotheses based on the Stokes number calculations, suggesting that particles with larger diameters increasingly being unable to follow the fluid streamlines.

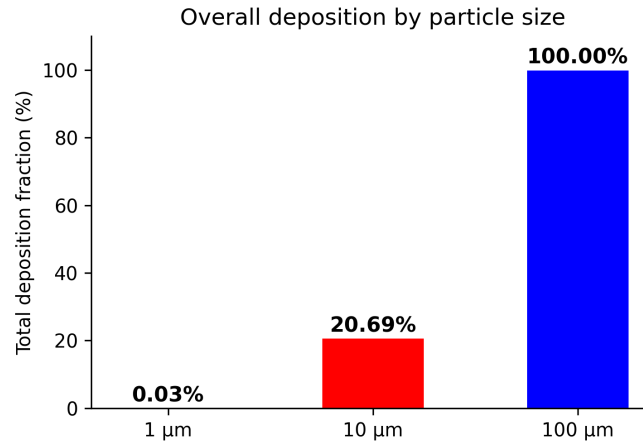


Figure11: Steady-State Trials, Overall deposition fraction by particles of size 1μm, 10μm, 100μm

The local deposition fractions by generation, shown in Figure 12a, 12b reveal distinct spatial deposition patterns for each parcel size. For the 100μm case, 99.2% of the particles deposited in the G1 walls. These particles have such high inertia that they impact the carina almost immediately upon entering the trachea, and none escape the domain. The 10μm case exhibits a distributed deposition pattern across all 6 generations, with G4 collecting the largest share, which may be due to the limitation of the 2D model allowing for the most walls in our domain to be in G4, or due to the flow profile changing from branching; however, a full lung model in 3D would need to be performed to verify that. The 1μm particles deposited in negligible quantities across all generations. At this size, the Stokes number is much less than 1, therefore the particles follow the streamlines and escape as expected.

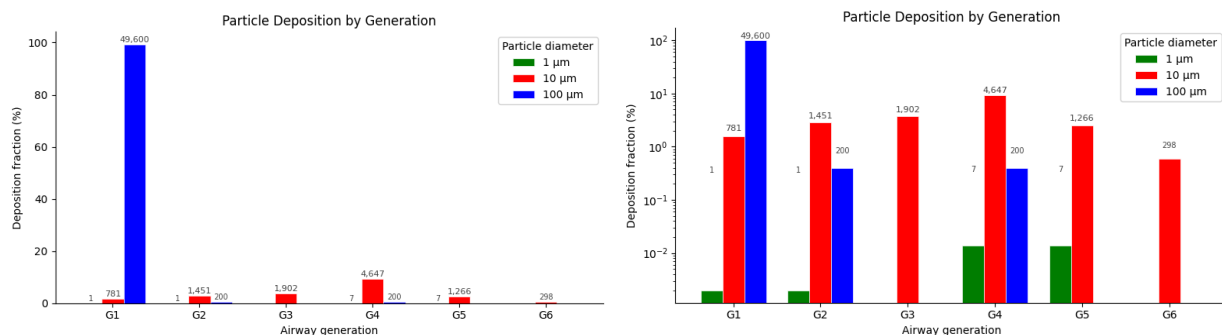


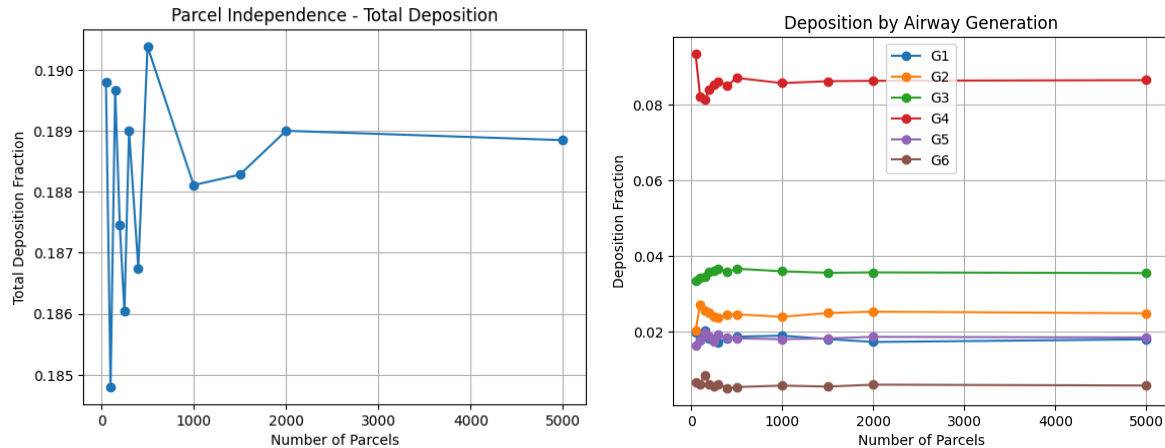
Figure 12a, 12b: Steady-State Trials, Local Particle Deposition Fraction by Generation for Particles of Size 1μm, 10μm, 100μm, represented in standard axis and log axis for easier visualization

3.1.2 Verification

All data, calculations, tables, and plots for the verification section can be found within the associated sections in the Appendix.

3.1.2.1 Parcel Independence

A parcel independence study was conducted on the coarse mesh using $10\mu\text{m}$ particles with 100 DRW tries, varying the number of injection streams across 12 values (50, 100, 150, 200, 250, 300, 400, 500, 1000, 1500, 2000, 5000). The purpose of this study was to establish that the parcel sample size is sufficiently large enough to produce deposition statistics representative of the true particle behavior, independent of stream count. As shown in Figure 13a, the total deposition fraction oscillates between ~ 0.180 to 0.195 , a narrow enough region to be attributed to the stochastic nature of the DRW model. However, to confirm parcel independence the per-generation deposition fractions were also examined, as seen in Figure 13b, which shows that all 6 generations converge by 500 streams. Based on these results, a stream count of 500 was adopted for all steady-state simulations performed on the fine mesh.



Figures 13a and 13b: Parcel Independence Study Total Deposition Fraction and Local/Per-Generation Deposition

3.1.2.2 Grid Independence

The GCI results in Table 7 demonstrate strong grid independence for the flow quantities. The inlet total pressure and total pressure drop (mass-weighted average) both achieved GCI_{21} values below 0.15%.

It should be noted that in the previous iteration of the analysis, presented during the final presentation, an area-weighted *static* pressure was accidentally used as the scalar quantity, producing an apparent jump between mesh results. The static pressure quantity does not account for the kinetic energy of the flow, making it a misleading quantity. The finalized analysis uses mass-flow weighted total pressure, which includes

static and dynamic pressure contributions; capturing the true irreversible viscous loss through the domain. The resulting values for total inlet and total pressure drop show a physically consistent progression that resolves the previous concern.

The overall deposition fraction exhibited oscillatory convergence with $GCI_{21} = \sim 7\%$, which is acceptable considering the stochasticity within the DPM model due to the DRW. Similar behaviors can be observed in the local generation deposition fractions too, with the most evident one being in G1 with a $GCI_{21} = \sim 75\%$ followed by G4 with $GCI_{21} = \sim 16\%$. This behavior can again be attributed to stochasticity of the Lagrangian DPM, rather than a dependence on the grid. As aforementioned, further studies varying the DRW tries will likely provide some further clarity.

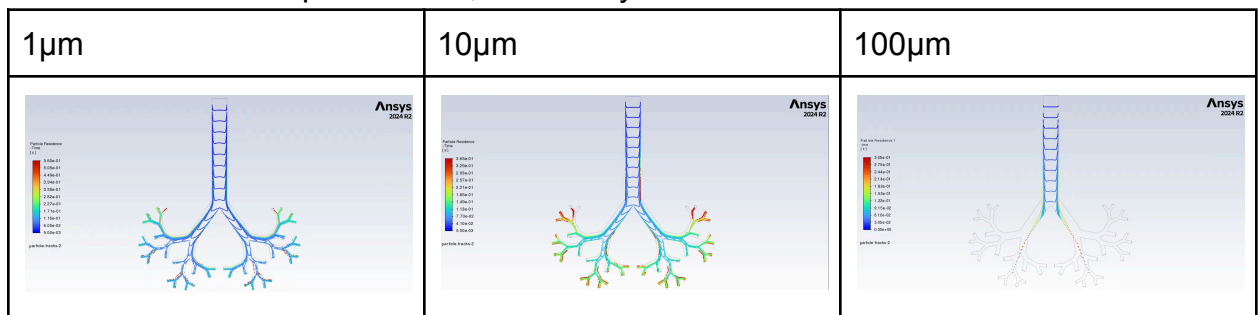
Table 7: Grid Independence Calculations based on Çelik et al. [14]

Variable	f1 (fine)	f2 (medium)	f3 (coarse)	Convergence	p	GCI_{21} (%)	GCI_{32} (%)	AR	e_{ext} (%)
Inlet total pressure [Pa]	4.2616	4.2199	4.0497	Monotonic	4.17	0.14	1.37	1.01	0.11
Total pressure drop [Pa]	2.6519	2.6657	2.4378	Oscillatory	7.47	0.01	0.72	1.09	0.01
Overall deposition fraction	0.2069	0.225	0.1875	Oscillatory	1.71	7.02	23.74	1.32	5.95
Dep. fraction G1	0.0156	0.0209	0.0178	Oscillatory	0.81	75.3	53.2	0.45	151.3
Dep. fraction G2	0.029	0.0285	0.0245	Monotonic	5.86	0.09	2.28	1.02	0.07
Dep. fraction G3	0.038	0.0406	0.0356	Oscillatory	1.59	5.91	19.2	1.35	4.96
Dep. fraction G4	0.0929	0.1041	0.0854	Oscillatory	1.18	16.3	40.9	1.31	15
Dep. fraction G5	0.0253	0.0243	0.0193	Monotonic	4.7	0.4	5.56	1.04	0.32
Dep. fraction G6	0.006	0.0067	0.0049	Oscillatory	2.18	6.33	26.4	1.26	5.34

3.2 Transient Flow with Discrete Phase

3.2.1 Particle Flow and Residence Times

Table 8: Contours of particle flow, colored by residence time



3.2.2 Particle Accretion Rates

Table 9: Graphs of particle accretion rate over the length of the simulation

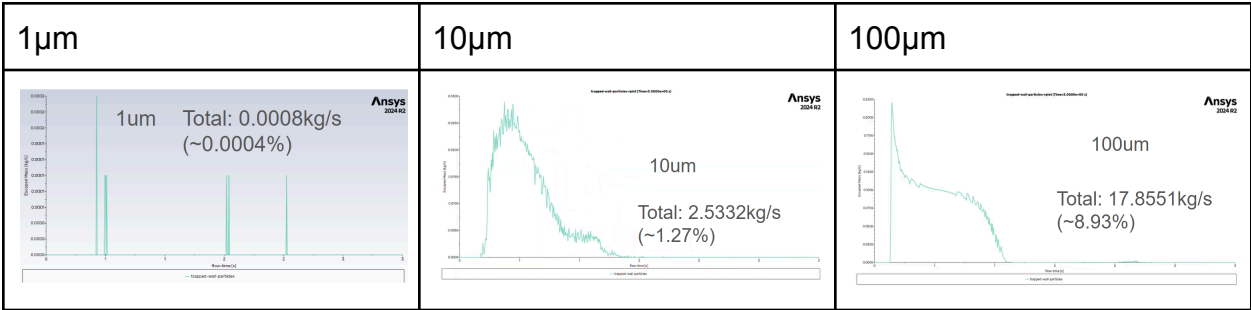
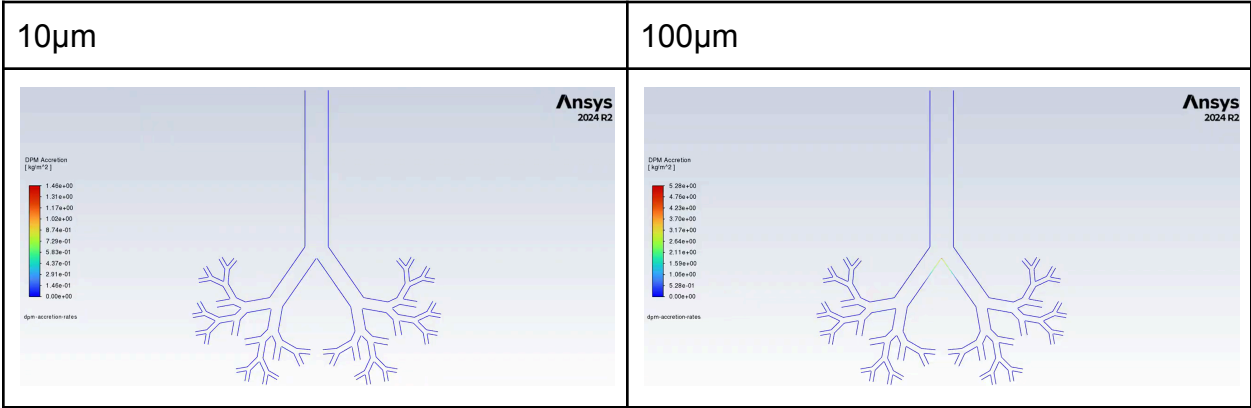


Table 10: Contours of particle accretion on lung walls



The contour for the 1µm particles is not shown because there was no meaningful accretion, as shown in Table 11.

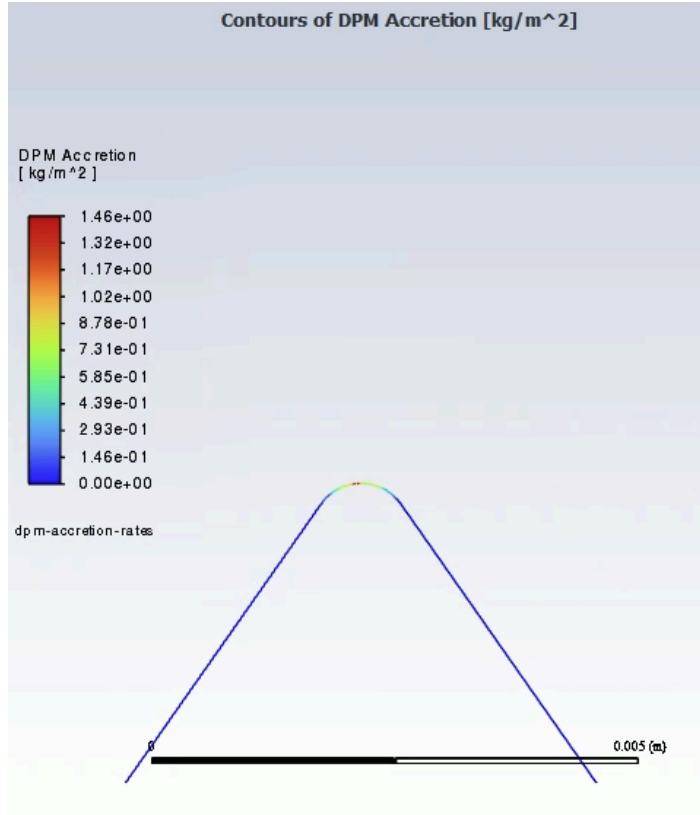


Figure 14: Zoomed in view of particle deposition contour centered on the carina (transition between G0 and G1) for 10 μ m

4. Discussion

4.1 Steady State

The velocity profile found in Case 1 (without particles) was qualitatively validated by comparing to existing literature contours [6,15,16]. Our flow conditions successfully replicated these results, giving us confidence that we can move on to adding particles. Quantitative validation was primarily evaluated using the 10 μ m particles, which exhibit a transitional behavior ($Stk \approx 1$) and distributed across all six generations. Our steady-state deposition fractions were compared against 3D computational data [6], supported by the idealized tracheobronchial experimental trends observed by Oldham [16]. The 2D model successfully captured the relative deposition trends seen in the literature, specifically highlighting a significant deposition spike at Generation 4 (9.12% in our model vs. 4.57% in literature), driven by airway narrowing and increased inertial impaction. While the relative trends align well, the absolute deposition values in our middle generations are noticeably higher than the 3D literature values. In a realistic 3D airway, complex secondary flows develop and help distribute particles or carry them

around the bifurcations. Because our model is 2D, we lose those flow characteristics. This discrepancy is a direct result of our 2D geometry assumption.

Airway Generation	Our 10 μm Fine Mesh Results (DF %)	Literature 10 μm (Simulation) [6]	Literature 3 μm (Simulation) [6]
G1 (Bifurcation 1)	1.70%	3.49%	0.55%
G2 (Bifurcation 2)	3.01%	0.96%	0.46%
G3 (Bifurcation 3)	3.78%	1.57%	0.48%
G4 (Bifurcation 4)	9.12%	4.57%	1.05%
G5 (Bifurcation 5)	2.48%	3.12%	0.89%
G6 (Bifurcation 6)	0.65%	4.21%	0.83%

Table 12: Steady-State Validation: 10 μm Particle Deposition Fractions vs. Literatures

4.2 Transient

The results were as expected from the theory and followed the trends set by the steady state case. There was a significant increase in the rate of particle deposition as the diameter of the particle increased.

The transient results followed the steady state findings until the first exhalation. At that point, the particles reversed direction, giving them more opportunity to collide and deposit on the airway walls. Due to the dominance of inertial forces in the 100 μm case, the introduction of an exhalation did not impact the deposition, as those particles are dominated by inertial forces, not airflow.

5. Conclusions and Future Steps

This report successfully quantified and compared the airflow dynamics and local deposition efficiencies of microparticles in a simplified 2D tracheobronchial tree model under both steady-state and transient sinusoidal breathing conditions. Future studies should transition to a fully 3D, asymmetric lung geometry to properly capture out-of-plane secondary flows that naturally mitigate blunt impaction at the carinal ridges.

6. References

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7. Appendices

7.1 Appendix A: Steady State Particle Size Data with Ultra-Fine, 500 Parcels/Streams

```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model  
Specific)*PushButton1(Discrete Phase)")  
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")  
(cx-gui-do cx-set-list-selections "Sample Trajectories*Table1*Table3*List1(Release from  
Injections)" '( 0))  
(cx-gui-do cx-activate-item "Sample Trajectories*Table1*Table3*List1(Release from  
Injections)")  
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 50000, escaped = 49984, trapped = 16

```
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton2(Close)")
```

```
adjoint/      mesh/      report/  
close-fluent  parallel/  solve/  
define/       plot/      surface/  
display/      preferences/ views/  
file/         print-license-usage
```

```
> report/dpm-summary
```

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Trapped	walls-g1	7	1	1.714e+00	1.714e+00	1.714e+00	0.000e+00
injection-1micr	34497	injection-1micr	34497				
Trapped	walls-g2	8	1	6.900e-02	6.900e-02	6.900e-02	0.000e+00
injection-1micr	37370	injection-1micr	37370				
Trapped	walls-g4	10	7	4.023e-01	1.471e+00	1.024e+00	3.345e-01
injection-1micr	5401	injection-1micr	29496				

Trapped	walls-g5	11	7	4.567e-01	1.254e+00	8.522e-01	2.859e-01
injection-1micr	3654 injection-1micr	46416					
Escaped	outlet-g6-1	14	1571	1.406e-01	2.828e+00	5.016e-01	4.586e-01
injection-1micr	8500 injection-1micr	8993					
Escaped	outlet-g6-2	15	534	1.373e-01	3.397e+00	2.758e-01	4.067e-01
injection-1micr	12035 injection-1micr	6427					
Escaped	outlet-g6-3	16	1382	1.375e-01	6.435e-01	2.354e-01	6.698e-02
injection-1micr	7738 injection-1micr	5622					
Escaped	outlet-g6-4	17	824	1.368e-01	1.346e+00	1.865e-01	5.949e-02
injection-1micr	20644 injection-1micr	1169					
Escaped	outlet-g6-5	18	873	1.281e-01	1.289e+00	2.234e-01	1.362e-01
injection-1micr	22117 injection-1micr	1740					
Escaped	outlet-g6-6	19	383	1.269e-01	4.992e-01	1.593e-01	4.500e-02
injection-1micr	8673 injection-1micr	11106					
Escaped	outlet-g6-7	20	433	1.186e-01	6.021e-01	1.429e-01	4.588e-02
injection-1micr	12391 injection-1micr	8961					
Escaped	outlet-g6-8	21	743	1.176e-01	8.441e-01	1.621e-01	7.570e-02
injection-1micr	12190 injection-1micr	8182					
Escaped	outlet-g6-9	22	524	1.087e-01	5.452e-01	1.335e-01	3.515e-02
injection-1micr	16582 injection-1micr	31956					
Escaped	outlet-g6-10	23	367	1.098e-01	5.289e-01	1.294e-01	3.248e-02
injection-1micr	25646 injection-1micr	13668					
Escaped	outlet-g6-11	24	653	1.176e-01	3.881e-01	1.329e-01	2.403e-02
injection-1micr	7646 injection-1micr	9085					
Escaped	outlet-g6-12	25	747	1.181e-01	2.390e+00	1.581e-01	1.031e-01
injection-1micr	25226 injection-1micr	28496					
Escaped	outlet-g6-13	26	756	1.204e-01	8.814e-01	1.539e-01	5.897e-02
injection-1micr	38967 injection-1micr	25343					
Escaped	outlet-g6-14	27	624	1.178e-01	3.159e-01	1.338e-01	2.388e-02
injection-1micr	43703 injection-1micr	22224					
Escaped	outlet-g6-15	28	327	1.098e-01	2.412e-01	1.257e-01	1.943e-02
injection-1micr	37403 injection-1micr	37048					
Escaped	outlet-g6-16	29	506	1.132e-01	5.187e-01	1.327e-01	3.080e-02
injection-1micr	25662 injection-1micr	31616					
Escaped	outlet-g6-17	30	753	1.214e-01	1.483e+00	1.756e-01	1.116e-01
injection-1micr	45989 injection-1micr	48081					
Escaped	outlet-g6-18	31	419	1.177e-01	9.325e-01	1.431e-01	5.163e-02
injection-1micr	37295 injection-1micr	43919					
Escaped	outlet-g6-19	32	430	1.276e-01	3.775e-01	1.559e-01	3.173e-02
injection-1micr	26646 injection-1micr	38993					

Escaped	outlet-g6-20	33	873	1.276e-01	1.095e+00	2.108e-01	1.153e-01
injection-1micr	39175	injection-1micr	32709				
Escaped	outlet-g6-21	34	845	1.384e-01	6.112e-01	1.837e-01	3.659e-02
injection-1micr	37775	injection-1micr	40514				
Escaped	outlet-g6-22	35	1579	1.351e-01	8.493e-01	2.244e-01	6.340e-02
injection-1micr	42362	injection-1micr	48561				
Escaped	outlet-g6-23	36	374	1.349e-01	4.412e+00	3.604e-01	6.081e-01
injection-1micr	38142	injection-1micr	37838				
Escaped	outlet-g6-24	37	1622	1.442e-01	2.622e+00	4.921e-01	
4.374e-01	injection-1micr	39797	injection-1micr	46358			
Escaped	outlet-g5-1	38	1830	1.023e-01	9.846e-01	1.266e-01	3.704e-02
injection-1micr	17402	injection-1micr	23113				
Escaped	outlet-g5-2	39	1824	1.038e-01	1.993e+00	1.935e-01	1.526e-01
injection-1micr	15324	injection-1micr	15563				
Escaped	outlet-g5-3	40	1872	1.017e-01	2.293e+00	1.977e-01	1.675e-01
injection-1micr	41487	injection-1micr	45005				
Escaped	outlet-g5-4	41	1817	1.024e-01	9.035e-01	1.262e-01	3.090e-02
injection-1micr	27571	injection-1micr	25811				
Escaped	outlet-g4-1	42	2254	8.864e-02	1.689e+00	1.287e-01	8.399e-02
injection-1micr	18728	injection-1micr	18338				
Escaped	outlet-g4-2	43	4064	8.255e-02	1.510e+00	9.380e-02	2.839e-02
injection-1micr	17342	injection-1micr	3798				
Escaped	outlet-g4-3	44	3239	8.014e-02	2.168e-01	9.041e-02	8.235e-03
injection-1micr	15639	injection-1micr	14329				
Escaped	outlet-g4-4	45	2826	7.991e-02	5.190e-01	9.077e-02	1.537e-02
injection-1micr	20572	injection-1micr	12403				
Escaped	outlet-g4-5	46	2767	8.086e-02	7.805e-01	9.108e-02	1.942e-02
injection-1micr	43527	injection-1micr	26735				
Escaped	outlet-g4-6	47	3141	8.040e-02	2.581e-01	9.102e-02	9.923e-03
injection-1micr	19333	injection-1micr	44630				
Escaped	outlet-g4-7	48	4014	8.274e-02	7.469e-01	9.404e-02	2.456e-02
injection-1micr	28636	injection-1micr	47934				
Escaped	outlet-g4-8	49	2194	8.632e-02	1.653e+00	1.278e-01	8.266e-02
injection-1micr	28064	injection-1micr	35666				

(*)- Mass Transfer Summary -(*)

Fate	Zone	Zone	Mass Flow (kg/s)		
	Name	Id	Initial	Final	Change

Trapped	walls-g1	7	2.000e-13	2.000e-13	0.000e+00
Trapped	walls-g2	8	2.000e-13	2.000e-13	0.000e+00
Trapped	walls-g4	10	1.400e-12	1.400e-12	0.000e+00
Trapped	walls-g5	11	1.400e-12	1.400e-12	0.000e+00
Escaped	outlet-g6-1	14	3.142e-10	3.142e-10	0.000e+00
Escaped	outlet-g6-2	15	1.068e-10	1.068e-10	0.000e+00
Escaped	outlet-g6-3	16	2.764e-10	2.764e-10	0.000e+00
Escaped	outlet-g6-4	17	1.648e-10	1.648e-10	0.000e+00
Escaped	outlet-g6-5	18	1.746e-10	1.746e-10	0.000e+00
Escaped	outlet-g6-6	19	7.660e-11	7.660e-11	0.000e+00
Escaped	outlet-g6-7	20	8.660e-11	8.660e-11	0.000e+00
Escaped	outlet-g6-8	21	1.486e-10	1.486e-10	0.000e+00
Escaped	outlet-g6-9	22	1.048e-10	1.048e-10	0.000e+00
Escaped	outlet-g6-10	23	7.340e-11	7.340e-11	0.000e+00
Escaped	outlet-g6-11	24	1.306e-10	1.306e-10	0.000e+00
Escaped	outlet-g6-12	25	1.494e-10	1.494e-10	0.000e+00
Escaped	outlet-g6-13	26	1.512e-10	1.512e-10	0.000e+00
Escaped	outlet-g6-14	27	1.248e-10	1.248e-10	0.000e+00
Escaped	outlet-g6-15	28	6.540e-11	6.540e-11	0.000e+00
Escaped	outlet-g6-16	29	1.012e-10	1.012e-10	0.000e+00
Escaped	outlet-g6-17	30	1.506e-10	1.506e-10	0.000e+00
Escaped	outlet-g6-18	31	8.380e-11	8.380e-11	0.000e+00
Escaped	outlet-g6-19	32	8.600e-11	8.600e-11	0.000e+00
Escaped	outlet-g6-20	33	1.746e-10	1.746e-10	0.000e+00
Escaped	outlet-g6-21	34	1.690e-10	1.690e-10	0.000e+00
Escaped	outlet-g6-22	35	3.158e-10	3.158e-10	0.000e+00
Escaped	outlet-g6-23	36	7.480e-11	7.480e-11	0.000e+00
Escaped	outlet-g6-24	37	3.244e-10	3.244e-10	0.000e+00
Escaped	outlet-g5-1	38	3.660e-10	3.660e-10	0.000e+00
Escaped	outlet-g5-2	39	3.648e-10	3.648e-10	0.000e+00
Escaped	outlet-g5-3	40	3.744e-10	3.744e-10	0.000e+00
Escaped	outlet-g5-4	41	3.634e-10	3.634e-10	0.000e+00
Escaped	outlet-g4-1	42	4.508e-10	4.508e-10	0.000e+00
Escaped	outlet-g4-2	43	8.128e-10	8.128e-10	0.000e+00
Escaped	outlet-g4-3	44	6.478e-10	6.478e-10	0.000e+00
Escaped	outlet-g4-4	45	5.652e-10	5.652e-10	0.000e+00
Escaped	outlet-g4-5	46	5.534e-10	5.534e-10	0.000e+00
Escaped	outlet-g4-6	47	6.282e-10	6.282e-10	0.000e+00
Escaped	outlet-g4-7	48	8.028e-10	8.028e-10	0.000e+00
Escaped	outlet-g4-8	49	4.388e-10	4.388e-10	0.000e+00

Net	1.000e-08	1.000e-08	0.000e+00
-----	-----------	-----------	-----------

```
> (cx-gui-do cx-activate-item
"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface
Integrals)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1 3
4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
34 35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1 2
3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
34 35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1 3
4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
34 35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

inlet	4.2615679
-------	-----------

```
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '())
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 3 4
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

outlet-g4-1	1.1612432
outlet-g4-2	1.7586683
outlet-g4-3	2.1326607

outlet-g4-4	2.3090998
outlet-g4-5	2.3109999
outlet-g4-6	2.1379088
outlet-g4-7	1.7591435
outlet-g4-8	1.160557
outlet-g5-1	1.186605
outlet-g5-2	1.1178511
outlet-g5-3	1.1183198
outlet-g5-4	1.186365
outlet-g6-1	0.49648957
outlet-g6-10	2.1134732
outlet-g6-11	2.0753322
outlet-g6-12	2.0604963
outlet-g6-13	2.0608151
outlet-g6-14	2.0757107
outlet-g6-15	2.1141049
outlet-g6-16	2.1143213
outlet-g6-17	1.7865428
outlet-g6-18	1.7855513
outlet-g6-19	1.7047645
outlet-g6-2	0.50153705
outlet-g6-20	1.699661
outlet-g6-21	0.54012823
outlet-g6-22	0.54654743
outlet-g6-23	0.5003054
outlet-g6-24	0.4953133
outlet-g6-3	0.54753172
outlet-g6-4	0.54123598
outlet-g6-5	1.700937
outlet-g6-6	1.7061205
outlet-g6-7	1.7867454
outlet-g6-8	1.787809
outlet-g6-9	2.1130588

Net 1.6097149

(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton2(Cancel)")
/file/stop-transcript

Performance Timer for 225 iterations on 4 compute nodes
Average wall-clock time per iteration: 3.055 sec

Global reductions per iteration:	378 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	12133 messages
Data transfer per iteration:	1.789 MB
LE solves per iteration:	3 solves
LE wall-clock time per iteration:	2.562 sec (83.8%)
LE global solves per iteration:	8 solves
LE global wall-clock time per iteration:	0.003 sec (0.1%)
LE global matrix maximum size:	96
AMG cycles per iteration:	8.591 cycles
Relaxation sweeps per iteration:	3930 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.004 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	406
DPM updates per iteration:	0.0400 updates
DPM wall-clock time per iteration:	2.935 sec (96.1%)

Total wall-clock time:	687.391 sec
Total dpm solve time:	660.459 sec
Total dpm i/o time:	0.000 sec

—
Transcript Start Time: 18:27:11, 15 Mar 2026 Central Daylight Time

Current Directory:

"C:\Users\kaan\Desktop\Lung-Flow-Steady-NoP\lung-2024-final_files\dp0\FFF-4\Fluent"

Build Time: May 13 2024 11:28:30 EST

Executable Path: C:/PROGRA~1/ANSYSI~1/v242/fluent/ntbin/win64/fluent.exe

ID	Hostname	Core	O.S.	PID	Vendor
n3	Borekci-X1	4/14	Windows-x64	15564	Intel(R) Core(TM) Ultra 7 165U
n2	Borekci-X1	3/14	Windows-x64	16216	Intel(R) Core(TM) Ultra 7 165U
n1	Borekci-X1	2/14	Windows-x64	15860	Intel(R) Core(TM) Ultra 7 165U
n0*	Borekci-X1	1/14	Windows-x64	20636	Intel(R) Core(TM) Ultra 7 165U
host	Borekci-X1		Windows-x64	10356	Intel(R) Core(TM) Ultra 7 165U

MPI Option Selected: intel

Selected system interconnect: default

Cortex Process ID: 32784

License Server Path: 1055@rho.mcc.northwestern.edu

```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model
Specific)*PushButton1(Discrete Phase)")
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 50000, escaped = 39655, trapped = 10345

```
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton2(Close)")
```

```
(cx-gui-do cx-activate-item
```

```
"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface
Integrals)")
```

```
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton2(Cancel)")
```

```
(cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model
Specific)*PushButton1(Discrete Phase)")
```

adjoint/	mesh/	report/
close-fluent	parallel/	solve/
define/	plot/	surface/
display/	preferences/	views/
file/	print-license-usage	

```
> report/dpm-summary
```

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Trapped	walls-g1	7	781	4.476e-02	7.354e-01	6.358e-02	3.446e-02
injection-10mic	23999	injection-10mic	48923				

Trapped	walls-g2	8	1451	6.190e-02	1.220e-01	7.172e-02	1.144e-02
injection-10mic 47230	injection-10mic 34484						
Trapped	walls-g3	9	1902	6.913e-02	4.282e-01	7.958e-02	1.461e-02
injection-10mic 24078	injection-10mic 8164						
Trapped	walls-g4	10	4647	7.460e-02	6.493e-01	9.369e-02	3.866e-02
injection-10mic 15639	injection-10mic 11211						
Trapped	walls-g5	11	1266	8.281e-02	4.424e-01	1.064e-01	4.783e-02
injection-10mic 41867	injection-10mic 19252						
Trapped	walls-g6	12	298	9.779e-02	8.655e-01	2.033e-01	1.231e-01
injection-10mic 12243	injection-10mic 44991						
Escaped	outlet-g6-1	14	1101	1.455e-01	1.852e+00	4.501e-01	3.932e-01
injection-10mic 4655	injection-10mic 8728						
Escaped	outlet-g6-2	15	412	1.386e-01	3.986e-01	2.914e-01	7.998e-02
injection-10mic 8454	injection-10mic 196						
Escaped	outlet-g6-3	16	538	1.397e-01	4.363e-01	2.084e-01	3.067e-02
injection-10mic 36173	injection-10mic 3261						
Escaped	outlet-g6-4	17	715	1.396e-01	6.238e-01	1.905e-01	3.321e-02
injection-10mic 11049	injection-10mic 13146						
Escaped	outlet-g6-5	18	712	1.249e-01	4.044e-01	1.741e-01	4.722e-02
injection-10mic 23163	injection-10mic 3912						
Escaped	outlet-g6-6	19	433	1.263e-01	2.987e-01	1.491e-01	1.857e-02
injection-10mic 9616	injection-10mic 8054						
Escaped	outlet-g6-7	20	306	1.172e-01	3.343e-01	1.392e-01	2.537e-02
injection-10mic 8811	injection-10mic 2373						
Escaped	outlet-g6-8	21	616	1.213e-01	3.881e-01	1.504e-01	3.331e-02
injection-10mic 10256	injection-10mic 3720						
Escaped	outlet-g6-9	22	524	1.146e-01	2.240e-01	1.391e-01	2.290e-02
injection-10mic 18627	injection-10mic 14089						
Escaped	outlet-g6-10	23	505	1.093e-01	2.466e-01	1.309e-01	2.425e-02
injection-10mic 20149	injection-10mic 12768						
Escaped	outlet-g6-11	24	600	1.177e-01	4.897e-01	1.344e-01	2.598e-02
injection-10mic 13992	injection-10mic 14250						
Escaped	outlet-g6-12	25	398	1.255e-01	7.356e-01	1.471e-01	3.194e-02
injection-10mic 5600	injection-10mic 6383						
Escaped	outlet-g6-13	26	363	1.285e-01	2.934e-01	1.463e-01	1.777e-02
injection-10mic 41575	injection-10mic 21070						
Escaped	outlet-g6-14	27	561	1.179e-01	2.695e-01	1.338e-01	1.926e-02
injection-10mic 35886	injection-10mic 36861						
Escaped	outlet-g6-15	28	437	1.086e-01	4.038e-01	1.322e-01	2.889e-02
injection-10mic 27289	injection-10mic 36563						

Escaped	outlet-g6-16	29	531	1.175e-01	2.241e-01	1.422e-01	2.423e-02
injection-10mic	19333	injection-10mic	25191				
Escaped	outlet-g6-17	30	606	1.196e-01	3.569e-01	1.526e-01	3.196e-02
injection-10mic	45895	injection-10mic	30981				
Escaped	outlet-g6-18	31	315	1.158e-01	2.054e-01	1.363e-01	1.817e-02
injection-10mic	37731	injection-10mic	35613				
Escaped	outlet-g6-19	32	445	1.232e-01	7.425e-01	1.541e-01	3.828e-02
injection-10mic	31678	injection-10mic	46463				
Escaped	outlet-g6-20	33	715	1.271e-01	4.146e-01	1.769e-01	4.960e-02
injection-10mic	32806	injection-10mic	48077				
Escaped	outlet-g6-21	34	704	1.369e-01	4.194e-01	1.881e-01	2.650e-02
injection-10mic	24357	injection-10mic	43422				
Escaped	outlet-g6-22	35	554	1.391e-01	5.779e-01	2.076e-01	3.588e-02
injection-10mic	21202	injection-10mic	43447				
Escaped	outlet-g6-23	36	440	1.413e-01	5.393e-01	2.862e-01	8.256e-02
injection-10mic	35094	injection-10mic	42264				
Escaped	outlet-g6-24	37	1074	1.416e-01	1.535e+00	4.452e-01	
3.933e-01	injection-10mic	45525	injection-10mic	49969			
Escaped	outlet-g5-1	38	1565	1.040e-01	5.333e-01	1.238e-01	1.690e-02
injection-10mic	10540	injection-10mic	16485				
Escaped	outlet-g5-2	39	1346	1.068e-01	1.478e+00	1.960e-01	1.252e-01
injection-10mic	5905	injection-10mic	9771				
Escaped	outlet-g5-3	40	1288	1.085e-01	1.487e+00	1.874e-01	1.124e-01
injection-10mic	36913	injection-10mic	34586				
Escaped	outlet-g5-4	41	1545	1.020e-01	3.798e-01	1.237e-01	1.751e-02
injection-10mic	21673	injection-10mic	26545				
Escaped	outlet-g4-1	42	2092	8.793e-02	7.064e-01	1.137e-01	3.066e-02
injection-10mic	7167	injection-10mic	9181				
Escaped	outlet-g4-2	43	3054	8.276e-02	2.570e-01	9.384e-02	1.061e-02
injection-10mic	12391	injection-10mic	7255				
Escaped	outlet-g4-3	44	2409	8.021e-02	2.013e-01	9.212e-02	1.139e-02
injection-10mic	11078	injection-10mic	42548				
Escaped	outlet-g4-4	45	2659	8.054e-02	3.780e-01	9.164e-02	1.506e-02
injection-10mic	5985	injection-10mic	30837				
Escaped	outlet-g4-5	46	2660	8.028e-02	5.069e-01	9.224e-02	1.658e-02
injection-10mic	43757	injection-10mic	29402				
Escaped	outlet-g4-6	47	2329	8.158e-02	2.042e-01	9.265e-02	1.260e-02
injection-10mic	36536	injection-10mic	33131				
Escaped	outlet-g4-7	48	2983	8.252e-02	3.709e-01	9.366e-02	1.060e-02
injection-10mic	38512	injection-10mic	30574				

Escaped outlet-g4-8 49 2120 8.788e-02 4.851e-01 1.141e-01 2.553e-02
injection-10mic 26641 injection-10mic 48549

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Trapped	walls-g1	7	1.562e-10	1.562e-10	0.000e+00
Trapped	walls-g2	8	2.902e-10	2.902e-10	0.000e+00
Trapped	walls-g3	9	3.804e-10	3.804e-10	0.000e+00
Trapped	walls-g4	10	9.294e-10	9.294e-10	0.000e+00
Trapped	walls-g5	11	2.532e-10	2.532e-10	0.000e+00
Trapped	walls-g6	12	5.960e-11	5.960e-11	0.000e+00
Escaped	outlet-g6-1	14	2.202e-10	2.202e-10	0.000e+00
Escaped	outlet-g6-2	15	8.240e-11	8.240e-11	0.000e+00
Escaped	outlet-g6-3	16	1.076e-10	1.076e-10	0.000e+00
Escaped	outlet-g6-4	17	1.430e-10	1.430e-10	0.000e+00
Escaped	outlet-g6-5	18	1.424e-10	1.424e-10	0.000e+00
Escaped	outlet-g6-6	19	8.660e-11	8.660e-11	0.000e+00
Escaped	outlet-g6-7	20	6.120e-11	6.120e-11	0.000e+00
Escaped	outlet-g6-8	21	1.232e-10	1.232e-10	0.000e+00
Escaped	outlet-g6-9	22	1.048e-10	1.048e-10	0.000e+00
Escaped	outlet-g6-10	23	1.010e-10	1.010e-10	0.000e+00
Escaped	outlet-g6-11	24	1.200e-10	1.200e-10	0.000e+00
Escaped	outlet-g6-12	25	7.960e-11	7.960e-11	0.000e+00
Escaped	outlet-g6-13	26	7.260e-11	7.260e-11	0.000e+00
Escaped	outlet-g6-14	27	1.122e-10	1.122e-10	0.000e+00
Escaped	outlet-g6-15	28	8.740e-11	8.740e-11	0.000e+00
Escaped	outlet-g6-16	29	1.062e-10	1.062e-10	0.000e+00
Escaped	outlet-g6-17	30	1.212e-10	1.212e-10	0.000e+00
Escaped	outlet-g6-18	31	6.300e-11	6.300e-11	0.000e+00
Escaped	outlet-g6-19	32	8.900e-11	8.900e-11	0.000e+00
Escaped	outlet-g6-20	33	1.430e-10	1.430e-10	0.000e+00
Escaped	outlet-g6-21	34	1.408e-10	1.408e-10	0.000e+00
Escaped	outlet-g6-22	35	1.108e-10	1.108e-10	0.000e+00
Escaped	outlet-g6-23	36	8.800e-11	8.800e-11	0.000e+00
Escaped	outlet-g6-24	37	2.148e-10	2.148e-10	0.000e+00
Escaped	outlet-g5-1	38	3.130e-10	3.130e-10	0.000e+00
Escaped	outlet-g5-2	39	2.692e-10	2.692e-10	0.000e+00

Escaped	outlet-g5-3	40	2.576e-10	2.576e-10	0.000e+00
Escaped	outlet-g5-4	41	3.090e-10	3.090e-10	0.000e+00
Escaped	outlet-g4-1	42	4.184e-10	4.184e-10	0.000e+00
Escaped	outlet-g4-2	43	6.108e-10	6.108e-10	0.000e+00
Escaped	outlet-g4-3	44	4.818e-10	4.818e-10	0.000e+00
Escaped	outlet-g4-4	45	5.318e-10	5.318e-10	0.000e+00
Escaped	outlet-g4-5	46	5.320e-10	5.320e-10	0.000e+00
Escaped	outlet-g4-6	47	4.658e-10	4.658e-10	0.000e+00
Escaped	outlet-g4-7	48	5.966e-10	5.966e-10	0.000e+00
Escaped	outlet-g4-8	49	4.240e-10	4.240e-10	0.000e+00

-----	-----	-----	-----	-----	-----
Net			1.000e-08	1.000e-08	0.000e+00

```
> (cx-gui-do cx-activate-item
"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface
Integrals)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

-----	-----
inlet	4.2615679

```
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '())
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 3 4
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

-----	-----
outlet-g4-1	1.1612432
outlet-g4-2	1.7586683
outlet-g4-3	2.1326607
outlet-g4-4	2.3090998
outlet-g4-5	2.3109999
outlet-g4-6	2.1379088
outlet-g4-7	1.7591435

outlet-g4-8	1.160557
outlet-g5-1	1.186605
outlet-g5-2	1.1178511
outlet-g5-3	1.1183198
outlet-g5-4	1.186365
outlet-g6-1	0.49648957
outlet-g6-10	2.1134732
outlet-g6-11	2.0753322
outlet-g6-12	2.0604963
outlet-g6-13	2.0608151
outlet-g6-14	2.0757107
outlet-g6-15	2.1141049
outlet-g6-16	2.1143213
outlet-g6-17	1.7865428
outlet-g6-18	1.7855513
outlet-g6-19	1.7047645
outlet-g6-2	0.50153705
outlet-g6-20	1.699661
outlet-g6-21	0.54012823
outlet-g6-22	0.54654743
outlet-g6-23	0.5003054
outlet-g6-24	0.4953133
outlet-g6-3	0.54753172
outlet-g6-4	0.54123598
outlet-g6-5	1.700937
outlet-g6-6	1.7061205
outlet-g6-7	1.7867454
outlet-g6-8	1.787809
outlet-g6-9	2.1130588

Net 1.6097149

(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton2(Cancel)")
/file/stop-transcript

Performance Timer for 225 iterations on 4 compute nodes

Average wall-clock time per iteration:	3.055 sec
Global reductions per iteration:	378 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	12133 messages
Data transfer per iteration:	1.789 MB

LE solves per iteration:	3 solves
LE wall-clock time per iteration:	2.562 sec (83.8%)
LE global solves per iteration:	8 solves
LE global wall-clock time per iteration:	0.003 sec (0.1%)
LE global matrix maximum size:	96
AMG cycles per iteration:	8.591 cycles
Relaxation sweeps per iteration:	3930 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.004 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	401
DPM updates per iteration:	0.0267 updates
DPM wall-clock time per iteration:	1.945 sec (63.7%)
Total wall-clock time:	687.391 sec
Total dpm solve time:	437.650 sec
Total dpm i/o time:	0.000 sec

Transcript Stop Time: 18:30:11, 15 Mar 2026 Central Daylight Time
Total Transcript Time: 3 Minute 0 Second.

—

```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model
Specific)*PushButton1(Discrete Phase)")
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")
(cx-gui-do cx-set-list-selections "Sample Trajectories*Table1*Table3*List1(Release from
Injections)" '( 0))
(cx-gui-do cx-activate-item "Sample Trajectories*Table1*Table3*List1(Release from
Injections)")
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 50000, trapped = 50000

```
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton2(Close)")
```

adjoint/	mesh/	report/
close-fluent	parallel/	solve/

```

define/      plot/      surface/
display/     preferences/ views/
file/        print-license-usage

```

```
> report/dpm-summary
```

Fate Index	Zone Injection, Index	Zone Number Id	Elapsed Time (s)				Injection, Min
	Name		Min	Max	Avg	Std Dev	
Max							
Trapped	walls-g1	7	49600	4.563e-02	1.555e-01	5.274e-02	1.254e-02
injection-100mi	16407	injection-100mi	49770				
Trapped	walls-g2	8	200	3.495e-01	3.495e-01	3.495e-01	1.879e-05
injection-100mi	25	injection-100mi	49968				
Trapped	walls-g4	10	200	2.665e-01	2.666e-01	2.666e-01	1.922e-05
injection-100mi	147	injection-100mi	49806				

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Trapped	walls-g1	7	9.920e-09	9.920e-09	0.000e+00
Trapped	walls-g2	8	4.000e-11	4.000e-11	0.000e+00
Trapped	walls-g4	10	4.000e-11	4.000e-11	0.000e+00
Net			1.000e-08	1.000e-08	0.000e+00

```

> (cx-gui-do cx-activate-item
"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface
Integrals)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1 3
4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
34 35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")

```

Mass-Weighted Average
Total Pressure [Pa]

```
-----
inlet 4.2615679
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '())
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 3 4
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average
Total Pressure [Pa]

```
-----
outlet-g4-1 1.1612432
outlet-g4-2 1.7586683
outlet-g4-3 2.1326607
outlet-g4-4 2.3090998
outlet-g4-5 2.3109999
outlet-g4-6 2.1379088
outlet-g4-7 1.7591435
outlet-g4-8 1.160557
outlet-g5-1 1.186605
outlet-g5-2 1.1178511
outlet-g5-3 1.1183198
outlet-g5-4 1.186365
outlet-g6-1 0.49648957
outlet-g6-10 2.1134732
outlet-g6-11 2.0753322
outlet-g6-12 2.0604963
outlet-g6-13 2.0608151
outlet-g6-14 2.0757107
outlet-g6-15 2.1141049
outlet-g6-16 2.1143213
outlet-g6-17 1.7865428
outlet-g6-18 1.7855513
outlet-g6-19 1.7047645
outlet-g6-2 0.50153705
```


outlet-g6-20	1.699661
outlet-g6-21	0.54012823
outlet-g6-22	0.54654743
outlet-g6-23	0.5003054
outlet-g6-24	0.4953133
outlet-g6-3	0.54753172
outlet-g6-4	0.54123598
outlet-g6-5	1.700937
outlet-g6-6	1.7061205
outlet-g6-7	1.7867454
outlet-g6-8	1.787809
outlet-g6-9	2.1130588

Net	1.6097149
-----	-----------

/file/stop-transcript

Performance Timer for 225 iterations on 4 compute nodes

Average wall-clock time per iteration:	3.055 sec
Global reductions per iteration:	378 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	12133 messages
Data transfer per iteration:	1.789 MB
LE solves per iteration:	3 solves
LE wall-clock time per iteration:	2.562 sec (83.8%)
LE global solves per iteration:	8 solves
LE global wall-clock time per iteration:	0.003 sec (0.1%)
LE global matrix maximum size:	96
AMG cycles per iteration:	8.591 cycles
Relaxation sweeps per iteration:	3930 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.004 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	406
DPM updates per iteration:	0.0444 updates
DPM wall-clock time per iteration:	2.983 sec (97.6%)

Total wall-clock time:	687.391 sec
Total dpm solve time:	671.134 sec
Total dpm i/o time:	0.000 sec

7.1.1 Associated Python Code

All Particles, Ultra-Fine, 500 Streams, 100 Discrete Random Walks Results Comparison

```
In [75]: import re
import pandas as pd
```

file:///Users/kaanborekci/Desktop/CFD-FinalProject/ANALYSIS/parcel-and-grid-independence.html

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parcel-and-grid-independence

3/16/26, 6:23 PM

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib

# matplotlib.rcParams.update({
#     "font.family": "serif",
#     "font.serif": ["Times New Roman"],
#     "font.size": 12,
#     "axes.labelsize": 13,
#     "axes.titlesize": 14,
#     "xtick.labelsize": 11,
#     "ytick.labelsize": 11,
#     "legend.fontsize": 11,
#     "figure.dpi": 200,
# })

# Parser (handles both "escaped = X, trapped = Y" and "trapped = Y" on

def parse_dpm_file(filepath):
    with open(filepath, encoding="utf-8", errors="replace") as f:
        lines = f.readlines()

    tracked = escaped = trapped = None
    particle_rows = []
    reading_particle_table = False

    for raw_line in lines:
        line = raw_line.strip()

        if "number tracked" in line:
            m = re.search(r"number tracked = (\d+)", line)
            if m:
                tracked = int(m.group(1))
            m2 = re.search(r"escaped = (\d+)", line)
            escaped = int(m2.group(1)) if m2 else 0
            m3 = re.search(r"trapped = (\d+)", line)
            trapped = int(m3.group(1)) if m3 else 0

        if line.startswith("Fate") and "Elapsed Time" in line:
            reading_particle_table = True
            continue

        if "Mass Transfer Summary" in line:
            reading_particle_table = False
            continue

        if reading_particle_table and line.startswith("Trapped"):
            parts = line.split()
            if len(parts) >= 4:
                particle_rows.append({
                    "zone": parts[1],
```

file:///Users/kaanborekci/Desktop/CFD-FinalProject/ANALYSIS/parcel-and-grid-independence.html

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```

        "count": int(parts[3]),
    })

    # Aggregate by generation
    gen_dep = {}
    for row in particle_rows:
        m = re.search(r"g(\d+)", row["zone"])
        if m:
            g = int(m.group(1))
            gen_dep[g] = gen_dep.get(g, 0) + row["count"]

    return {
        "tracked": tracked,
        "escaped": escaped,
        "trapped": trapped,
        "total_dep_frac": trapped / tracked if tracked else 0,
        "gen_dep_counts": gen_dep,
        "gen_dep_fracs": {g: c / tracked for g, c in gen_dep.items()},
    }

# Parse all three files
files = {
    "1 μm": "/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/ALL-PA
    "10 μm": "/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/ALL-PA
    "100 μm": "/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/ALL-PA
}

data = {label: parse_dpm_file(fp) for label, fp in files.items()}

# Summary table
all_gens = sorted(set(g for d in data.values() for g in d["gen_dep_fra

rows = []
for label, d in data.items():
    row = {
        "Particle size": label,
        "Tracked": d["tracked"],
        "Escaped": d["escaped"],
        "Trapped": d["trapped"],
        "Total dep. (%)": round(d["total_dep_frac"] * 100, 2),
    }
    for g in all_gens:
        row[f"G{g} dep. (%)"] = round(d["gen_dep_fracs"].get(g, 0) * 1
    rows.append(row)

summary_df = pd.DataFrame(rows)
print("=" * 80)
print("DEPOSITION SUMMARY TABLE")
print("=" * 80)

```

```

print(summary_df.to_string(index=False))
print()

# Also print a transposed view that's easier to read
print("=" * 80)
print("DEPOSITION BY GENERATION (% of tracked)")
print("=" * 80)
gen_table = []
for g in all_gens:
    row = {"Generation": f"G{g}"}
    for label, d in data.items():
        frac = d["gen_dep_fracs"].get(g, 0)
        row[label] = f"{frac*100:.4f}" if frac > 0 else "0"
    gen_table.append(row)

# Add total row
total_row = {"Generation": "Total"}
for label, d in data.items():
    total_row[label] = f"{d['total_dep_frac']*100:.2f}"
gen_table.append(total_row)

gen_df = pd.DataFrame(gen_table)
print(gen_df.to_string(index=False))

# Bar plot: deposition fraction by generation for each particle size
fig, ax = plt.subplots(figsize=(8, 4.5))

generations = [f"G{g}" for g in all_gens]
x = np.arange(len(generations))
width = 0.25

colors = {
    "1  $\mu\text{m}$ ": "green",
    "10  $\mu\text{m}$ ": "red",
    "100  $\mu\text{m}$ ": "blue",
}

for i, (label, d) in enumerate(data.items()):
    vals = [d["gen_dep_fracs"].get(g, 0) * 100 for g in all_gens]
    bars = ax.bar(x + (i - 1) * width, vals, width, label=label,
                  color=colors[label], edgecolor="white", linewidth=0.

    # Add count labels on top of bars (only if > 0)
    for bar, g in zip(bars, all_gens):
        count = d["gen_dep_counts"].get(g, 0)
        if count > 0 and bar.get_height() > 0:
            fontsize = 7 if bar.get_height() < 1 else 8
            ax.text(bar.get_x() + bar.get_width() / 2, bar.get_height(
                f"{count:}", ha="center", va="bottom", fontsize=f
                color="#444")

```

```

ax.set_xlabel("Airway generation")
ax.set_ylabel("Deposition fraction (%)")
ax.set_title("Particle deposition by generation – ultra-fine mesh (688
ax.set_xticks(x)
ax.set_xticklabels(generations)
ax.legend(title="Particle diameter", frameon=True, edgecolor="#ccc")
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
# ax.set_ylim(0, ax.get_ylim()[1] * 1.15)
ax.set_yscale("log")

plt.tight_layout()
plt.savefig("/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/ALL-PART
plt.savefig("/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/ALL-PART
print("\nPlots saved.")

# Stacked/total deposition bar
fig2, ax2 = plt.subplots(figsize=(5, 3.5))

labels = list(data.keys())
totals = [data[l]["total_dep_frac"] * 100 for l in labels]
bar_colors = [colors[l] for l in labels]

bars2 = ax2.bar(labels, totals, color=bar_colors, edgecolor="white", l

for bar, val in zip(bars2, totals):
    ax2.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.5
             f"{val:.2f}%", ha="center", va="bottom", fontsize=11, fon

ax2.set_ylabel("Total deposition fraction (%)")
ax2.set_title("Overall deposition by particle size")
ax2.spines["top"].set_visible(False)
ax2.spines["right"].set_visible(False)
ax2.set_ylim(0, 110)

plt.tight_layout()
plt.savefig("/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/ALL-PART
plt.savefig("/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/ALL-PART
print("Total deposition plot saved.")

```

=====

DEPOSITION SUMMARY TABLE

=====

Particle size	Tracked	Escaped	Trapped	Total dep. (%)	G1 dep. (%)
G2 dep. (%)	G3 dep. (%)	G4 dep. (%)	G5 dep. (%)	G6 dep. (%)	
1 μm	50000	49984	16	0.03	0.002
0.002	0.000	0.014	0.014	0.000	
10 μm	50000	39655	10345	20.69	1.562
2.902	3.804	9.294	2.532	0.596	
100 μm	50000	0	50000	100.00	99.200
0.400	0.000	0.400	0.000	0.000	

=====

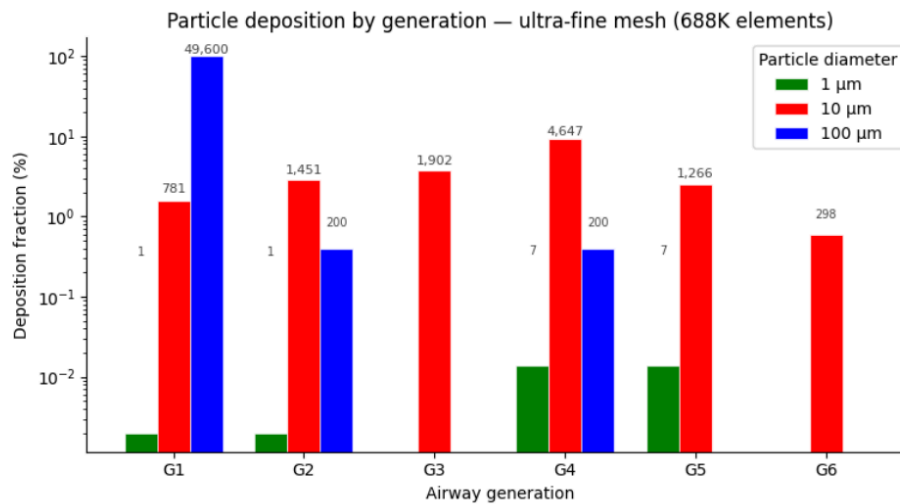
DEPOSITION BY GENERATION (% of tracked)

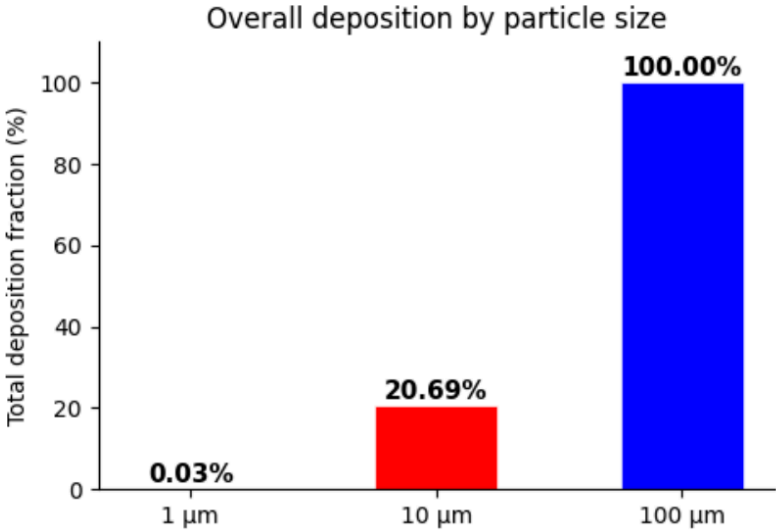
=====

Generation	1 μm	10 μm	100 μm
G1	0.0020	1.5620	99.2000
G2	0.0020	2.9020	0.4000
G3	0	3.8040	0
G4	0.0140	9.2940	0.4000
G5	0.0140	2.5320	0
G6	0	0.5960	0
Total	0.03	20.69	100.00

Plots saved.

Total deposition plot saved.





```
In [73]: summary_df
```

Out[73]:

	Particle size	Tracked	Escaped	Trapped	Total dep. (%)	G1 dep. (%)	G2 dep. (%)	G3 dep. (%)	G4 dep. (%)
0	1 μm	50000	49984	16	0.03	0.002	0.002	0.000	0.014
1	10 μm	50000	39655	10345	20.69	1.562	2.902	3.804	9.294
2	100 μm	50000	0	50000	100.00	99.200	0.400	0.000	0.400

```
In [ ]:
```


7.2 Appendix B: Parcel Independence Data

Transcript Start Time: 21:32:01, 05 Mar 2026 Central Standard Time

Current Directory:

"C:\Users\kaan\Desktop\Lung-Flow-Steady-NoP\lung-2024-final_files\dp0\FFF-2\Fluent"

Build Time: May 13 2024 11:28:30 EST

Executable Path: C:/PROGRA~1/ANSYSI~1/v242/fluent/ntbin/win64/fluent.exe

```
-----
ID   Hostname   Core O.S.      PID   Vendor
-----
n3   Borekci-X1  4/14 Windows-x64 24008 Intel(R) Core(TM) Ultra 7 165U
n2   Borekci-X1  3/14 Windows-x64 30148 Intel(R) Core(TM) Ultra 7 165U
n1   Borekci-X1  2/14 Windows-x64 27216 Intel(R) Core(TM) Ultra 7 165U
n0*  Borekci-X1  1/14 Windows-x64 31564 Intel(R) Core(TM) Ultra 7 165U
host Borekci-X1      Windows-x64 38772 Intel(R) Core(TM) Ultra 7 165U
```

MPI Option Selected: intel

Selected system interconnect: default

Cortex Process ID: 37784

License Server Path: 1055@rho.mcc.northwestern.edu

> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 5000, escaped = 4051, trapped = 949

report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,	
Index	Injection, Index	Name	Id	Min	Max	Avg	Std Dev	Min
Max								

Trapped	walls-g1	7	99	4.719e-02	2.109e-01	9.478e-02	5.480e-02	
injection-10mic	1346 injection-10mic	2091						

Trapped	walls-g2	8	101	6.351e-02	1.251e-01	7.551e-02	1.617e-02
injection-10mic	4427	injection-10mic	3448				
Trapped	walls-g3	9	167	7.078e-02	1.191e-01	8.246e-02	8.203e-03
injection-10mic	3332	injection-10mic	584				
Trapped	walls-g4	10	468	7.804e-02	3.907e+00	1.165e-01	2.607e-01
injection-10mic	3528	injection-10mic	2571				
Trapped	walls-g5	11	82	8.807e-02	4.659e-01	1.206e-01	6.417e-02
injection-10mic	4022	injection-10mic	348				
Trapped	walls-g6	12	32	1.121e-01	6.688e-01	2.181e-01	1.146e-01
injection-10mic	2941	injection-10mic	4897				
Escaped	outlet-g6-1	14	106	1.548e-01	1.311e+00	2.901e-01	1.914e-01
injection-10mic	1844	injection-10mic	157				
Escaped	outlet-g6-2	15	127	1.504e-01	1.523e+00	3.372e-01	1.278e-01
injection-10mic	528	injection-10mic	1618				
Escaped	outlet-g6-3	16	21	1.463e-01	1.702e+00	2.579e-01	3.242e-01
injection-10mic	697	injection-10mic	188				
Escaped	outlet-g6-4	17	45	1.453e-01	1.727e+00	2.368e-01	2.573e-01
injection-10mic	1278	injection-10mic	254				
Escaped	outlet-g6-5	18	76	1.371e-01	3.630e-01	1.706e-01	3.849e-02
injection-10mic	608	injection-10mic	251				
Escaped	outlet-g6-6	19	43	1.338e-01	7.887e-01	1.739e-01	1.022e-01
injection-10mic	3592	injection-10mic	259				
Escaped	outlet-g6-7	20	46	1.246e-01	6.564e-01	1.549e-01	8.063e-02
injection-10mic	1257	injection-10mic	317				
Escaped	outlet-g6-8	21	74	1.277e-01	2.929e-01	1.616e-01	3.613e-02
injection-10mic	588	injection-10mic	2088				
Escaped	outlet-g6-9	22	49	1.231e-01	2.219e-01	1.480e-01	2.361e-02
injection-10mic	3819	injection-10mic	2086				
Escaped	outlet-g6-10	23	42	1.163e-01	2.639e-01	1.361e-01	3.036e-02
injection-10mic	2179	injection-10mic	1526				
Escaped	outlet-g6-11	24	55	1.220e-01	2.961e-01	1.418e-01	3.190e-02
injection-10mic	988	injection-10mic	1276				
Escaped	outlet-g6-12	25	31	1.389e-01	2.172e-01	1.485e-01	1.427e-02
injection-10mic	2336	injection-10mic	2356				
Escaped	outlet-g6-13	26	30	1.283e-01	1.877e-01	1.470e-01	1.100e-02
injection-10mic	3617	injection-10mic	4020				
Escaped	outlet-g6-14	27	66	1.230e-01	2.359e-01	1.376e-01	2.108e-02
injection-10mic	3137	injection-10mic	3809				
Escaped	outlet-g6-15	28	42	1.183e-01	2.561e-01	1.412e-01	3.108e-02
injection-10mic	3346	injection-10mic	3375				

Escaped	outlet-g6-16	29	62	1.235e-01	2.187e-01	1.450e-01	2.512e-02
injection-10mic	3275	injection-10mic	4340				
Escaped	outlet-g6-17	30	93	1.272e-01	1.174e+00	1.804e-01	1.159e-01
injection-10mic	2791	injection-10mic	4845				
Escaped	outlet-g6-18	31	34	1.257e-01	3.299e-01	1.432e-01	3.359e-02
injection-10mic	2316	injection-10mic	4355				
Escaped	outlet-g6-19	32	39	1.390e-01	2.163e-01	1.569e-01	1.783e-02
injection-10mic	3266	injection-10mic	4336				
Escaped	outlet-g6-20	33	94	1.414e-01	7.855e-01	1.930e-01	9.004e-02
injection-10mic	4410	injection-10mic	4884				
Escaped	outlet-g6-21	34	43	1.504e-01	1.323e+00	2.393e-01	2.337e-01
injection-10mic	3833	injection-10mic	4804				
Escaped	outlet-g6-22	35	24	1.429e-01	7.667e-01	2.025e-01	1.189e-01
injection-10mic	4558	injection-10mic	4282				
Escaped	outlet-g6-23	36	119	1.493e-01	1.105e+00	3.429e-01	9.547e-02
injection-10mic	3668	injection-10mic	3233				
Escaped	outlet-g6-24	37	122	1.477e-01	9.814e+00	3.552e-01	8.738e-01
injection-10mic	3888	injection-10mic	4136				
Escaped	outlet-g5-1	38	123	1.074e-01	3.125e+00	1.576e-01	2.804e-01
injection-10mic	1340	injection-10mic	2184				
Escaped	outlet-g5-2	39	111	1.171e-01	9.090e+00	2.769e-01	8.739e-01
injection-10mic	1805	injection-10mic	3144				
Escaped	outlet-g5-3	40	107	1.254e-01	2.058e+00	2.047e-01	1.961e-01
injection-10mic	3656	injection-10mic	2072				
Escaped	outlet-g5-4	41	139	1.058e-01	1.407e+00	1.367e-01	1.090e-01
injection-10mic	3749	injection-10mic	3240				
Escaped	outlet-g4-1	42	180	9.243e-02	2.605e+00	1.334e-01	1.908e-01
injection-10mic	2073	injection-10mic	250				
Escaped	outlet-g4-2	43	335	8.646e-02	1.031e+00	1.019e-01	6.464e-02
injection-10mic	1547	injection-10mic	187				
Escaped	outlet-g4-3	44	236	8.570e-02	1.996e-01	9.621e-02	1.530e-02
injection-10mic	2275	injection-10mic	2209				
Escaped	outlet-g4-4	45	278	8.554e-02	2.152e-01	9.626e-02	1.565e-02
injection-10mic	766	injection-10mic	2885				
Escaped	outlet-g4-5	46	319	8.456e-02	4.920e+00	1.107e-01	2.701e-01
injection-10mic	3683	injection-10mic	2532				
Escaped	outlet-g4-6	47	244	8.606e-02	1.898e-01	9.652e-02	1.559e-02
injection-10mic	2221	injection-10mic	3849				
Escaped	outlet-g4-7	48	341	8.642e-02	1.506e-01	9.679e-02	1.025e-02
injection-10mic	3623	injection-10mic	2927				

Escaped outlet-g4-8 49 155 8.921e-02 1.030e+00 1.260e-01 8.379e-02
injection-10mic 3308 injection-10mic 4247

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Trapped	walls-g1	7	1.980e-10	1.980e-10	0.000e+00
Trapped	walls-g2	8	2.020e-10	2.020e-10	0.000e+00
Trapped	walls-g3	9	3.340e-10	3.340e-10	0.000e+00
Trapped	walls-g4	10	9.360e-10	9.360e-10	0.000e+00
Trapped	walls-g5	11	1.640e-10	1.640e-10	0.000e+00
Trapped	walls-g6	12	6.400e-11	6.400e-11	0.000e+00
Escaped	outlet-g6-1	14	2.120e-10	2.120e-10	0.000e+00
Escaped	outlet-g6-2	15	2.540e-10	2.540e-10	0.000e+00
Escaped	outlet-g6-3	16	4.200e-11	4.200e-11	0.000e+00
Escaped	outlet-g6-4	17	9.000e-11	9.000e-11	0.000e+00
Escaped	outlet-g6-5	18	1.520e-10	1.520e-10	0.000e+00
Escaped	outlet-g6-6	19	8.600e-11	8.600e-11	0.000e+00
Escaped	outlet-g6-7	20	9.200e-11	9.200e-11	0.000e+00
Escaped	outlet-g6-8	21	1.480e-10	1.480e-10	0.000e+00
Escaped	outlet-g6-9	22	9.800e-11	9.800e-11	0.000e+00
Escaped	outlet-g6-10	23	8.400e-11	8.400e-11	0.000e+00
Escaped	outlet-g6-11	24	1.100e-10	1.100e-10	0.000e+00
Escaped	outlet-g6-12	25	6.200e-11	6.200e-11	0.000e+00
Escaped	outlet-g6-13	26	6.000e-11	6.000e-11	0.000e+00
Escaped	outlet-g6-14	27	1.320e-10	1.320e-10	0.000e+00
Escaped	outlet-g6-15	28	8.400e-11	8.400e-11	0.000e+00
Escaped	outlet-g6-16	29	1.240e-10	1.240e-10	0.000e+00
Escaped	outlet-g6-17	30	1.860e-10	1.860e-10	0.000e+00
Escaped	outlet-g6-18	31	6.800e-11	6.800e-11	0.000e+00
Escaped	outlet-g6-19	32	7.800e-11	7.800e-11	0.000e+00
Escaped	outlet-g6-20	33	1.880e-10	1.880e-10	0.000e+00
Escaped	outlet-g6-21	34	8.600e-11	8.600e-11	0.000e+00
Escaped	outlet-g6-22	35	4.800e-11	4.800e-11	0.000e+00
Escaped	outlet-g6-23	36	2.380e-10	2.380e-10	0.000e+00
Escaped	outlet-g6-24	37	2.440e-10	2.440e-10	0.000e+00
Escaped	outlet-g5-1	38	2.460e-10	2.460e-10	0.000e+00
Escaped	outlet-g5-2	39	2.220e-10	2.220e-10	0.000e+00

Escaped	outlet-g5-3	40	2.140e-10	2.140e-10	0.000e+00
Escaped	outlet-g5-4	41	2.780e-10	2.780e-10	0.000e+00
Escaped	outlet-g4-1	42	3.600e-10	3.600e-10	0.000e+00
Escaped	outlet-g4-2	43	6.700e-10	6.700e-10	0.000e+00
Escaped	outlet-g4-3	44	4.720e-10	4.720e-10	0.000e+00
Escaped	outlet-g4-4	45	5.560e-10	5.560e-10	0.000e+00
Escaped	outlet-g4-5	46	6.380e-10	6.380e-10	0.000e+00
Escaped	outlet-g4-6	47	4.880e-10	4.880e-10	0.000e+00
Escaped	outlet-g4-7	48	6.820e-10	6.820e-10	0.000e+00
Escaped	outlet-g4-8	49	3.100e-10	3.100e-10	0.000e+00

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Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time:	0.000 sec
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Transcript Stop Time: 21:33:04, 05 Mar 2026 Central Standard Time

Total Transcript Time: 1 Minute 3 Second.

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> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 10000, escaped = 8151, trapped = 1848, incomplete = 1

report/dpm-summary

Fate Index	Zone Injection, Index	Zone Number Name Id	Elapsed Time (s)				Injection, Min
		Min	Max	Avg	Std Dev		
Max							

Incomplete		1	2.241e+01	2.241e+01	2.241e+01	0.000e+00	
injection-10micron	360 injection-10micron	360					
Trapped	walls-g1	7 178	4.620e-02	2.160e-01	8.167e-02	4.853e-02	
injection-10mic	5328 injection-10mic	3955					
Trapped	walls-g2	8 270	6.326e-02	1.318e-01	7.343e-02	1.380e-02	
injection-10mic	3261 injection-10mic	4664					
Trapped	walls-g3	9 343	7.263e-02	1.153e+00	8.929e-02	7.644e-02	
injection-10mic	5739 injection-10mic	843					
Trapped	walls-g4	10 822	7.777e-02	1.986e+00	1.112e-01	1.471e-01	
injection-10mic	7082 injection-10mic	6346					
Trapped	walls-g5	11 175	8.680e-02	1.284e+00	1.348e-01	1.328e-01	
injection-10mic	1788 injection-10mic	9538					
Trapped	walls-g6	12 60	1.055e-01	6.561e-01	1.990e-01	9.528e-02	
injection-10mic	5365 injection-10mic	399					
Escaped	outlet-g6-1	14 198	1.430e-01	8.141e+00	3.131e-01	5.704e-01	
injection-10mic	1813 injection-10mic	2996					
Escaped	outlet-g6-2	15 133	1.417e-01	3.615e-01	3.176e-01	7.810e-02	
injection-10mic	4042 injection-10mic	49					
Escaped	outlet-g6-3	16 129	1.439e-01	6.799e-01	1.954e-01	4.887e-02	
injection-10mic	1381 injection-10mic	279					
Escaped	outlet-g6-4	17 90	1.435e-01	2.665e+00	2.431e-01	3.115e-01	
injection-10mic	3991 injection-10mic	272					
Escaped	outlet-g6-5	18 189	1.373e-01	1.037e+01	2.446e-01	7.420e-01	
injection-10mic	2415 injection-10mic	487					
Escaped	outlet-g6-6	19 78	1.362e-01	2.209e-01	1.579e-01	1.789e-02	
injection-10mic	1985 injection-10mic	751					

Escaped	outlet-g6-7	20	80	1.269e-01	5.877e-01	1.487e-01	5.325e-02
injection-10mic	2136	injection-10mic	1349				
Escaped	outlet-g6-8	21	147	1.264e-01	4.046e-01	1.653e-01	4.276e-02
injection-10mic	5303	injection-10mic	367				
Escaped	outlet-g6-9	22	105	1.234e-01	2.266e-01	1.441e-01	2.534e-02
injection-10mic	2805	injection-10mic	4390				
Escaped	outlet-g6-10	23	101	1.152e-01	2.777e-01	1.380e-01	3.147e-02
injection-10mic	2591	injection-10mic	1569				
Escaped	outlet-g6-11	24	146	1.221e-01	6.223e-01	1.402e-01	5.026e-02
injection-10mic	3294	injection-10mic	5832				
Escaped	outlet-g6-12	25	78	1.272e-01	3.179e-01	1.486e-01	2.123e-02
injection-10mic	4845	injection-10mic	1811				
Escaped	outlet-g6-13	26	81	1.371e-01	1.920e-01	1.480e-01	1.111e-02
injection-10mic	6840	injection-10mic	5469				
Escaped	outlet-g6-14	27	155	1.226e-01	5.529e-01	1.393e-01	3.740e-02
injection-10mic	8032	injection-10mic	7931				
Escaped	outlet-g6-15	28	84	1.142e-01	2.965e-01	1.381e-01	3.462e-02
injection-10mic	6093	injection-10mic	3641				
Escaped	outlet-g6-16	29	99	1.227e-01	2.195e-01	1.472e-01	2.561e-02
injection-10mic	6490	injection-10mic	7945				
Escaped	outlet-g6-17	30	162	1.281e-01	3.542e-01	1.619e-01	3.955e-02
injection-10mic	7521	injection-10mic	8179				
Escaped	outlet-g6-18	31	80	1.234e-01	2.772e+00	1.768e-01	2.931e-01
injection-10mic	7054	injection-10mic	7611				
Escaped	outlet-g6-19	32	75	1.326e-01	3.608e-01	1.633e-01	3.699e-02
injection-10mic	4108	injection-10mic	8827				
Escaped	outlet-g6-20	33	214	1.307e-01	3.962e+00	2.247e-01	3.273e-01
injection-10mic	8848	injection-10mic	8728				
Escaped	outlet-g6-21	34	79	1.490e-01	2.671e+00	2.240e-01	2.865e-01
injection-10mic	8722	injection-10mic	9540				
Escaped	outlet-g6-22	35	142	1.435e-01	2.570e-01	1.904e-01	1.726e-02
injection-10mic	4472	injection-10mic	9624				
Escaped	outlet-g6-23	36	133	1.553e-01	2.081e+00	3.343e-01	1.702e-01
injection-10mic	7507	injection-10mic	5660				
Escaped	outlet-g6-24	37	179	1.538e-01	4.553e+00	3.210e-01	3.978e-01
injection-10mic	7895	injection-10mic	9453				
Escaped	outlet-g5-1	38	266	1.083e-01	7.351e+00	2.074e-01	6.760e-01
injection-10mic	2899	injection-10mic	3384				
Escaped	outlet-g5-2	39	211	1.152e-01	9.134e+00	2.732e-01	8.718e-01
injection-10mic	3315	injection-10mic	4916				

Escaped	outlet-g5-3	40	214	1.151e-01	1.331e+00	1.969e-01	1.177e-01
injection-10mic 4597	injection-10mic 4199						
Escaped	outlet-g5-4	41	291	1.074e-01	7.628e+00	2.050e-01	5.609e-01
injection-10mic 6277	injection-10mic 8750						
Escaped	outlet-g4-1	42	360	9.108e-02	1.177e+00	1.240e-01	7.992e-02
injection-10mic 3319	injection-10mic 491						
Escaped	outlet-g4-2	43	625	8.626e-02	3.312e+00	1.070e-01	1.711e-01
injection-10mic 4021	injection-10mic 1580						
Escaped	outlet-g4-3	44	471	8.505e-02	1.789e-01	9.467e-02	1.126e-02
injection-10mic 3160	injection-10mic 2110						
Escaped	outlet-g4-4	45	631	8.446e-02	6.954e-01	9.652e-02	2.708e-02
injection-10mic 3956	injection-10mic 1348						
Escaped	outlet-g4-5	46	629	8.447e-02	2.680e-01	9.635e-02	1.712e-02
injection-10mic 8146	injection-10mic 7980						
Escaped	outlet-g4-6	47	525	8.423e-02	1.930e-01	9.521e-02	1.235e-02
injection-10mic 8225	injection-10mic 8473						
Escaped	outlet-g4-7	48	621	8.609e-02	8.250e-01	9.901e-02	4.004e-02
injection-10mic 6292	injection-10mic 9658						
Escaped	outlet-g4-8	49	350	9.122e-02	4.399e+00	1.649e-01	3.061e-01
injection-10mic 7948	injection-10mic 9156						

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Incomplete			1.000e-12	1.000e-12	0.000e+00
Trapped	walls-g1	7	1.780e-10	1.780e-10	0.000e+00
Trapped	walls-g2	8	2.700e-10	2.700e-10	0.000e+00
Trapped	walls-g3	9	3.430e-10	3.430e-10	0.000e+00
Trapped	walls-g4	10	8.220e-10	8.220e-10	0.000e+00
Trapped	walls-g5	11	1.750e-10	1.750e-10	0.000e+00
Trapped	walls-g6	12	6.000e-11	6.000e-11	0.000e+00
Escaped	outlet-g6-1	14	1.980e-10	1.980e-10	0.000e+00
Escaped	outlet-g6-2	15	1.330e-10	1.330e-10	0.000e+00
Escaped	outlet-g6-3	16	1.290e-10	1.290e-10	0.000e+00
Escaped	outlet-g6-4	17	9.000e-11	9.000e-11	0.000e+00
Escaped	outlet-g6-5	18	1.890e-10	1.890e-10	0.000e+00
Escaped	outlet-g6-6	19	7.800e-11	7.800e-11	0.000e+00
Escaped	outlet-g6-7	20	8.000e-11	8.000e-11	0.000e+00

Escaped	outlet-g6-8	21	1.470e-10	1.470e-10	0.000e+00
Escaped	outlet-g6-9	22	1.050e-10	1.050e-10	0.000e+00
Escaped	outlet-g6-10	23	1.010e-10	1.010e-10	0.000e+00
Escaped	outlet-g6-11	24	1.460e-10	1.460e-10	0.000e+00
Escaped	outlet-g6-12	25	7.800e-11	7.800e-11	0.000e+00
Escaped	outlet-g6-13	26	8.100e-11	8.100e-11	0.000e+00
Escaped	outlet-g6-14	27	1.550e-10	1.550e-10	0.000e+00
Escaped	outlet-g6-15	28	8.400e-11	8.400e-11	0.000e+00
Escaped	outlet-g6-16	29	9.900e-11	9.900e-11	0.000e+00
Escaped	outlet-g6-17	30	1.620e-10	1.620e-10	0.000e+00
Escaped	outlet-g6-18	31	8.000e-11	8.000e-11	0.000e+00
Escaped	outlet-g6-19	32	7.500e-11	7.500e-11	0.000e+00
Escaped	outlet-g6-20	33	2.140e-10	2.140e-10	0.000e+00
Escaped	outlet-g6-21	34	7.900e-11	7.900e-11	0.000e+00
Escaped	outlet-g6-22	35	1.420e-10	1.420e-10	0.000e+00
Escaped	outlet-g6-23	36	1.330e-10	1.330e-10	0.000e+00
Escaped	outlet-g6-24	37	1.790e-10	1.790e-10	0.000e+00
Escaped	outlet-g5-1	38	2.660e-10	2.660e-10	0.000e+00
Escaped	outlet-g5-2	39	2.110e-10	2.110e-10	0.000e+00
Escaped	outlet-g5-3	40	2.140e-10	2.140e-10	0.000e+00
Escaped	outlet-g5-4	41	2.910e-10	2.910e-10	0.000e+00
Escaped	outlet-g4-1	42	3.600e-10	3.600e-10	0.000e+00
Escaped	outlet-g4-2	43	6.250e-10	6.250e-10	0.000e+00
Escaped	outlet-g4-3	44	4.710e-10	4.710e-10	0.000e+00
Escaped	outlet-g4-4	45	6.310e-10	6.310e-10	0.000e+00
Escaped	outlet-g4-5	46	6.290e-10	6.290e-10	0.000e+00
Escaped	outlet-g4-6	47	5.250e-10	5.250e-10	0.000e+00
Escaped	outlet-g4-7	48	6.210e-10	6.210e-10	0.000e+00
Escaped	outlet-g4-8	49	3.500e-10	3.500e-10	0.000e+00

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Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB

LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time: 0.000 sec

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> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 15000, escaped = 12155, trapped = 2845

report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,	
Index	Injection, Index	Name	Id	Min	Max	Avg	Std Dev	Min
Max								

Trapped	walls-g1	7	302	4.623e-02	2.169e-01	8.228e-02	4.966e-02	
injection-10mic 8844	injection-10mic 6031							
Trapped	walls-g2	8	382	6.346e-02	2.811e-01	7.555e-02	2.007e-02	
injection-10mic 10604	injection-10mic 10934							
Trapped	walls-g3	9	518	7.265e-02	1.721e+00	9.119e-02	9.392e-02	
injection-10mic 6876	injection-10mic 575							
Trapped	walls-g4	10	1221	7.722e-02	1.360e+01	1.145e-01	4.085e-01	
injection-10mic 2879	injection-10mic 6076							
Trapped	walls-g5	11	295	8.676e-02	3.145e+00	1.272e-01	1.833e-01	
injection-10mic 5244	injection-10mic 3759							

Trapped	walls-g6	12	127	1.096e-01	6.981e-01	2.192e-01	8.459e-02
injection-10mic	7970	injection-10mic	2997				
Escaped	outlet-g6-1	14	296	1.564e-01	8.739e+00	3.192e-01	5.449e-01
injection-10mic	1711	injection-10mic	1327				
Escaped	outlet-g6-2	15	249	1.454e-01	2.465e+00	3.178e-01	1.515e-01
injection-10mic	3829	injection-10mic	4546				
Escaped	outlet-g6-3	16	68	1.399e-01	2.118e+00	2.175e-01	2.424e-01
injection-10mic	5980	injection-10mic	3460				
Escaped	outlet-g6-4	17	177	1.449e-01	1.659e+00	2.312e-01	1.612e-01
injection-10mic	3186	injection-10mic	828				
Escaped	outlet-g6-5	18	291	1.348e-01	9.075e-01	1.890e-01	7.874e-02
injection-10mic	1699	injection-10mic	804				
Escaped	outlet-g6-6	19	132	1.330e-01	1.119e+00	1.692e-01	9.817e-02
injection-10mic	4281	injection-10mic	677				
Escaped	outlet-g6-7	20	138	1.238e-01	1.101e+00	1.518e-01	8.347e-02
injection-10mic	3440	injection-10mic	411				
Escaped	outlet-g6-8	21	233	1.266e-01	4.367e+00	1.836e-01	2.782e-01
injection-10mic	2859	injection-10mic	1444				
Escaped	outlet-g6-9	22	150	1.218e-01	2.242e-01	1.464e-01	2.582e-02
injection-10mic	5328	injection-10mic	7265				
Escaped	outlet-g6-10	23	136	1.159e-01	3.330e-01	1.387e-01	3.641e-02
injection-10mic	5353	injection-10mic	4559				
Escaped	outlet-g6-11	24	210	1.223e-01	2.590e-01	1.371e-01	2.049e-02
injection-10mic	4467	injection-10mic	7520				
Escaped	outlet-g6-12	25	109	1.265e-01	1.801e-01	1.484e-01	9.286e-03
injection-10mic	4211	injection-10mic	4854				
Escaped	outlet-g6-13	26	97	1.314e-01	3.256e-01	1.509e-01	2.570e-02
injection-10mic	10676	injection-10mic	12603				
Escaped	outlet-g6-14	27	192	1.234e-01	2.674e-01	1.369e-01	1.974e-02
injection-10mic	11887	injection-10mic	10099				
Escaped	outlet-g6-15	28	115	1.152e-01	3.642e-01	1.375e-01	3.975e-02
injection-10mic	9172	injection-10mic	9960				
Escaped	outlet-g6-16	29	148	1.225e-01	2.281e-01	1.479e-01	2.678e-02
injection-10mic	10711	injection-10mic	10910				
Escaped	outlet-g6-17	30	223	1.284e-01	5.121e-01	1.631e-01	4.045e-02
injection-10mic	12162	injection-10mic	6967				
Escaped	outlet-g6-18	31	142	1.231e-01	2.282e-01	1.448e-01	2.087e-02
injection-10mic	8616	injection-10mic	14346				
Escaped	outlet-g6-19	32	112	1.355e-01	4.038e-01	1.640e-01	3.661e-02
injection-10mic	11141	injection-10mic	13170				

Escaped	outlet-g6-20	33	287	1.372e-01	1.768e+00	2.089e-01	1.291e-01
injection-10mic	13995	injection-10mic	10165				
Escaped	outlet-g6-21	34	144	1.429e-01	1.751e+00	2.225e-01	1.942e-01
injection-10mic	9044	injection-10mic	14560				
Escaped	outlet-g6-22	35	113	1.395e-01	2.053e+00	2.213e-01	1.956e-01
injection-10mic	11413	injection-10mic	12238				
Escaped	outlet-g6-23	36	249	1.463e-01	1.561e+00	3.159e-01	1.032e-01
injection-10mic	12713	injection-10mic	14453				
Escaped	outlet-g6-24	37	282	1.438e-01	2.345e+00	3.351e-01	3.147e-01
injection-10mic	4310	injection-10mic	11122				
Escaped	outlet-g5-1	38	427	1.091e-01	1.206e+01	1.794e-01	6.384e-01
injection-10mic	4626	injection-10mic	7697				
Escaped	outlet-g5-2	39	335	1.109e-01	7.971e+00	2.909e-01	6.530e-01
injection-10mic	5131	injection-10mic	9480				
Escaped	outlet-g5-3	40	338	1.177e-01	6.540e+00	2.270e-01	4.129e-01
injection-10mic	7095	injection-10mic	10177				
Escaped	outlet-g5-4	41	407	1.071e-01	5.820e+00	1.767e-01	4.125e-01
injection-10mic	8520	injection-10mic	7287				
Escaped	outlet-g4-1	42	529	9.156e-02	2.680e+00	1.372e-01	1.829e-01
injection-10mic	5367	injection-10mic	1307				
Escaped	outlet-g4-2	43	922	8.545e-02	1.483e+00	1.004e-01	6.222e-02
injection-10mic	6488	injection-10mic	1770				
Escaped	outlet-g4-3	44	739	8.490e-02	1.863e-01	9.487e-02	1.167e-02
injection-10mic	1806	injection-10mic	3324				
Escaped	outlet-g4-4	45	950	8.408e-02	2.982e-01	9.686e-02	1.765e-02
injection-10mic	6750	injection-10mic	2928				
Escaped	outlet-g4-5	46	956	8.516e-02	2.454e-01	9.679e-02	1.555e-02
injection-10mic	12040	injection-10mic	5724				
Escaped	outlet-g4-6	47	761	8.365e-02	1.829e-01	9.474e-02	1.121e-02
injection-10mic	12042	injection-10mic	8079				
Escaped	outlet-g4-7	48	978	8.646e-02	9.809e-01	9.730e-02	3.104e-02
injection-10mic	7548	injection-10mic	13829				
Escaped	outlet-g4-8	49	520	8.847e-02	1.079e+01	1.535e-01	4.907e-01
injection-10mic	11132	injection-10mic	13024				

(*)- Mass Transfer Summary -(*)

Fate	Zone	Zone	Mass Flow (kg/s)		
	Name	Id	Initial	Final	Change

Trapped	walls-g1	7	2.013e-10	2.013e-10	0.000e+00
Trapped	walls-g2	8	2.547e-10	2.547e-10	0.000e+00
Trapped	walls-g3	9	3.453e-10	3.453e-10	0.000e+00
Trapped	walls-g4	10	8.140e-10	8.140e-10	0.000e+00
Trapped	walls-g5	11	1.967e-10	1.967e-10	0.000e+00
Trapped	walls-g6	12	8.467e-11	8.467e-11	0.000e+00
Escaped	outlet-g6-1	14	1.973e-10	1.973e-10	0.000e+00
Escaped	outlet-g6-2	15	1.660e-10	1.660e-10	0.000e+00
Escaped	outlet-g6-3	16	4.533e-11	4.533e-11	0.000e+00
Escaped	outlet-g6-4	17	1.180e-10	1.180e-10	0.000e+00
Escaped	outlet-g6-5	18	1.940e-10	1.940e-10	0.000e+00
Escaped	outlet-g6-6	19	8.800e-11	8.800e-11	0.000e+00
Escaped	outlet-g6-7	20	9.200e-11	9.200e-11	0.000e+00
Escaped	outlet-g6-8	21	1.553e-10	1.553e-10	0.000e+00
Escaped	outlet-g6-9	22	1.000e-10	1.000e-10	0.000e+00
Escaped	outlet-g6-10	23	9.067e-11	9.067e-11	0.000e+00
Escaped	outlet-g6-11	24	1.400e-10	1.400e-10	0.000e+00
Escaped	outlet-g6-12	25	7.267e-11	7.267e-11	0.000e+00
Escaped	outlet-g6-13	26	6.467e-11	6.467e-11	0.000e+00
Escaped	outlet-g6-14	27	1.280e-10	1.280e-10	0.000e+00
Escaped	outlet-g6-15	28	7.667e-11	7.667e-11	0.000e+00
Escaped	outlet-g6-16	29	9.867e-11	9.867e-11	0.000e+00
Escaped	outlet-g6-17	30	1.487e-10	1.487e-10	0.000e+00
Escaped	outlet-g6-18	31	9.467e-11	9.467e-11	0.000e+00
Escaped	outlet-g6-19	32	7.467e-11	7.467e-11	0.000e+00
Escaped	outlet-g6-20	33	1.913e-10	1.913e-10	0.000e+00
Escaped	outlet-g6-21	34	9.600e-11	9.600e-11	0.000e+00
Escaped	outlet-g6-22	35	7.533e-11	7.533e-11	0.000e+00
Escaped	outlet-g6-23	36	1.660e-10	1.660e-10	0.000e+00
Escaped	outlet-g6-24	37	1.880e-10	1.880e-10	0.000e+00
Escaped	outlet-g5-1	38	2.847e-10	2.847e-10	0.000e+00
Escaped	outlet-g5-2	39	2.233e-10	2.233e-10	0.000e+00
Escaped	outlet-g5-3	40	2.253e-10	2.253e-10	0.000e+00
Escaped	outlet-g5-4	41	2.713e-10	2.713e-10	0.000e+00
Escaped	outlet-g4-1	42	3.527e-10	3.527e-10	0.000e+00
Escaped	outlet-g4-2	43	6.147e-10	6.147e-10	0.000e+00
Escaped	outlet-g4-3	44	4.927e-10	4.927e-10	0.000e+00
Escaped	outlet-g4-4	45	6.333e-10	6.333e-10	0.000e+00
Escaped	outlet-g4-5	46	6.373e-10	6.373e-10	0.000e+00
Escaped	outlet-g4-6	47	5.073e-10	5.073e-10	0.000e+00

Escaped	outlet-g4-7	48	6.520e-10	6.520e-10	0.000e+00
Escaped	outlet-g4-8	49	3.467e-10	3.467e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time:	0.000 sec
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> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 20000, escaped = 16250, trapped = 3749, incomplete = 1
report/dpm-summaryt

non-fatal Error: invalid command [dpm-summaryt]

> report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Incomplete		1	1.388e+01	1.388e+01	1.388e+01	0.000e+00	
injection-10micron 1212	injection-10micron 1212						
Trapped	walls-g1	7	362	4.688e-02	2.161e-01	8.369e-02	5.082e-02
injection-10mic 7138	injection-10mic 13508						
Trapped	walls-g2	8	499	6.342e-02	2.917e-01	7.394e-02	1.695e-02
injection-10mic 4467	injection-10mic 4868						
Trapped	walls-g3	9	714	7.249e-02	1.077e+01	1.042e-01	4.069e-01
injection-10mic 16181	injection-10mic 17093						
Trapped	walls-g4	10	1680	7.752e-02	1.790e+00	9.956e-02	7.410e-02
injection-10mic 9259	injection-10mic 10930						
Trapped	walls-g5	11	373	8.645e-02	5.836e+00	1.341e-01	3.032e-01
injection-10mic 5419	injection-10mic 5543						
Trapped	walls-g6	12	121	1.036e-01	7.305e-01	1.842e-01	8.834e-02
injection-10mic 17665	injection-10mic 18556						
Escaped	outlet-g6-1	14	406	1.476e-01	2.769e+00	3.123e-01	3.036e-01
injection-10mic 2182	injection-10mic 775						
Escaped	outlet-g6-2	15	274	1.406e-01	1.399e+00	3.150e-01	1.025e-01
injection-10mic 7950	injection-10mic 1307						
Escaped	outlet-g6-3	16	182	1.389e-01	7.695e-01	1.938e-01	5.819e-02
injection-10mic 10874	injection-10mic 1091						
Escaped	outlet-g6-4	17	204	1.446e-01	7.630e+00	2.871e-01	6.172e-01
injection-10mic 4136	injection-10mic 950						
Escaped	outlet-g6-5	18	392	1.373e-01	1.559e+00	2.043e-01	1.215e-01
injection-10mic 3937	injection-10mic 4587						
Escaped	outlet-g6-6	19	164	1.303e-01	2.582e+00	1.827e-01	2.106e-01
injection-10mic 1463	injection-10mic 505						
Escaped	outlet-g6-7	20	160	1.231e-01	4.392e-01	1.460e-01	3.166e-02
injection-10mic 2583	injection-10mic 1388						
Escaped	outlet-g6-8	21	287	1.262e-01	1.194e+00	1.669e-01	7.544e-02
injection-10mic 2706	injection-10mic 319						

Escaped	outlet-g6-9	22	220	1.228e-01	2.228e-01	1.411e-01	2.067e-02
injection-10mic	10850	injection-10mic	2974				
Escaped	outlet-g6-10	23	208	1.151e-01	2.894e-01	1.383e-01	3.456e-02
injection-10mic	6658	injection-10mic	11287				
Escaped	outlet-g6-11	24	291	1.223e-01	3.876e-01	1.370e-01	2.422e-02
injection-10mic	4073	injection-10mic	12854				
Escaped	outlet-g6-12	25	143	1.326e-01	1.804e-01	1.480e-01	9.164e-03
injection-10mic	7382	injection-10mic	3760				
Escaped	outlet-g6-13	26	150	1.377e-01	3.018e-01	1.490e-01	1.481e-02
injection-10mic	14914	injection-10mic	15354				
Escaped	outlet-g6-14	27	253	1.218e-01	3.919e-01	1.375e-01	2.505e-02
injection-10mic	17071	injection-10mic	15403				
Escaped	outlet-g6-15	28	189	1.150e-01	3.450e-01	1.369e-01	3.304e-02
injection-10mic	13970	injection-10mic	11925				
Escaped	outlet-g6-16	29	205	1.226e-01	2.215e-01	1.451e-01	2.366e-02
injection-10mic	11133	injection-10mic	12162				
Escaped	outlet-g6-17	30	303	1.271e-01	3.457e-01	1.631e-01	4.033e-02
injection-10mic	14031	injection-10mic	19181				
Escaped	outlet-g6-18	31	187	1.238e-01	6.211e-01	1.485e-01	4.644e-02
injection-10mic	16263	injection-10mic	19687				
Escaped	outlet-g6-19	32	135	1.312e-01	1.479e+00	1.725e-01	1.330e-01
injection-10mic	18005	injection-10mic	18633				
Escaped	outlet-g6-20	33	394	1.340e-01	1.364e+00	2.038e-01	1.231e-01
injection-10mic	13534	injection-10mic	18831				
Escaped	outlet-g6-21	34	209	1.420e-01	7.262e-01	1.932e-01	6.432e-02
injection-10mic	14091	injection-10mic	15958				
Escaped	outlet-g6-22	35	190	1.416e-01	1.904e+00	2.156e-01	1.761e-01
injection-10mic	17003	injection-10mic	16553				
Escaped	outlet-g6-23	36	284	1.455e-01	7.326e-01	3.142e-01	9.001e-02
injection-10mic	16874	injection-10mic	19482				
Escaped	outlet-g6-24	37	375	1.465e-01	5.759e+00	3.206e-01	3.896e-01
injection-10mic	15830	injection-10mic	19611				
Escaped	outlet-g5-1	38	564	1.072e-01	9.501e+00	1.637e-01	4.869e-01
injection-10mic	7565	injection-10mic	9288				
Escaped	outlet-g5-2	39	458	1.106e-01	1.481e+01	2.358e-01	7.500e-01
injection-10mic	9182	injection-10mic	7301				
Escaped	outlet-g5-3	40	448	1.137e-01	4.716e+00	1.972e-01	2.317e-01
injection-10mic	14586	injection-10mic	10621				
Escaped	outlet-g5-4	41	541	1.062e-01	2.940e+00	1.425e-01	1.538e-01
injection-10mic	15170	injection-10mic	14375				

Escaped	outlet-g4-1	42	689	9.200e-02	1.992e+00	1.398e-01	1.475e-01
injection-10mic 3555	injection-10mic 540						
Escaped	outlet-g4-2	43	1266	8.568e-02	2.665e+00	1.014e-01	8.941e-02
injection-10mic 8042	injection-10mic 507						
Escaped	outlet-g4-3	44	1008	8.470e-02	1.951e-01	9.499e-02	1.210e-02
injection-10mic 9179	injection-10mic 7516						
Escaped	outlet-g4-4	45	1293	8.332e-02	4.040e+00	1.048e-01	1.624e-01
injection-10mic 3801	injection-10mic 16931						
Escaped	outlet-g4-5	46	1224	8.432e-02	1.674e+00	9.767e-02	4.795e-02
injection-10mic 13802	injection-10mic 9926						
Escaped	outlet-g4-6	47	975	8.498e-02	1.895e-01	9.565e-02	1.302e-02
injection-10mic 13465	injection-10mic 12910						
Escaped	outlet-g4-7	48	1293	8.483e-02	1.284e+00	9.905e-02	4.192e-02
injection-10mic 12106	injection-10mic 19372						
Escaped	outlet-g4-8	49	686	9.072e-02	3.429e+00	1.474e-01	1.953e-01
injection-10mic 12284	injection-10mic 18976						

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)	Initial	Final	Change
Incomplete				5.000e-13	5.000e-13	0.000e+00
Trapped	walls-g1	7		1.810e-10	1.810e-10	0.000e+00
Trapped	walls-g2	8		2.495e-10	2.495e-10	0.000e+00
Trapped	walls-g3	9		3.570e-10	3.570e-10	0.000e+00
Trapped	walls-g4	10		8.400e-10	8.400e-10	0.000e+00
Trapped	walls-g5	11		1.865e-10	1.865e-10	0.000e+00
Trapped	walls-g6	12		6.050e-11	6.050e-11	0.000e+00
Escaped	outlet-g6-1	14		2.030e-10	2.030e-10	0.000e+00
Escaped	outlet-g6-2	15		1.370e-10	1.370e-10	0.000e+00
Escaped	outlet-g6-3	16		9.100e-11	9.100e-11	0.000e+00
Escaped	outlet-g6-4	17		1.020e-10	1.020e-10	0.000e+00
Escaped	outlet-g6-5	18		1.960e-10	1.960e-10	0.000e+00
Escaped	outlet-g6-6	19		8.200e-11	8.200e-11	0.000e+00
Escaped	outlet-g6-7	20		8.000e-11	8.000e-11	0.000e+00
Escaped	outlet-g6-8	21		1.435e-10	1.435e-10	0.000e+00
Escaped	outlet-g6-9	22		1.100e-10	1.100e-10	0.000e+00
Escaped	outlet-g6-10	23		1.040e-10	1.040e-10	0.000e+00
Escaped	outlet-g6-11	24		1.455e-10	1.455e-10	0.000e+00

Escaped	outlet-g6-12	25	7.150e-11	7.150e-11	0.000e+00
Escaped	outlet-g6-13	26	7.500e-11	7.500e-11	0.000e+00
Escaped	outlet-g6-14	27	1.265e-10	1.265e-10	0.000e+00
Escaped	outlet-g6-15	28	9.450e-11	9.450e-11	0.000e+00
Escaped	outlet-g6-16	29	1.025e-10	1.025e-10	0.000e+00
Escaped	outlet-g6-17	30	1.515e-10	1.515e-10	0.000e+00
Escaped	outlet-g6-18	31	9.350e-11	9.350e-11	0.000e+00
Escaped	outlet-g6-19	32	6.750e-11	6.750e-11	0.000e+00
Escaped	outlet-g6-20	33	1.970e-10	1.970e-10	0.000e+00
Escaped	outlet-g6-21	34	1.045e-10	1.045e-10	0.000e+00
Escaped	outlet-g6-22	35	9.500e-11	9.500e-11	0.000e+00
Escaped	outlet-g6-23	36	1.420e-10	1.420e-10	0.000e+00
Escaped	outlet-g6-24	37	1.875e-10	1.875e-10	0.000e+00
Escaped	outlet-g5-1	38	2.820e-10	2.820e-10	0.000e+00
Escaped	outlet-g5-2	39	2.290e-10	2.290e-10	0.000e+00
Escaped	outlet-g5-3	40	2.240e-10	2.240e-10	0.000e+00
Escaped	outlet-g5-4	41	2.705e-10	2.705e-10	0.000e+00
Escaped	outlet-g4-1	42	3.445e-10	3.445e-10	0.000e+00
Escaped	outlet-g4-2	43	6.330e-10	6.330e-10	0.000e+00
Escaped	outlet-g4-3	44	5.040e-10	5.040e-10	0.000e+00
Escaped	outlet-g4-4	45	6.465e-10	6.465e-10	0.000e+00
Escaped	outlet-g4-5	46	6.120e-10	6.120e-10	0.000e+00
Escaped	outlet-g4-6	47	4.875e-10	4.875e-10	0.000e+00
Escaped	outlet-g4-7	48	6.465e-10	6.465e-10	0.000e+00
Escaped	outlet-g4-8	49	3.430e-10	3.430e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)

LE global matrix maximum size: 0
 AMG cycles per iteration: 0.000 cycles
 Relaxation sweeps per iteration: 0 sweeps
 Relaxation exchanges per iteration: 0 exchanges
 LE early protections (stall) per iteration: 0.000 times
 LE early protections (divergence) per iteration: 0.000 times
 Total SVARS touched: 388

Total wall-clock time: 0.000 sec

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> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 25000, escaped = 20346, trapped = 4651, incomplete = 3

report/dpm-summary

Fate Index	Zone Injection, Index	Zone Number	Elapsed Time (s)				Injection, Min
	Name	Id	Min	Max	Avg	Std Dev	
Max							

Incomplete		3	1.372e+01	2.270e+01	1.675e+01	4.205e+00	
	injection-10micron 1155	injection-10micron 12453					
Trapped	walls-g1	7	450	4.517e-02	2.177e-01	8.542e-02	5.124e-02
	injection-10mic 13301	injection-10mic 12939					
Trapped	walls-g2	8	596	6.185e-02	1.320e-01	7.449e-02	1.462e-02
	injection-10mic 3332	injection-10mic 12712					
Trapped	walls-g3	9	904	7.184e-02	8.888e+00	9.507e-02	3.055e-01
	injection-10mic 3801	injection-10mic 718					
Trapped	walls-g4	10	2134	7.665e-02	6.031e+00	1.013e-01	1.405e-01
	injection-10mic 13069	injection-10mic 17636					
Trapped	walls-g5	11	431	8.642e-02	2.498e+00	1.292e-01	1.447e-01
	injection-10mic 11885	injection-10mic 4546					
Trapped	walls-g6	12	136	1.085e-01	7.364e-01	2.028e-01	8.579e-02
	injection-10mic 14661	injection-10mic 23343					

Escaped	outlet-g6-1	14	480	1.484e-01	6.359e+00	3.216e-01	3.896e-01
injection-10mic	3391	injection-10mic	12200				
Escaped	outlet-g6-2	15	305	1.423e-01	1.283e+00	2.994e-01	9.999e-02
injection-10mic	2449	injection-10mic	7269				
Escaped	outlet-g6-3	16	292	1.405e-01	1.404e+00	2.116e-01	1.186e-01
injection-10mic	5853	injection-10mic	3738				
Escaped	outlet-g6-4	17	271	1.405e-01	1.233e+00	2.040e-01	1.269e-01
injection-10mic	11259	injection-10mic	10874				
Escaped	outlet-g6-5	18	496	1.388e-01	8.443e-01	1.864e-01	6.816e-02
injection-10mic	3514	injection-10mic	638				
Escaped	outlet-g6-6	19	182	1.311e-01	7.414e-01	1.666e-01	5.481e-02
injection-10mic	12250	injection-10mic	695				
Escaped	outlet-g6-7	20	216	1.236e-01	6.458e-01	1.494e-01	4.544e-02
injection-10mic	5896	injection-10mic	1307				
Escaped	outlet-g6-8	21	354	1.261e-01	4.975e-01	1.623e-01	4.351e-02
injection-10mic	8670	injection-10mic	825				
Escaped	outlet-g6-9	22	270	1.203e-01	2.266e-01	1.450e-01	2.320e-02
injection-10mic	6750	injection-10mic	17496				
Escaped	outlet-g6-10	23	252	1.155e-01	2.769e-01	1.355e-01	2.858e-02
injection-10mic	11354	injection-10mic	10331				
Escaped	outlet-g6-11	24	323	1.223e-01	3.212e-01	1.352e-01	1.723e-02
injection-10mic	6191	injection-10mic	4854				
Escaped	outlet-g6-12	25	210	1.285e-01	2.452e-01	1.491e-01	1.337e-02
injection-10mic	4173	injection-10mic	4312				
Escaped	outlet-g6-13	26	184	1.267e-01	4.010e-01	1.510e-01	2.512e-02
injection-10mic	20433	injection-10mic	19529				
Escaped	outlet-g6-14	27	302	1.230e-01	3.434e-01	1.366e-01	2.343e-02
injection-10mic	20013	injection-10mic	18006				
Escaped	outlet-g6-15	28	245	1.145e-01	3.495e-01	1.340e-01	2.816e-02
injection-10mic	17179	injection-10mic	19943				
Escaped	outlet-g6-16	29	239	1.217e-01	2.211e-01	1.438e-01	2.347e-02
injection-10mic	9112	injection-10mic	11944				
Escaped	outlet-g6-17	30	347	1.234e-01	1.187e+00	1.650e-01	7.626e-02
injection-10mic	19436	injection-10mic	23559				
Escaped	outlet-g6-18	31	219	1.229e-01	2.874e-01	1.448e-01	2.267e-02
injection-10mic	15426	injection-10mic	19317				
Escaped	outlet-g6-19	32	193	1.335e-01	4.378e-01	1.630e-01	3.728e-02
injection-10mic	13045	injection-10mic	24321				
Escaped	outlet-g6-20	33	509	1.354e-01	1.605e+00	2.003e-01	1.150e-01
injection-10mic	18925	injection-10mic	19705				

Escaped	outlet-g6-21	34	267	1.381e-01	1.655e+00	2.159e-01	1.731e-01
injection-10mic	19121	injection-10mic	23392				
Escaped	outlet-g6-22	35	293	1.425e-01	8.819e-01	2.001e-01	6.184e-02
injection-10mic	22781	injection-10mic	23153				
Escaped	outlet-g6-23	36	291	1.445e-01	8.399e+00	3.399e-01	4.815e-01
injection-10mic	21409	injection-10mic	24399				
Escaped	outlet-g6-24	37	561	1.531e-01	5.296e+00	3.395e-01	3.974e-01
injection-10mic	5752	injection-10mic	21953				
Escaped	outlet-g5-1	38	692	1.053e-01	9.177e+00	1.567e-01	4.254e-01
injection-10mic	8097	injection-10mic	16985				
Escaped	outlet-g5-2	39	545	1.128e-01	6.112e+00	2.174e-01	3.413e-01
injection-10mic	6402	injection-10mic	11480				
Escaped	outlet-g5-3	40	539	1.133e-01	5.721e+00	2.416e-01	4.116e-01
injection-10mic	9939	injection-10mic	4891				
Escaped	outlet-g5-4	41	696	1.063e-01	1.043e+01	1.803e-01	5.424e-01
injection-10mic	20341	injection-10mic	17089				
Escaped	outlet-g4-1	42	858	9.115e-02	2.630e+00	1.407e-01	1.919e-01
injection-10mic	4714	injection-10mic	1166				
Escaped	outlet-g4-2	43	1658	8.566e-02	1.868e+00	9.861e-02	5.941e-02
injection-10mic	4467	injection-10mic	1580				
Escaped	outlet-g4-3	44	1202	8.416e-02	1.991e-01	9.482e-02	1.196e-02
injection-10mic	8234	injection-10mic	5932				
Escaped	outlet-g4-4	45	1626	8.430e-02	2.359e+00	9.752e-02	5.919e-02
injection-10mic	16827	injection-10mic	22413				
Escaped	outlet-g4-5	46	1590	8.508e-02	1.853e+01	1.111e-01	4.708e-01
injection-10mic	18747	injection-10mic	19745				
Escaped	outlet-g4-6	47	1167	8.468e-02	1.883e-01	9.505e-02	1.189e-02
injection-10mic	19220	injection-10mic	10356				
Escaped	outlet-g4-7	48	1614	8.603e-02	1.764e+00	9.844e-02	4.737e-02
injection-10mic	21434	injection-10mic	23218				
Escaped	outlet-g4-8	49	858	9.067e-02	9.545e+00	1.399e-01	3.405e-01
injection-10mic	17133	injection-10mic	24400				

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)
			Initial Final Change
Incomplete			1.200e-12 1.200e-12 0.000e+00
Trapped	walls-g1	7	1.800e-10 1.800e-10 0.000e+00

Trapped	walls-g2	8	2.384e-10	2.384e-10	0.000e+00
Trapped	walls-g3	9	3.616e-10	3.616e-10	0.000e+00
Trapped	walls-g4	10	8.536e-10	8.536e-10	0.000e+00
Trapped	walls-g5	11	1.724e-10	1.724e-10	0.000e+00
Trapped	walls-g6	12	5.440e-11	5.440e-11	0.000e+00
Escaped	outlet-g6-1	14	1.920e-10	1.920e-10	0.000e+00
Escaped	outlet-g6-2	15	1.220e-10	1.220e-10	0.000e+00
Escaped	outlet-g6-3	16	1.168e-10	1.168e-10	0.000e+00
Escaped	outlet-g6-4	17	1.084e-10	1.084e-10	0.000e+00
Escaped	outlet-g6-5	18	1.984e-10	1.984e-10	0.000e+00
Escaped	outlet-g6-6	19	7.280e-11	7.280e-11	0.000e+00
Escaped	outlet-g6-7	20	8.640e-11	8.640e-11	0.000e+00
Escaped	outlet-g6-8	21	1.416e-10	1.416e-10	0.000e+00
Escaped	outlet-g6-9	22	1.080e-10	1.080e-10	0.000e+00
Escaped	outlet-g6-10	23	1.008e-10	1.008e-10	0.000e+00
Escaped	outlet-g6-11	24	1.292e-10	1.292e-10	0.000e+00
Escaped	outlet-g6-12	25	8.400e-11	8.400e-11	0.000e+00
Escaped	outlet-g6-13	26	7.360e-11	7.360e-11	0.000e+00
Escaped	outlet-g6-14	27	1.208e-10	1.208e-10	0.000e+00
Escaped	outlet-g6-15	28	9.800e-11	9.800e-11	0.000e+00
Escaped	outlet-g6-16	29	9.560e-11	9.560e-11	0.000e+00
Escaped	outlet-g6-17	30	1.388e-10	1.388e-10	0.000e+00
Escaped	outlet-g6-18	31	8.760e-11	8.760e-11	0.000e+00
Escaped	outlet-g6-19	32	7.720e-11	7.720e-11	0.000e+00
Escaped	outlet-g6-20	33	2.036e-10	2.036e-10	0.000e+00
Escaped	outlet-g6-21	34	1.068e-10	1.068e-10	0.000e+00
Escaped	outlet-g6-22	35	1.172e-10	1.172e-10	0.000e+00
Escaped	outlet-g6-23	36	1.164e-10	1.164e-10	0.000e+00
Escaped	outlet-g6-24	37	2.244e-10	2.244e-10	0.000e+00
Escaped	outlet-g5-1	38	2.768e-10	2.768e-10	0.000e+00
Escaped	outlet-g5-2	39	2.180e-10	2.180e-10	0.000e+00
Escaped	outlet-g5-3	40	2.156e-10	2.156e-10	0.000e+00
Escaped	outlet-g5-4	41	2.784e-10	2.784e-10	0.000e+00
Escaped	outlet-g4-1	42	3.432e-10	3.432e-10	0.000e+00
Escaped	outlet-g4-2	43	6.632e-10	6.632e-10	0.000e+00
Escaped	outlet-g4-3	44	4.808e-10	4.808e-10	0.000e+00
Escaped	outlet-g4-4	45	6.504e-10	6.504e-10	0.000e+00
Escaped	outlet-g4-5	46	6.360e-10	6.360e-10	0.000e+00
Escaped	outlet-g4-6	47	4.668e-10	4.668e-10	0.000e+00
Escaped	outlet-g4-7	48	6.456e-10	6.456e-10	0.000e+00

Escaped	outlet-g4-8	49	3.432e-10	3.432e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript]

non-fatal Error: invalid command [stop-transcript]]

> \f

non-fatal Error: invalid command [\]

ile/stop-transcript

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time:	0.000 sec
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```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model
Specific)*PushButton1(Discrete Phase)")
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 30000, escaped = 24328, trapped = 5670, incomplete = 2

report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Incomplete		2	1.442e+01	2.251e+01	1.846e+01	4.045e+00	
injection-10micron	25102	injection-10micron	1878				
Trapped	walls-g1	7	513	4.534e-02	2.157e-01	8.340e-02	5.062e-02
injection-10mic	7867	injection-10mic	10036				
Trapped	walls-g2	8	713	6.389e-02	3.888e-01	7.516e-02	1.886e-02
injection-10mic	10366	injection-10mic	17680				
Trapped	walls-g3	9	1099	7.191e-02	1.306e+00	8.731e-02	5.897e-02
injection-10mic	15908	injection-10mic	1270				
Trapped	walls-g4	10	2587	7.662e-02	9.397e+00	1.028e-01	1.944e-01
injection-10mic	5093	injection-10mic	15038				
Trapped	walls-g5	11	575	8.679e-02	1.275e+00	1.203e-01	7.818e-02
injection-10mic	12770	injection-10mic	11148				
Trapped	walls-g6	12	183	1.085e-01	1.132e+00	2.071e-01	9.311e-02
injection-10mic	10368	injection-10mic	16813				
Escaped	outlet-g6-1	14	580	1.414e-01	9.500e+00	3.459e-01	5.375e-01
injection-10mic	10986	injection-10mic	1327				
Escaped	outlet-g6-2	15	410	1.405e-01	1.062e+00	3.041e-01	8.393e-02
injection-10mic	19674	injection-10mic	2533				
Escaped	outlet-g6-3	16	209	1.404e-01	4.086e+00	2.191e-01	2.899e-01
injection-10mic	6641	injection-10mic	1155				

Escaped	outlet-g6-4	17	373	1.450e-01	2.587e+00	2.093e-01	1.746e-01
injection-10mic	14647	injection-10mic	1365				
Escaped	outlet-g6-5	18	559	1.317e-01	7.046e-01	1.860e-01	6.191e-02
injection-10mic	11071	injection-10mic	635				
Escaped	outlet-g6-6	19	276	1.308e-01	2.744e-01	1.552e-01	1.933e-02
injection-10mic	8928	injection-10mic	1189				
Escaped	outlet-g6-7	20	247	1.235e-01	4.315e+00	1.761e-01	3.259e-01
injection-10mic	8300	injection-10mic	1166				
Escaped	outlet-g6-8	21	461	1.258e-01	4.247e+00	1.839e-01	2.274e-01
injection-10mic	7632	injection-10mic	1742				
Escaped	outlet-g6-9	22	323	1.218e-01	2.257e-01	1.424e-01	2.221e-02
injection-10mic	10714	injection-10mic	17829				
Escaped	outlet-g6-10	23	309	1.153e-01	3.544e-01	1.387e-01	3.829e-02
injection-10mic	7869	injection-10mic	20141				
Escaped	outlet-g6-11	24	413	1.226e-01	3.316e-01	1.367e-01	2.149e-02
injection-10mic	5127	injection-10mic	4145				
Escaped	outlet-g6-12	25	215	1.299e-01	3.488e-01	1.506e-01	2.309e-02
injection-10mic	17726	injection-10mic	11749				
Escaped	outlet-g6-13	26	206	1.344e-01	3.868e-01	1.489e-01	1.895e-02
injection-10mic	26668	injection-10mic	21816				
Escaped	outlet-g6-14	27	405	1.227e-01	3.575e-01	1.358e-01	2.114e-02
injection-10mic	14991	injection-10mic	18178				
Escaped	outlet-g6-15	28	305	1.147e-01	3.540e-01	1.362e-01	3.112e-02
injection-10mic	26406	injection-10mic	24697				
Escaped	outlet-g6-16	29	300	1.219e-01	2.257e-01	1.425e-01	2.281e-02
injection-10mic	20608	injection-10mic	24418				
Escaped	outlet-g6-17	30	464	1.241e-01	7.789e-01	1.639e-01	4.831e-02
injection-10mic	23226	injection-10mic	27012				
Escaped	outlet-g6-18	31	265	1.241e-01	8.961e-01	1.509e-01	5.809e-02
injection-10mic	27089	injection-10mic	29035				
Escaped	outlet-g6-19	32	263	1.345e-01	1.988e+00	1.725e-01	1.293e-01
injection-10mic	22330	injection-10mic	28569				
Escaped	outlet-g6-20	33	607	1.320e-01	1.265e+00	1.968e-01	9.404e-02
injection-10mic	20696	injection-10mic	29349				
Escaped	outlet-g6-21	34	305	1.420e-01	1.513e+00	2.211e-01	1.795e-01
injection-10mic	20434	injection-10mic	29142				
Escaped	outlet-g6-22	35	293	1.394e-01	1.640e+00	2.051e-01	1.205e-01
injection-10mic	22225	injection-10mic	29157				
Escaped	outlet-g6-23	36	418	1.436e-01	1.343e+00	3.174e-01	1.223e-01
injection-10mic	17780	injection-10mic	29377				

Escaped	outlet-g6-24	37	609	1.463e-01	5.080e+00	3.147e-01	3.600e-01
injection-10mic 20917 injection-10mic 28560							
Escaped	outlet-g5-1	38	844	1.039e-01	5.870e+00	1.532e-01	2.921e-01
injection-10mic 8033 injection-10mic 4285							
Escaped	outlet-g5-2	39	654	1.088e-01	7.132e+00	2.121e-01	3.200e-01
injection-10mic 6214 injection-10mic 7148							
Escaped	outlet-g5-3	40	605	1.124e-01	4.723e+00	2.240e-01	3.559e-01
injection-10mic 20007 injection-10mic 19574							
Escaped	outlet-g5-4	41	840	1.068e-01	1.033e+01	1.607e-01	4.478e-01
injection-10mic 26295 injection-10mic 8502							
Escaped	outlet-g4-1	42	1125	9.022e-02	2.679e+00	1.321e-01	1.352e-01
injection-10mic 3749 injection-10mic 11534							
Escaped	outlet-g4-2	43	1952	8.472e-02	1.710e+00	9.756e-02	4.472e-02
injection-10mic 3855 injection-10mic 639							
Escaped	outlet-g4-3	44	1385	8.480e-02	9.650e-01	9.572e-02	2.640e-02
injection-10mic 10747 injection-10mic 1891							
Escaped	outlet-g4-4	45	1908	8.475e-02	4.611e+00	1.015e-01	1.209e-01
injection-10mic 3595 injection-10mic 15792							
Escaped	outlet-g4-5	46	1834	8.246e-02	1.097e+01	1.071e-01	2.813e-01
injection-10mic 21322 injection-10mic 19901							
Escaped	outlet-g4-6	47	1454	8.443e-02	1.907e-01	9.544e-02	1.346e-02
injection-10mic 23439 injection-10mic 19152							
Escaped	outlet-g4-7	48	1924	8.531e-02	2.716e+00	1.007e-01	8.427e-02
injection-10mic 14158 injection-10mic 26650							
Escaped	outlet-g4-8	49	988	9.142e-02	3.755e+00	1.382e-01	1.777e-01
injection-10mic 19195 injection-10mic 26700							

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Incomplete			6.667e-13	6.667e-13	0.000e+00
Trapped	walls-g1	7	1.710e-10	1.710e-10	0.000e+00
Trapped	walls-g2	8	2.377e-10	2.377e-10	0.000e+00
Trapped	walls-g3	9	3.663e-10	3.663e-10	0.000e+00
Trapped	walls-g4	10	8.623e-10	8.623e-10	0.000e+00
Trapped	walls-g5	11	1.917e-10	1.917e-10	0.000e+00
Trapped	walls-g6	12	6.100e-11	6.100e-11	0.000e+00
Escaped	outlet-g6-1	14	1.933e-10	1.933e-10	0.000e+00

Escaped	outlet-g6-2	15	1.367e-10	1.367e-10	0.000e+00
Escaped	outlet-g6-3	16	6.967e-11	6.967e-11	0.000e+00
Escaped	outlet-g6-4	17	1.243e-10	1.243e-10	0.000e+00
Escaped	outlet-g6-5	18	1.863e-10	1.863e-10	0.000e+00
Escaped	outlet-g6-6	19	9.200e-11	9.200e-11	0.000e+00
Escaped	outlet-g6-7	20	8.233e-11	8.233e-11	0.000e+00
Escaped	outlet-g6-8	21	1.537e-10	1.537e-10	0.000e+00
Escaped	outlet-g6-9	22	1.077e-10	1.077e-10	0.000e+00
Escaped	outlet-g6-10	23	1.030e-10	1.030e-10	0.000e+00
Escaped	outlet-g6-11	24	1.377e-10	1.377e-10	0.000e+00
Escaped	outlet-g6-12	25	7.167e-11	7.167e-11	0.000e+00
Escaped	outlet-g6-13	26	6.867e-11	6.867e-11	0.000e+00
Escaped	outlet-g6-14	27	1.350e-10	1.350e-10	0.000e+00
Escaped	outlet-g6-15	28	1.017e-10	1.017e-10	0.000e+00
Escaped	outlet-g6-16	29	1.000e-10	1.000e-10	0.000e+00
Escaped	outlet-g6-17	30	1.547e-10	1.547e-10	0.000e+00
Escaped	outlet-g6-18	31	8.833e-11	8.833e-11	0.000e+00
Escaped	outlet-g6-19	32	8.767e-11	8.767e-11	0.000e+00
Escaped	outlet-g6-20	33	2.023e-10	2.023e-10	0.000e+00
Escaped	outlet-g6-21	34	1.017e-10	1.017e-10	0.000e+00
Escaped	outlet-g6-22	35	9.767e-11	9.767e-11	0.000e+00
Escaped	outlet-g6-23	36	1.393e-10	1.393e-10	0.000e+00
Escaped	outlet-g6-24	37	2.030e-10	2.030e-10	0.000e+00
Escaped	outlet-g5-1	38	2.813e-10	2.813e-10	0.000e+00
Escaped	outlet-g5-2	39	2.180e-10	2.180e-10	0.000e+00
Escaped	outlet-g5-3	40	2.017e-10	2.017e-10	0.000e+00
Escaped	outlet-g5-4	41	2.800e-10	2.800e-10	0.000e+00
Escaped	outlet-g4-1	42	3.750e-10	3.750e-10	0.000e+00
Escaped	outlet-g4-2	43	6.507e-10	6.507e-10	0.000e+00
Escaped	outlet-g4-3	44	4.617e-10	4.617e-10	0.000e+00
Escaped	outlet-g4-4	45	6.360e-10	6.360e-10	0.000e+00
Escaped	outlet-g4-5	46	6.113e-10	6.113e-10	0.000e+00
Escaped	outlet-g4-6	47	4.847e-10	4.847e-10	0.000e+00
Escaped	outlet-g4-7	48	6.413e-10	6.413e-10	0.000e+00
Escaped	outlet-g4-8	49	3.293e-10	3.293e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time: 0.000 sec

—

>

adjoint/	mesh/	report/
close-fluent	parallel/	solve/
define/	plot/	surface/
display/	preferences/	views/
file/	print-license-usage	

> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 40000, escaped = 32527, trapped = 7470, incomplete = 3

report/dpm-summary

Fate Index	Zone Injection, Index	Zone Number	Elapsed Time (s)	Injection,
---------------	--------------------------	-------------	------------------	------------

	Name	Id	Min	Max	Avg	Std Dev	Min
Max	-----						
Incomplete		3	1.370e+01	1.449e+01	1.418e+01	3.414e-01	
injection-10micron	1045	injection-10micron	39669				
Trapped	walls-g1	7	726	4.419e-02	2.160e-01	8.436e-02	4.953e-02
injection-10mic	13069	injection-10mic	29409				
Trapped	walls-g2	8	976	6.339e-02	3.502e-01	7.529e-02	1.753e-02
injection-10mic	26992	injection-10mic	11206				
Trapped	walls-g3	9	1429	7.211e-02	1.552e+00	8.596e-02	5.533e-02
injection-10mic	33020	injection-10mic	37564				
Trapped	walls-g4	10	3406	7.640e-02	3.759e+00	1.017e-01	9.931e-02
injection-10mic	22664	injection-10mic	24783				
Trapped	walls-g5	11	735	8.642e-02	7.724e-01	1.200e-01	6.853e-02
injection-10mic	30692	injection-10mic	35704				
Trapped	walls-g6	12	198	1.062e-01	9.459e-01	2.047e-01	1.078e-01
injection-10mic	5588	injection-10mic	1398				
Escaped	outlet-g6-1	14	800	1.457e-01	2.910e+00	3.063e-01	2.598e-01
injection-10mic	9753	injection-10mic	11259				
Escaped	outlet-g6-2	15	445	1.439e-01	2.195e+00	3.146e-01	1.517e-01
injection-10mic	3377	injection-10mic	4194				
Escaped	outlet-g6-3	16	452	1.376e-01	1.693e+00	2.054e-01	1.023e-01
injection-10mic	6298	injection-10mic	2302				
Escaped	outlet-g6-4	17	485	1.423e-01	7.226e+00	2.232e-01	3.540e-01
injection-10mic	23202	injection-10mic	3822				
Escaped	outlet-g6-5	18	758	1.318e-01	1.869e+00	1.954e-01	1.101e-01
injection-10mic	11625	injection-10mic	1679				
Escaped	outlet-g6-6	19	335	1.314e-01	3.520e-01	1.593e-01	2.430e-02
injection-10mic	18389	injection-10mic	5054				
Escaped	outlet-g6-7	20	329	1.223e-01	2.824e-01	1.432e-01	1.972e-02
injection-10mic	7376	injection-10mic	4689				
Escaped	outlet-g6-8	21	585	1.238e-01	1.646e+00	1.706e-01	9.637e-02
injection-10mic	14184	injection-10mic	1133				
Escaped	outlet-g6-9	22	425	1.218e-01	2.237e-01	1.445e-01	2.375e-02
injection-10mic	9056	injection-10mic	17736				
Escaped	outlet-g6-10	23	378	1.151e-01	3.388e-01	1.381e-01	3.418e-02
injection-10mic	15553	injection-10mic	7626				
Escaped	outlet-g6-11	24	535	1.229e-01	2.696e-01	1.358e-01	1.757e-02
injection-10mic	12254	injection-10mic	11369				

Escaped	outlet-g6-12	25	307	1.262e-01	3.900e-01	1.491e-01	2.034e-02
injection-10mic	10924	injection-10mic	7629				
Escaped	outlet-g6-13	26	306	1.259e-01	4.505e-01	1.494e-01	2.171e-02
injection-10mic	22001	injection-10mic	18912				
Escaped	outlet-g6-14	27	584	1.229e-01	6.257e-01	1.382e-01	3.237e-02
injection-10mic	18738	injection-10mic	22579				
Escaped	outlet-g6-15	28	390	1.135e-01	3.541e-01	1.363e-01	3.603e-02
injection-10mic	22037	injection-10mic	24884				
Escaped	outlet-g6-16	29	402	1.209e-01	2.247e-01	1.440e-01	2.381e-02
injection-10mic	33275	injection-10mic	28281				
Escaped	outlet-g6-17	30	623	1.259e-01	1.587e+00	1.714e-01	8.672e-02
injection-10mic	31692	injection-10mic	38521				
Escaped	outlet-g6-18	31	338	1.209e-01	8.325e-01	1.466e-01	4.387e-02
injection-10mic	32908	injection-10mic	37534				
Escaped	outlet-g6-19	32	320	1.312e-01	6.858e+00	1.852e-01	3.795e-01
injection-10mic	27834	injection-10mic	32913				
Escaped	outlet-g6-20	33	754	1.325e-01	1.487e+00	1.963e-01	1.053e-01
injection-10mic	28913	injection-10mic	38361				
Escaped	outlet-g6-21	34	451	1.384e-01	9.419e+00	2.219e-01	4.539e-01
injection-10mic	30409	injection-10mic	32959				
Escaped	outlet-g6-22	35	456	1.394e-01	1.498e+00	2.034e-01	9.147e-02
injection-10mic	27005	injection-10mic	39238				
Escaped	outlet-g6-23	36	451	1.392e-01	1.015e+01	3.408e-01	4.838e-01
injection-10mic	32388	injection-10mic	38169				
Escaped	outlet-g6-24	37	798	1.442e-01	2.517e+00	3.124e-01	2.557e-01
injection-10mic	30034	injection-10mic	39118				
Escaped	outlet-g5-1	38	1024	1.066e-01	8.234e+00	1.762e-01	4.912e-01
injection-10mic	12943	injection-10mic	20166				
Escaped	outlet-g5-2	39	933	1.070e-01	7.004e+00	2.282e-01	4.436e-01
injection-10mic	7308	injection-10mic	15733				
Escaped	outlet-g5-3	40	832	1.088e-01	1.482e+01	2.663e-01	8.196e-01
injection-10mic	20027	injection-10mic	11787				
Escaped	outlet-g5-4	41	1124	1.073e-01	1.379e+01	1.804e-01	6.239e-01
injection-10mic	29450	injection-10mic	19474				
Escaped	outlet-g4-1	42	1400	8.773e-02	4.202e+00	1.409e-01	2.001e-01
injection-10mic	13059	injection-10mic	1532				
Escaped	outlet-g4-2	43	2644	8.492e-02	1.876e+00	9.866e-02	4.702e-02
injection-10mic	10257	injection-10mic	1229				
Escaped	outlet-g4-3	44	1860	8.359e-02	1.923e-01	9.542e-02	1.331e-02
injection-10mic	13180	injection-10mic	21809				

Escaped	outlet-g4-4	45	2541	8.335e-02	2.570e+00	9.834e-02	6.408e-02
injection-10mic	21807	injection-10mic	33376				
Escaped	outlet-g4-5	46	2591	8.380e-02	2.101e+01	1.149e-01	4.734e-01
injection-10mic	34014	injection-10mic	19208				
Escaped	outlet-g4-6	47	1893	8.500e-02	1.927e-01	9.523e-02	1.264e-02
injection-10mic	29180	injection-10mic	26143				
Escaped	outlet-g4-7	48	2570	8.481e-02	1.668e+00	9.984e-02	5.577e-02
injection-10mic	11178	injection-10mic	39550				
Escaped	outlet-g4-8	49	1408	9.128e-02	4.769e+00	1.411e-01	2.037e-01
injection-10mic	25267	injection-10mic	39209				

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)			
			Initial	Final	Change	
Incomplete			7.500e-13	7.500e-13	0.000e+00	
Trapped	walls-g1	7	1.815e-10	1.815e-10	0.000e+00	
Trapped	walls-g2	8	2.440e-10	2.440e-10	0.000e+00	
Trapped	walls-g3	9	3.572e-10	3.572e-10	0.000e+00	
Trapped	walls-g4	10	8.515e-10	8.515e-10	0.000e+00	
Trapped	walls-g5	11	1.837e-10	1.837e-10	0.000e+00	
Trapped	walls-g6	12	4.950e-11	4.950e-11	0.000e+00	
Escaped	outlet-g6-1	14	2.000e-10	2.000e-10	0.000e+00	
Escaped	outlet-g6-2	15	1.112e-10	1.112e-10	0.000e+00	
Escaped	outlet-g6-3	16	1.130e-10	1.130e-10	0.000e+00	
Escaped	outlet-g6-4	17	1.212e-10	1.212e-10	0.000e+00	
Escaped	outlet-g6-5	18	1.895e-10	1.895e-10	0.000e+00	
Escaped	outlet-g6-6	19	8.375e-11	8.375e-11	0.000e+00	
Escaped	outlet-g6-7	20	8.225e-11	8.225e-11	0.000e+00	
Escaped	outlet-g6-8	21	1.462e-10	1.462e-10	0.000e+00	
Escaped	outlet-g6-9	22	1.062e-10	1.062e-10	0.000e+00	
Escaped	outlet-g6-10	23	9.450e-11	9.450e-11	0.000e+00	
Escaped	outlet-g6-11	24	1.337e-10	1.337e-10	0.000e+00	
Escaped	outlet-g6-12	25	7.675e-11	7.675e-11	0.000e+00	
Escaped	outlet-g6-13	26	7.650e-11	7.650e-11	0.000e+00	
Escaped	outlet-g6-14	27	1.460e-10	1.460e-10	0.000e+00	
Escaped	outlet-g6-15	28	9.750e-11	9.750e-11	0.000e+00	
Escaped	outlet-g6-16	29	1.005e-10	1.005e-10	0.000e+00	
Escaped	outlet-g6-17	30	1.557e-10	1.557e-10	0.000e+00	

Escaped	outlet-g6-18	31	8.450e-11	8.450e-11	0.000e+00
Escaped	outlet-g6-19	32	8.000e-11	8.000e-11	0.000e+00
Escaped	outlet-g6-20	33	1.885e-10	1.885e-10	0.000e+00
Escaped	outlet-g6-21	34	1.127e-10	1.127e-10	0.000e+00
Escaped	outlet-g6-22	35	1.140e-10	1.140e-10	0.000e+00
Escaped	outlet-g6-23	36	1.127e-10	1.127e-10	0.000e+00
Escaped	outlet-g6-24	37	1.995e-10	1.995e-10	0.000e+00
Escaped	outlet-g5-1	38	2.560e-10	2.560e-10	0.000e+00
Escaped	outlet-g5-2	39	2.332e-10	2.332e-10	0.000e+00
Escaped	outlet-g5-3	40	2.080e-10	2.080e-10	0.000e+00
Escaped	outlet-g5-4	41	2.810e-10	2.810e-10	0.000e+00
Escaped	outlet-g4-1	42	3.500e-10	3.500e-10	0.000e+00
Escaped	outlet-g4-2	43	6.610e-10	6.610e-10	0.000e+00
Escaped	outlet-g4-3	44	4.650e-10	4.650e-10	0.000e+00
Escaped	outlet-g4-4	45	6.353e-10	6.353e-10	0.000e+00
Escaped	outlet-g4-5	46	6.478e-10	6.478e-10	0.000e+00
Escaped	outlet-g4-6	47	4.733e-10	4.733e-10	0.000e+00
Escaped	outlet-g4-7	48	6.425e-10	6.425e-10	0.000e+00
Escaped	outlet-g4-8	49	3.520e-10	3.520e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times

Total SVARS touched: 388

Total wall-clock time: 0.000 sec

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> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 50000, escaped = 40477, trapped = 9519, incomplete = 4

report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Incomplete		4	1.366e+01	1.438e+01	1.403e+01	3.318e-01	
injection-10micron	1367 injection-10micron	49362					
Trapped	walls-g1	7	931	4.581e-02	1.779e+00	8.571e-02	7.568e-02
injection-10mic	16581 injection-10mic	48735					
Trapped	walls-g2	8	1226	6.299e-02	2.921e-01	7.537e-02	1.660e-02
injection-10mic	31558 injection-10mic	39970					
Trapped	walls-g3	9	1831	7.181e-02	1.357e+00	8.577e-02	4.740e-02
injection-10mic	28816 injection-10mic	48340					
Trapped	walls-g4	10	4360	7.714e-02	1.068e+01	1.074e-01	2.409e-01
injection-10mic	38699 injection-10mic	2216					
Trapped	walls-g5	11	908	8.510e-02	9.137e-01	1.149e-01	5.588e-02
injection-10mic	19406 injection-10mic	46768					
Trapped	walls-g6	12	263	1.067e-01	3.476e+00	2.177e-01	2.722e-01
injection-10mic	29476 injection-10mic	32913					
Escaped	outlet-g6-1	14	1025	1.449e-01	2.695e+00	3.060e-01	2.454e-01
injection-10mic	5604 injection-10mic	5162					
Escaped	outlet-g6-2	15	571	1.391e-01	7.251e+00	3.286e-01	3.220e-01
injection-10mic	9678 injection-10mic	4136					
Escaped	outlet-g6-3	16	479	1.419e-01	1.130e+00	1.972e-01	5.632e-02
injection-10mic	18035 injection-10mic	1032					

Escaped	outlet-g6-4	17	574	1.419e-01	2.800e+00	2.210e-01	1.997e-01
injection-10mic	14989	injection-10mic	6886				
Escaped	outlet-g6-5	18	923	1.369e-01	3.071e+00	1.911e-01	1.310e-01
injection-10mic	5510	injection-10mic	25339				
Escaped	outlet-g6-6	19	399	1.306e-01	1.616e+00	1.647e-01	8.822e-02
injection-10mic	10816	injection-10mic	3430				
Escaped	outlet-g6-7	20	421	1.207e-01	4.029e+00	1.566e-01	1.990e-01
injection-10mic	7411	injection-10mic	3460				
Escaped	outlet-g6-8	21	717	1.254e-01	1.701e+00	1.683e-01	8.811e-02
injection-10mic	8520	injection-10mic	2013				
Escaped	outlet-g6-9	22	520	1.209e-01	2.252e-01	1.450e-01	2.447e-02
injection-10mic	4828	injection-10mic	29413				
Escaped	outlet-g6-10	23	548	1.147e-01	3.276e-01	1.374e-01	3.109e-02
injection-10mic	24429	injection-10mic	18155				
Escaped	outlet-g6-11	24	628	1.235e-01	5.537e-01	1.363e-01	2.815e-02
injection-10mic	33685	injection-10mic	21108				
Escaped	outlet-g6-12	25	393	1.244e-01	3.664e-01	1.505e-01	1.995e-02
injection-10mic	10251	injection-10mic	8213				
Escaped	outlet-g6-13	26	397	1.250e-01	4.519e-01	1.514e-01	2.868e-02
injection-10mic	33866	injection-10mic	35507				
Escaped	outlet-g6-14	27	653	1.220e-01	6.377e-01	1.386e-01	3.108e-02
injection-10mic	41109	injection-10mic	37650				
Escaped	outlet-g6-15	28	454	1.152e-01	3.376e-01	1.396e-01	3.821e-02
injection-10mic	38296	injection-10mic	29491				
Escaped	outlet-g6-16	29	532	1.216e-01	2.263e-01	1.446e-01	2.292e-02
injection-10mic	26241	injection-10mic	34921				
Escaped	outlet-g6-17	30	728	1.219e-01	2.861e+00	1.698e-01	1.235e-01
injection-10mic	39909	injection-10mic	48100				
Escaped	outlet-g6-18	31	429	1.235e-01	5.531e+00	1.582e-01	2.626e-01
injection-10mic	39742	injection-10mic	46677				
Escaped	outlet-g6-19	32	390	1.329e-01	3.368e+00	1.755e-01	1.891e-01
injection-10mic	27380	injection-10mic	44658				
Escaped	outlet-g6-20	33	942	1.361e-01	8.597e-01	1.970e-01	9.113e-02
injection-10mic	17430	injection-10mic	49032				
Escaped	outlet-g6-21	34	562	1.448e-01	3.841e+01	2.795e-01	
1.616e+00	injection-10mic	46660	injection-10mic	36862			
Escaped	outlet-g6-22	35	488	1.393e-01	8.928e+00	2.266e-01	4.141e-01
injection-10mic	37342	injection-10mic	46889				
Escaped	outlet-g6-23	36	557	1.460e-01	1.160e+01	3.539e-01	6.670e-01
injection-10mic	10887	injection-10mic	43978				

Escaped	outlet-g6-24	37	1006	1.444e-01	2.967e+00	3.081e-01	2.582e-01
	injection-10mic 45549						injection-10mic 40298
Escaped	outlet-g5-1	38	1349	1.044e-01	1.758e+01	1.706e-01	5.674e-01
	injection-10mic 42811						injection-10mic 16238
Escaped	outlet-g5-2	39	1139	1.106e-01	7.645e+00	2.121e-01	3.535e-01
	injection-10mic 35467						injection-10mic 30793
Escaped	outlet-g5-3	40	1051	1.118e-01	5.086e+00	2.140e-01	3.055e-01
	injection-10mic 22538						injection-10mic 34411
Escaped	outlet-g5-4	41	1361	1.075e-01	1.278e+01	1.889e-01	6.324e-01
	injection-10mic 37645						injection-10mic 31922
Escaped	outlet-g4-1	42	1746	8.999e-02	4.338e+00	1.297e-01	1.406e-01
	injection-10mic 11308						injection-10mic 1738
Escaped	outlet-g4-2	43	3278	8.488e-02	2.561e+00	1.026e-01	9.932e-02
	injection-10mic 12702						injection-10mic 2996
Escaped	outlet-g4-3	44	2396	8.401e-02	3.936e-01	9.493e-02	1.365e-02
	injection-10mic 14693						injection-10mic 14273
Escaped	outlet-g4-4	45	3176	8.363e-02	6.096e+00	1.013e-01	1.592e-01
	injection-10mic 13898						injection-10mic 11998
Escaped	outlet-g4-5	46	3220	8.293e-02	2.857e+00	9.820e-02	6.908e-02
	injection-10mic 29623						injection-10mic 16792
Escaped	outlet-g4-6	47	2459	8.434e-02	1.878e-01	9.506e-02	1.173e-02
	injection-10mic 35916						injection-10mic 43878
Escaped	outlet-g4-7	48	3259	8.461e-02	2.299e+00	9.990e-02	6.215e-02
	injection-10mic 37706						injection-10mic 38337
Escaped	outlet-g4-8	49	1707	9.099e-02	9.403e+00	1.490e-01	3.249e-01
	injection-10mic 24055						injection-10mic 44723

(*)- Mass Transfer Summary -(*)

Fate	Zone	Zone Id	Mass Flow (kg/s)		
	Name	Initial	Final	Change	
Incomplete			8.000e-13	8.000e-13	0.000e+00
Trapped	walls-g1	7	1.862e-10	1.862e-10	0.000e+00
Trapped	walls-g2	8	2.452e-10	2.452e-10	0.000e+00
Trapped	walls-g3	9	3.662e-10	3.662e-10	0.000e+00
Trapped	walls-g4	10	8.720e-10	8.720e-10	0.000e+00
Trapped	walls-g5	11	1.816e-10	1.816e-10	0.000e+00
Trapped	walls-g6	12	5.260e-11	5.260e-11	0.000e+00
Escaped	outlet-g6-1	14	2.050e-10	2.050e-10	0.000e+00

Escaped	outlet-g6-2	15	1.142e-10	1.142e-10	0.000e+00
Escaped	outlet-g6-3	16	9.580e-11	9.580e-11	0.000e+00
Escaped	outlet-g6-4	17	1.148e-10	1.148e-10	0.000e+00
Escaped	outlet-g6-5	18	1.846e-10	1.846e-10	0.000e+00
Escaped	outlet-g6-6	19	7.980e-11	7.980e-11	0.000e+00
Escaped	outlet-g6-7	20	8.420e-11	8.420e-11	0.000e+00
Escaped	outlet-g6-8	21	1.434e-10	1.434e-10	0.000e+00
Escaped	outlet-g6-9	22	1.040e-10	1.040e-10	0.000e+00
Escaped	outlet-g6-10	23	1.096e-10	1.096e-10	0.000e+00
Escaped	outlet-g6-11	24	1.256e-10	1.256e-10	0.000e+00
Escaped	outlet-g6-12	25	7.860e-11	7.860e-11	0.000e+00
Escaped	outlet-g6-13	26	7.940e-11	7.940e-11	0.000e+00
Escaped	outlet-g6-14	27	1.306e-10	1.306e-10	0.000e+00
Escaped	outlet-g6-15	28	9.080e-11	9.080e-11	0.000e+00
Escaped	outlet-g6-16	29	1.064e-10	1.064e-10	0.000e+00
Escaped	outlet-g6-17	30	1.456e-10	1.456e-10	0.000e+00
Escaped	outlet-g6-18	31	8.580e-11	8.580e-11	0.000e+00
Escaped	outlet-g6-19	32	7.800e-11	7.800e-11	0.000e+00
Escaped	outlet-g6-20	33	1.884e-10	1.884e-10	0.000e+00
Escaped	outlet-g6-21	34	1.124e-10	1.124e-10	0.000e+00
Escaped	outlet-g6-22	35	9.760e-11	9.760e-11	0.000e+00
Escaped	outlet-g6-23	36	1.114e-10	1.114e-10	0.000e+00
Escaped	outlet-g6-24	37	2.012e-10	2.012e-10	0.000e+00
Escaped	outlet-g5-1	38	2.698e-10	2.698e-10	0.000e+00
Escaped	outlet-g5-2	39	2.278e-10	2.278e-10	0.000e+00
Escaped	outlet-g5-3	40	2.102e-10	2.102e-10	0.000e+00
Escaped	outlet-g5-4	41	2.722e-10	2.722e-10	0.000e+00
Escaped	outlet-g4-1	42	3.492e-10	3.492e-10	0.000e+00
Escaped	outlet-g4-2	43	6.556e-10	6.556e-10	0.000e+00
Escaped	outlet-g4-3	44	4.792e-10	4.792e-10	0.000e+00
Escaped	outlet-g4-4	45	6.352e-10	6.352e-10	0.000e+00
Escaped	outlet-g4-5	46	6.440e-10	6.440e-10	0.000e+00
Escaped	outlet-g4-6	47	4.918e-10	4.918e-10	0.000e+00
Escaped	outlet-g4-7	48	6.518e-10	6.518e-10	0.000e+00
Escaped	outlet-g4-8	49	3.414e-10	3.414e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration: 0.000 sec
 Global reductions per iteration: 0 ops
 Global reductions time per iteration: 0.000 sec (0.0%)
 Message count per iteration: 0 messages
 Data transfer per iteration: 0.000 MB
 LE solves per iteration: 0 solves
 LE wall-clock time per iteration: 0.000 sec (0.0%)
 LE global solves per iteration: 0 solves
 LE global wall-clock time per iteration: 0.000 sec (0.0%)
 LE global matrix maximum size: 0
 AMG cycles per iteration: 0.000 cycles
 Relaxation sweeps per iteration: 0 sweeps
 Relaxation exchanges per iteration: 0 exchanges
 LE early protections (stall) per iteration: 0.000 times
 LE early protections (divergence) per iteration: 0.000 times
 Total SVARS touched: 388

Total wall-clock time: 0.000 sec

> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 50000, escaped = 40477, trapped = 9519, incomplete = 4

report/dpm-summary

Fate Index	Zone Injection, Index	Zone Number Id	Elapsed Time (s)				Injection, Min
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							
Incomplete	injection-10micron	4	1.366e+01	1.438e+01	1.403e+01	3.318e-01	
	1367 injection-10micron	49362					
Trapped	walls-g1	7	931	4.581e-02	1.779e+00	8.571e-02	7.568e-02
	injection-10mic	16581	injection-10mic	48735			

Trapped	walls-g2	8	1226	6.299e-02	2.921e-01	7.537e-02	1.660e-02
injection-10mic 31558	injection-10mic 39970						
Trapped	walls-g3	9	1831	7.181e-02	1.357e+00	8.577e-02	4.740e-02
injection-10mic 28816	injection-10mic 48340						
Trapped	walls-g4	10	4360	7.714e-02	1.068e+01	1.074e-01	2.409e-01
injection-10mic 38699	injection-10mic 2216						
Trapped	walls-g5	11	908	8.510e-02	9.137e-01	1.149e-01	5.588e-02
injection-10mic 19406	injection-10mic 46768						
Trapped	walls-g6	12	263	1.067e-01	3.476e+00	2.177e-01	2.722e-01
injection-10mic 29476	injection-10mic 32913						
Escaped	outlet-g6-1	14	1025	1.449e-01	2.695e+00	3.060e-01	2.454e-01
injection-10mic 5604	injection-10mic 5162						
Escaped	outlet-g6-2	15	571	1.391e-01	7.251e+00	3.286e-01	3.220e-01
injection-10mic 9678	injection-10mic 4136						
Escaped	outlet-g6-3	16	479	1.419e-01	1.130e+00	1.972e-01	5.632e-02
injection-10mic 18035	injection-10mic 1032						
Escaped	outlet-g6-4	17	574	1.419e-01	2.800e+00	2.210e-01	1.997e-01
injection-10mic 14989	injection-10mic 6886						
Escaped	outlet-g6-5	18	923	1.369e-01	3.071e+00	1.911e-01	1.310e-01
injection-10mic 5510	injection-10mic 25339						
Escaped	outlet-g6-6	19	399	1.306e-01	1.616e+00	1.647e-01	8.822e-02
injection-10mic 10816	injection-10mic 3430						
Escaped	outlet-g6-7	20	421	1.207e-01	4.029e+00	1.566e-01	1.990e-01
injection-10mic 7411	injection-10mic 3460						
Escaped	outlet-g6-8	21	717	1.254e-01	1.701e+00	1.683e-01	8.811e-02
injection-10mic 8520	injection-10mic 2013						
Escaped	outlet-g6-9	22	520	1.209e-01	2.252e-01	1.450e-01	2.447e-02
injection-10mic 4828	injection-10mic 29413						
Escaped	outlet-g6-10	23	548	1.147e-01	3.276e-01	1.374e-01	3.109e-02
injection-10mic 24429	injection-10mic 18155						
Escaped	outlet-g6-11	24	628	1.235e-01	5.537e-01	1.363e-01	2.815e-02
injection-10mic 33685	injection-10mic 21108						
Escaped	outlet-g6-12	25	393	1.244e-01	3.664e-01	1.505e-01	1.995e-02
injection-10mic 10251	injection-10mic 8213						
Escaped	outlet-g6-13	26	397	1.250e-01	4.519e-01	1.514e-01	2.868e-02
injection-10mic 33866	injection-10mic 35507						
Escaped	outlet-g6-14	27	653	1.220e-01	6.377e-01	1.386e-01	3.108e-02
injection-10mic 41109	injection-10mic 37650						
Escaped	outlet-g6-15	28	454	1.152e-01	3.376e-01	1.396e-01	3.821e-02
injection-10mic 38296	injection-10mic 29491						

Escaped	outlet-g6-16	29	532	1.216e-01	2.263e-01	1.446e-01	2.292e-02
injection-10mic	26241	injection-10mic	34921				
Escaped	outlet-g6-17	30	728	1.219e-01	2.861e+00	1.698e-01	1.235e-01
injection-10mic	39909	injection-10mic	48100				
Escaped	outlet-g6-18	31	429	1.235e-01	5.531e+00	1.582e-01	2.626e-01
injection-10mic	39742	injection-10mic	46677				
Escaped	outlet-g6-19	32	390	1.329e-01	3.368e+00	1.755e-01	1.891e-01
injection-10mic	27380	injection-10mic	44658				
Escaped	outlet-g6-20	33	942	1.361e-01	8.597e-01	1.970e-01	9.113e-02
injection-10mic	17430	injection-10mic	49032				
Escaped	outlet-g6-21	34	562	1.448e-01	3.841e+01	2.795e-01	
1.616e+00	injection-10mic	46660	injection-10mic	36862			
Escaped	outlet-g6-22	35	488	1.393e-01	8.928e+00	2.266e-01	4.141e-01
injection-10mic	37342	injection-10mic	46889				
Escaped	outlet-g6-23	36	557	1.460e-01	1.160e+01	3.539e-01	6.670e-01
injection-10mic	10887	injection-10mic	43978				
Escaped	outlet-g6-24	37	1006	1.444e-01	2.967e+00	3.081e-01	
2.582e-01	injection-10mic	45549	injection-10mic	40298			
Escaped	outlet-g5-1	38	1349	1.044e-01	1.758e+01	1.706e-01	5.674e-01
injection-10mic	42811	injection-10mic	16238				
Escaped	outlet-g5-2	39	1139	1.106e-01	7.645e+00	2.121e-01	3.535e-01
injection-10mic	35467	injection-10mic	30793				
Escaped	outlet-g5-3	40	1051	1.118e-01	5.086e+00	2.140e-01	3.055e-01
injection-10mic	22538	injection-10mic	34411				
Escaped	outlet-g5-4	41	1361	1.075e-01	1.278e+01	1.889e-01	6.324e-01
injection-10mic	37645	injection-10mic	31922				
Escaped	outlet-g4-1	42	1746	8.999e-02	4.338e+00	1.297e-01	1.406e-01
injection-10mic	11308	injection-10mic	1738				
Escaped	outlet-g4-2	43	3278	8.488e-02	2.561e+00	1.026e-01	9.932e-02
injection-10mic	12702	injection-10mic	2996				
Escaped	outlet-g4-3	44	2396	8.401e-02	3.936e-01	9.493e-02	1.365e-02
injection-10mic	14693	injection-10mic	14273				
Escaped	outlet-g4-4	45	3176	8.363e-02	6.096e+00	1.013e-01	1.592e-01
injection-10mic	13898	injection-10mic	11998				
Escaped	outlet-g4-5	46	3220	8.293e-02	2.857e+00	9.820e-02	6.908e-02
injection-10mic	29623	injection-10mic	16792				
Escaped	outlet-g4-6	47	2459	8.434e-02	1.878e-01	9.506e-02	1.173e-02
injection-10mic	35916	injection-10mic	43878				
Escaped	outlet-g4-7	48	3259	8.461e-02	2.299e+00	9.990e-02	6.215e-02
injection-10mic	37706	injection-10mic	38337				

Escaped outlet-g4-8 49 1707 9.099e-02 9.403e+00 1.490e-01 3.249e-01
injection-10mic 24055 injection-10mic 44723

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Incomplete			8.000e-13	8.000e-13	0.000e+00
Trapped	walls-g1	7	1.862e-10	1.862e-10	0.000e+00
Trapped	walls-g2	8	2.452e-10	2.452e-10	0.000e+00
Trapped	walls-g3	9	3.662e-10	3.662e-10	0.000e+00
Trapped	walls-g4	10	8.720e-10	8.720e-10	0.000e+00
Trapped	walls-g5	11	1.816e-10	1.816e-10	0.000e+00
Trapped	walls-g6	12	5.260e-11	5.260e-11	0.000e+00
Escaped	outlet-g6-1	14	2.050e-10	2.050e-10	0.000e+00
Escaped	outlet-g6-2	15	1.142e-10	1.142e-10	0.000e+00
Escaped	outlet-g6-3	16	9.580e-11	9.580e-11	0.000e+00
Escaped	outlet-g6-4	17	1.148e-10	1.148e-10	0.000e+00
Escaped	outlet-g6-5	18	1.846e-10	1.846e-10	0.000e+00
Escaped	outlet-g6-6	19	7.980e-11	7.980e-11	0.000e+00
Escaped	outlet-g6-7	20	8.420e-11	8.420e-11	0.000e+00
Escaped	outlet-g6-8	21	1.434e-10	1.434e-10	0.000e+00
Escaped	outlet-g6-9	22	1.040e-10	1.040e-10	0.000e+00
Escaped	outlet-g6-10	23	1.096e-10	1.096e-10	0.000e+00
Escaped	outlet-g6-11	24	1.256e-10	1.256e-10	0.000e+00
Escaped	outlet-g6-12	25	7.860e-11	7.860e-11	0.000e+00
Escaped	outlet-g6-13	26	7.940e-11	7.940e-11	0.000e+00
Escaped	outlet-g6-14	27	1.306e-10	1.306e-10	0.000e+00
Escaped	outlet-g6-15	28	9.080e-11	9.080e-11	0.000e+00
Escaped	outlet-g6-16	29	1.064e-10	1.064e-10	0.000e+00
Escaped	outlet-g6-17	30	1.456e-10	1.456e-10	0.000e+00
Escaped	outlet-g6-18	31	8.580e-11	8.580e-11	0.000e+00
Escaped	outlet-g6-19	32	7.800e-11	7.800e-11	0.000e+00
Escaped	outlet-g6-20	33	1.884e-10	1.884e-10	0.000e+00
Escaped	outlet-g6-21	34	1.124e-10	1.124e-10	0.000e+00
Escaped	outlet-g6-22	35	9.760e-11	9.760e-11	0.000e+00
Escaped	outlet-g6-23	36	1.114e-10	1.114e-10	0.000e+00
Escaped	outlet-g6-24	37	2.012e-10	2.012e-10	0.000e+00
Escaped	outlet-g5-1	38	2.698e-10	2.698e-10	0.000e+00

Escaped	outlet-g5-2	39	2.278e-10	2.278e-10	0.000e+00
Escaped	outlet-g5-3	40	2.102e-10	2.102e-10	0.000e+00
Escaped	outlet-g5-4	41	2.722e-10	2.722e-10	0.000e+00
Escaped	outlet-g4-1	42	3.492e-10	3.492e-10	0.000e+00
Escaped	outlet-g4-2	43	6.556e-10	6.556e-10	0.000e+00
Escaped	outlet-g4-3	44	4.792e-10	4.792e-10	0.000e+00
Escaped	outlet-g4-4	45	6.352e-10	6.352e-10	0.000e+00
Escaped	outlet-g4-5	46	6.440e-10	6.440e-10	0.000e+00
Escaped	outlet-g4-6	47	4.918e-10	4.918e-10	0.000e+00
Escaped	outlet-g4-7	48	6.518e-10	6.518e-10	0.000e+00
Escaped	outlet-g4-8	49	3.414e-10	3.414e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time:	0.000 sec
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```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model
Specific)*PushButton1(Discrete Phase)")
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 100000, escaped = 81181, trapped = 18811, incomplete = 8

report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Incomplete		8	1.362e+01	9.165e+01	2.361e+01	2.572e+01	
injection-10micron	10023	injection-10micron	90109				
Trapped	walls-g1	7	1888	4.544e-02	2.168e-01	8.109e-02	4.855e-02
injection-10mic	19813	injection-10mic	67754				
Trapped	walls-g2	8	2389	6.261e-02	1.601e+00	7.519e-02	3.628e-02
injection-10mic	34014	injection-10mic	8688				
Trapped	walls-g3	9	3594	7.206e-02	1.259e+00	8.505e-02	4.684e-02
injection-10mic	74130	injection-10mic	86317				
Trapped	walls-g4	10	8581	7.654e-02	9.332e+00	1.073e-01	2.012e-01
injection-10mic	72400	injection-10mic	40337				
Trapped	walls-g5	11	1794	8.557e-02	8.681e+00	1.265e-01	2.197e-01
injection-10mic	69216	injection-10mic	38140				
Trapped	walls-g6	12	565	1.026e-01	2.152e+00	2.159e-01	1.406e-01
injection-10mic	70586	injection-10mic	2917				
Escaped	outlet-g6-1	14	2051	1.460e-01	5.201e+00	3.104e-01	2.958e-01
injection-10mic	22908	injection-10mic	5775				
Escaped	outlet-g6-2	15	1037	1.402e-01	6.373e+00	3.225e-01	2.892e-01
injection-10mic	23832	injection-10mic	3067				
Escaped	outlet-g6-3	16	987	1.387e-01	2.803e+00	2.140e-01	1.725e-01
injection-10mic	23688	injection-10mic	26806				
Escaped	outlet-g6-4	17	1167	1.369e-01	6.811e+00	2.087e-01	2.343e-01
injection-10mic	30365	injection-10mic	4033				
Escaped	outlet-g6-5	18	1948	1.297e-01	1.679e+00	1.970e-01	1.029e-01
injection-10mic	66683	injection-10mic	1900				

Escaped	outlet-g6-6	19	789	1.306e-01	5.230e+00	1.710e-01	2.310e-01
injection-10mic 20413	injection-10mic 4393						
Escaped	outlet-g6-7	20	890	1.223e-01	1.931e+00	1.493e-01	7.723e-02
injection-10mic 31759	injection-10mic 26892						
Escaped	outlet-g6-8	21	1562	1.239e-01	1.886e+00	1.682e-01	8.395e-02
injection-10mic 23132	injection-10mic 31137						
Escaped	outlet-g6-9	22	1020	1.214e-01	2.258e-01	1.434e-01	2.382e-02
injection-10mic 44305	injection-10mic 21888						
Escaped	outlet-g6-10	23	1011	1.141e-01	3.322e-01	1.372e-01	3.412e-02
injection-10mic 30252	injection-10mic 35977						
Escaped	outlet-g6-11	24	1402	1.203e-01	4.672e-01	1.367e-01	2.289e-02
injection-10mic 43967	injection-10mic 40452						
Escaped	outlet-g6-12	25	742	1.258e-01	4.226e-01	1.506e-01	2.016e-02
injection-10mic 55269	injection-10mic 21455						
Escaped	outlet-g6-13	26	729	1.255e-01	5.628e-01	1.511e-01	2.569e-02
injection-10mic 67540	injection-10mic 84351						
Escaped	outlet-g6-14	27	1346	1.218e-01	5.380e-01	1.378e-01	2.649e-02
injection-10mic 47272	injection-10mic 86447						
Escaped	outlet-g6-15	28	964	1.144e-01	3.612e-01	1.378e-01	3.341e-02
injection-10mic 56288	injection-10mic 76882						
Escaped	outlet-g6-16	29	1026	1.210e-01	2.300e-01	1.435e-01	2.340e-02
injection-10mic 83184	injection-10mic 64563						
Escaped	outlet-g6-17	30	1493	1.229e-01	2.396e+00	1.753e-01	
1.400e-01	injection-10mic 85977						injection-10mic 97776
Escaped	outlet-g6-18	31	804	1.224e-01	1.155e+00	1.496e-01	6.071e-02
injection-10mic 86146	injection-10mic 88553						
Escaped	outlet-g6-19	32	735	1.296e-01	2.442e+00	1.725e-01	1.388e-01
injection-10mic 85436	injection-10mic 76796						
Escaped	outlet-g6-20	33	2000	1.312e-01	2.867e+00	2.038e-01	
1.450e-01	injection-10mic 76984						injection-10mic 96310
Escaped	outlet-g6-21	34	1096	1.390e-01	2.771e+00	2.097e-01	
1.800e-01	injection-10mic 69596						injection-10mic 89758
Escaped	outlet-g6-22	35	1082	1.352e-01	3.245e+00	2.138e-01	
1.739e-01	injection-10mic 61369						injection-10mic 48387
Escaped	outlet-g6-23	36	1052	1.420e-01	2.511e+00	3.127e-01	
1.400e-01	injection-10mic 47966						injection-10mic 95297
Escaped	outlet-g6-24	37	2089	1.455e-01	6.774e+00	3.156e-01	
3.533e-01	injection-10mic 89661						injection-10mic 92679
Escaped	outlet-g5-1	38	2796	1.049e-01	1.538e+01	1.610e-01	4.444e-01
injection-10mic 10566	injection-10mic 35985						

Escaped	outlet-g5-2	39	2233	1.089e-01	1.234e+01	2.358e-01	5.741e-01
injection-10mic 20977 injection-10mic 42730							
Escaped	outlet-g5-3	40	2178	1.061e-01	9.829e+00	2.156e-01	3.747e-01
injection-10mic 23059 injection-10mic 42444							
Escaped	outlet-g5-4	41	2790	1.039e-01	1.071e+01	1.485e-01	3.017e-01
injection-10mic 61376 injection-10mic 73816							
Escaped	outlet-g4-1	42	3402	8.980e-02	8.413e+00	1.431e-01	2.626e-01
injection-10mic 25573 injection-10mic 1236							
Escaped	outlet-g4-2	43	6418	8.461e-02	3.078e+00	9.974e-02	6.769e-02
injection-10mic 25209 injection-10mic 11259							
Escaped	outlet-g4-3	44	4691	8.358e-02	3.602e-01	9.483e-02	1.267e-02
injection-10mic 50347 injection-10mic 20873							
Escaped	outlet-g4-4	45	6436	8.232e-02	5.722e+00	1.028e-01	1.539e-01
injection-10mic 45826 injection-10mic 63599							
Escaped	outlet-g4-5	46	6404	8.313e-02	1.090e+01	1.043e-01	2.346e-01
injection-10mic 76937 injection-10mic 58918							
Escaped	outlet-g4-6	47	4818	8.234e-02	2.676e-01	9.526e-02	1.323e-02
injection-10mic 70482 injection-10mic 72050							
Escaped	outlet-g4-7	48	6503	8.496e-02	2.767e+00	9.968e-02	5.963e-02
injection-10mic 52471 injection-10mic 97445							
Escaped	outlet-g4-8	49	3490	9.011e-02	5.317e+00	1.378e-01	1.970e-01
injection-10mic 86562 injection-10mic 91798							

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Incomplete			8.000e-13	8.000e-13	0.000e+00
Trapped	walls-g1	7	1.888e-10	1.888e-10	0.000e+00
Trapped	walls-g2	8	2.389e-10	2.389e-10	0.000e+00
Trapped	walls-g3	9	3.594e-10	3.594e-10	0.000e+00
Trapped	walls-g4	10	8.581e-10	8.581e-10	0.000e+00
Trapped	walls-g5	11	1.794e-10	1.794e-10	0.000e+00
Trapped	walls-g6	12	5.650e-11	5.650e-11	0.000e+00
Escaped	outlet-g6-1	14	2.051e-10	2.051e-10	0.000e+00
Escaped	outlet-g6-2	15	1.037e-10	1.037e-10	0.000e+00
Escaped	outlet-g6-3	16	9.870e-11	9.870e-11	0.000e+00
Escaped	outlet-g6-4	17	1.167e-10	1.167e-10	0.000e+00
Escaped	outlet-g6-5	18	1.948e-10	1.948e-10	0.000e+00

Escaped	outlet-g6-6	19	7.890e-11	7.890e-11	0.000e+00
Escaped	outlet-g6-7	20	8.900e-11	8.900e-11	0.000e+00
Escaped	outlet-g6-8	21	1.562e-10	1.562e-10	0.000e+00
Escaped	outlet-g6-9	22	1.020e-10	1.020e-10	0.000e+00
Escaped	outlet-g6-10	23	1.011e-10	1.011e-10	0.000e+00
Escaped	outlet-g6-11	24	1.402e-10	1.402e-10	0.000e+00
Escaped	outlet-g6-12	25	7.420e-11	7.420e-11	0.000e+00
Escaped	outlet-g6-13	26	7.290e-11	7.290e-11	0.000e+00
Escaped	outlet-g6-14	27	1.346e-10	1.346e-10	0.000e+00
Escaped	outlet-g6-15	28	9.640e-11	9.640e-11	0.000e+00
Escaped	outlet-g6-16	29	1.026e-10	1.026e-10	0.000e+00
Escaped	outlet-g6-17	30	1.493e-10	1.493e-10	0.000e+00
Escaped	outlet-g6-18	31	8.040e-11	8.040e-11	0.000e+00
Escaped	outlet-g6-19	32	7.350e-11	7.350e-11	0.000e+00
Escaped	outlet-g6-20	33	2.000e-10	2.000e-10	0.000e+00
Escaped	outlet-g6-21	34	1.096e-10	1.096e-10	0.000e+00
Escaped	outlet-g6-22	35	1.082e-10	1.082e-10	0.000e+00
Escaped	outlet-g6-23	36	1.052e-10	1.052e-10	0.000e+00
Escaped	outlet-g6-24	37	2.089e-10	2.089e-10	0.000e+00
Escaped	outlet-g5-1	38	2.796e-10	2.796e-10	0.000e+00
Escaped	outlet-g5-2	39	2.233e-10	2.233e-10	0.000e+00
Escaped	outlet-g5-3	40	2.178e-10	2.178e-10	0.000e+00
Escaped	outlet-g5-4	41	2.790e-10	2.790e-10	0.000e+00
Escaped	outlet-g4-1	42	3.402e-10	3.402e-10	0.000e+00
Escaped	outlet-g4-2	43	6.418e-10	6.418e-10	0.000e+00
Escaped	outlet-g4-3	44	4.691e-10	4.691e-10	0.000e+00
Escaped	outlet-g4-4	45	6.436e-10	6.436e-10	0.000e+00
Escaped	outlet-g4-5	46	6.404e-10	6.404e-10	0.000e+00
Escaped	outlet-g4-6	47	4.818e-10	4.818e-10	0.000e+00
Escaped	outlet-g4-7	48	6.503e-10	6.503e-10	0.000e+00
Escaped	outlet-g4-8	49	3.490e-10	3.490e-10	0.000e+00

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Net	1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)

Message count per iteration: 0 messages
 Data transfer per iteration: 0.000 MB
 LE solves per iteration: 0 solves
 LE wall-clock time per iteration: 0.000 sec (0.0%)
 LE global solves per iteration: 0 solves
 LE global wall-clock time per iteration: 0.000 sec (0.0%)
 LE global matrix maximum size: 0
 AMG cycles per iteration: 0.000 cycles
 Relaxation sweeps per iteration: 0 sweeps
 Relaxation exchanges per iteration: 0 exchanges
 LE early protections (stall) per iteration: 0.000 times
 LE early protections (divergence) per iteration: 0.000 times
 Total SVARS touched: 388

 Total wall-clock time: 0.000 sec

—

>

adjoint/ mesh/ report/
 close-fluent parallel/ solve/
 define/ plot/ surface/
 display/ preferences/ views/
 file/ print-license-usage

> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 150000, escaped = 121748, trapped = 28242, incomplete = 10

report/dpm-summary

Fate Index	Zone Injection, Index	Zone Number Id	Elapsed Time (s)				Injection, Min
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Incomplete		10	1.369e+01	9.409e+01	2.204e+01	2.402e+01	
injection-10micron 3173	injection-10micron 141841						
Trapped	walls-g1	7	2698	4.475e-02	2.265e-01	8.303e-02	5.057e-02
injection-10mic 105454	injection-10mic 90292						
Trapped	walls-g2	8	3732	6.323e-02	3.642e-01	7.481e-02	1.642e-02
injection-10mic 95082	injection-10mic 146854						
Trapped	walls-g3	9	5331	7.179e-02	1.086e+01	8.702e-02	1.565e-01
injection-10mic 105684	injection-10mic 144793						
Trapped	walls-g4	10	12948	7.656e-02	8.571e+00	1.057e-01	1.732e-01
injection-10mic 57190	injection-10mic 15011						
Trapped	walls-g5	11	2724	8.642e-02	4.664e+00	1.257e-01	1.390e-01
injection-10mic 105240	injection-10mic 85364						
Trapped	walls-g6	12	809	1.044e-01	1.856e+00	2.072e-01	1.140e-01
injection-10mic 125290	injection-10mic 36332						
Escaped	outlet-g6-1	14	2992	1.466e-01	1.245e+01	3.268e-01	4.303e-01
injection-10mic 11801	injection-10mic 2216						
Escaped	outlet-g6-2	15	1622	1.371e-01	7.646e+00	3.161e-01	2.206e-01
injection-10mic 69325	injection-10mic 5161						
Escaped	outlet-g6-3	16	1482	1.362e-01	2.592e+00	2.084e-01	1.251e-01
injection-10mic 42353	injection-10mic 10106						
Escaped	outlet-g6-4	17	1745	1.409e-01	7.083e+00	2.124e-01	2.239e-01
injection-10mic 40440	injection-10mic 79529						
Escaped	outlet-g6-5	18	2909	1.282e-01	3.465e+00	1.931e-01	1.100e-01
injection-10mic 43225	injection-10mic 2105						
Escaped	outlet-g6-6	19	1191	1.303e-01	3.462e+00	1.670e-01	1.265e-01
injection-10mic 30660	injection-10mic 2955						
Escaped	outlet-g6-7	20	1272	1.216e-01	1.553e+00	1.488e-01	6.408e-02
injection-10mic 56040	injection-10mic 19728						
Escaped	outlet-g6-8	21	2248	1.234e-01	1.390e+00	1.652e-01	6.605e-02
injection-10mic 36474	injection-10mic 15563						
Escaped	outlet-g6-9	22	1580	1.195e-01	2.249e-01	1.437e-01	2.344e-02
injection-10mic 66877	injection-10mic 39498						
Escaped	outlet-g6-10	23	1480	1.132e-01	3.430e-01	1.386e-01	3.405e-02
injection-10mic 16747	injection-10mic 72702						
Escaped	outlet-g6-11	24	2138	1.216e-01	6.239e-01	1.372e-01	2.895e-02
injection-10mic 38552	injection-10mic 110898						
Escaped	outlet-g6-12	25	1145	1.220e-01	4.113e-01	1.501e-01	1.928e-02
injection-10mic 33649	injection-10mic 24521						
Escaped	outlet-g6-13	26	1099	1.264e-01	4.258e-01	1.495e-01	1.717e-02
injection-10mic 124310	injection-10mic 64952						

Escaped	outlet-g6-14	27	2006	1.213e-01	5.753e-01	1.370e-01	2.467e-02
injection-10mic	60099	injection-10mic	112955				
Escaped	outlet-g6-15	28	1425	1.129e-01	3.524e-01	1.371e-01	3.211e-02
injection-10mic	104059	injection-10mic	102155				
Escaped	outlet-g6-16	29	1484	1.204e-01	2.278e-01	1.453e-01	2.452e-02
injection-10mic	95760	injection-10mic	82285				
Escaped	outlet-g6-17	30	2236	1.236e-01	1.951e+00	1.691e-01	
8.457e-02	injection-10mic	129933	injection-10mic	130588			
Escaped	outlet-g6-18	31	1229	1.209e-01	3.212e+00	1.513e-01	
1.149e-01	injection-10mic	95345	injection-10mic	145662			
Escaped	outlet-g6-19	32	1223	1.296e-01	1.340e+00	1.635e-01	
7.334e-02	injection-10mic	110288	injection-10mic	144941			
Escaped	outlet-g6-20	33	2959	1.286e-01	5.456e+00	2.009e-01	
1.606e-01	injection-10mic	113348	injection-10mic	145090			
Escaped	outlet-g6-21	34	1571	1.373e-01	5.939e+00	2.107e-01	
2.228e-01	injection-10mic	104551	injection-10mic	140616			
Escaped	outlet-g6-22	35	1549	1.361e-01	2.941e+00	2.060e-01	
1.205e-01	injection-10mic	66113	injection-10mic	135495			
Escaped	outlet-g6-23	36	1642	1.409e-01	6.682e+00	3.166e-01	
2.404e-01	injection-10mic	98180	injection-10mic	122704			
Escaped	outlet-g6-24	37	3202	1.421e-01	3.633e+01	3.360e-01	
7.417e-01	injection-10mic	113950	injection-10mic	132543			
Escaped	outlet-g5-1	38	4187	1.055e-01	9.626e+00	1.550e-01	3.361e-01
injection-10mic	24720	injection-10mic	16463				
Escaped	outlet-g5-2	39	3367	1.096e-01	1.123e+01	2.206e-01	3.874e-01
injection-10mic	76358	injection-10mic	26206				
Escaped	outlet-g5-3	40	3160	1.079e-01	1.901e+01	2.237e-01	5.690e-01
injection-10mic	101430	injection-10mic	126182				
Escaped	outlet-g5-4	41	4186	1.061e-01	9.093e+00	1.615e-01	3.992e-01
injection-10mic	95505	injection-10mic	135762				
Escaped	outlet-g4-1	42	5200	8.923e-02	8.334e+00	1.399e-01	2.298e-01
injection-10mic	73715	injection-10mic	22881				
Escaped	outlet-g4-2	43	9746	8.498e-02	3.591e+00	1.006e-01	7.961e-02
injection-10mic	70228	injection-10mic	3898				
Escaped	outlet-g4-3	44	7271	8.345e-02	1.979e-01	9.493e-02	1.222e-02
injection-10mic	72405	injection-10mic	60471				
Escaped	outlet-g4-4	45	9435	8.320e-02	1.126e+01	1.017e-01	1.731e-01
injection-10mic	62366	injection-10mic	59738				
Escaped	outlet-g4-5	46	9521	8.310e-02	6.676e+00	9.985e-02	1.208e-01
injection-10mic	121946	injection-10mic	87889				

Escaped	outlet-g4-6	47	7256	8.396e-02	2.027e-01	9.497e-02	1.236e-02
injection-10mic	101764	injection-10mic	80622				
Escaped	outlet-g4-7	48	9747	8.463e-02	2.984e+00	9.981e-02	6.277e-02
injection-10mic	99656	injection-10mic	105982				
Escaped	outlet-g4-8	49	5243	8.936e-02	5.869e+00	1.385e-01	1.934e-01
injection-10mic	109241	injection-10mic	147521				

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow Initial	Mass Flow Final	Mass Flow Change
Incomplete			6.667e-13	6.667e-13	0.000e+00
Trapped	walls-g1	7	1.799e-10	1.799e-10	0.000e+00
Trapped	walls-g2	8	2.488e-10	2.488e-10	0.000e+00
Trapped	walls-g3	9	3.554e-10	3.554e-10	0.000e+00
Trapped	walls-g4	10	8.632e-10	8.632e-10	0.000e+00
Trapped	walls-g5	11	1.816e-10	1.816e-10	0.000e+00
Trapped	walls-g6	12	5.393e-11	5.393e-11	0.000e+00
Escaped	outlet-g6-1	14	1.995e-10	1.995e-10	0.000e+00
Escaped	outlet-g6-2	15	1.081e-10	1.081e-10	0.000e+00
Escaped	outlet-g6-3	16	9.880e-11	9.880e-11	0.000e+00
Escaped	outlet-g6-4	17	1.163e-10	1.163e-10	0.000e+00
Escaped	outlet-g6-5	18	1.939e-10	1.939e-10	0.000e+00
Escaped	outlet-g6-6	19	7.940e-11	7.940e-11	0.000e+00
Escaped	outlet-g6-7	20	8.480e-11	8.480e-11	0.000e+00
Escaped	outlet-g6-8	21	1.499e-10	1.499e-10	0.000e+00
Escaped	outlet-g6-9	22	1.053e-10	1.053e-10	0.000e+00
Escaped	outlet-g6-10	23	9.867e-11	9.867e-11	0.000e+00
Escaped	outlet-g6-11	24	1.425e-10	1.425e-10	0.000e+00
Escaped	outlet-g6-12	25	7.633e-11	7.633e-11	0.000e+00
Escaped	outlet-g6-13	26	7.327e-11	7.327e-11	0.000e+00
Escaped	outlet-g6-14	27	1.337e-10	1.337e-10	0.000e+00
Escaped	outlet-g6-15	28	9.500e-11	9.500e-11	0.000e+00
Escaped	outlet-g6-16	29	9.893e-11	9.893e-11	0.000e+00
Escaped	outlet-g6-17	30	1.491e-10	1.491e-10	0.000e+00
Escaped	outlet-g6-18	31	8.193e-11	8.193e-11	0.000e+00
Escaped	outlet-g6-19	32	8.153e-11	8.153e-11	0.000e+00
Escaped	outlet-g6-20	33	1.973e-10	1.973e-10	0.000e+00
Escaped	outlet-g6-21	34	1.047e-10	1.047e-10	0.000e+00

Escaped	outlet-g6-22	35	1.033e-10	1.033e-10	0.000e+00
Escaped	outlet-g6-23	36	1.095e-10	1.095e-10	0.000e+00
Escaped	outlet-g6-24	37	2.135e-10	2.135e-10	0.000e+00
Escaped	outlet-g5-1	38	2.791e-10	2.791e-10	0.000e+00
Escaped	outlet-g5-2	39	2.245e-10	2.245e-10	0.000e+00
Escaped	outlet-g5-3	40	2.107e-10	2.107e-10	0.000e+00
Escaped	outlet-g5-4	41	2.791e-10	2.791e-10	0.000e+00
Escaped	outlet-g4-1	42	3.467e-10	3.467e-10	0.000e+00
Escaped	outlet-g4-2	43	6.497e-10	6.497e-10	0.000e+00
Escaped	outlet-g4-3	44	4.847e-10	4.847e-10	0.000e+00
Escaped	outlet-g4-4	45	6.290e-10	6.290e-10	0.000e+00
Escaped	outlet-g4-5	46	6.347e-10	6.347e-10	0.000e+00
Escaped	outlet-g4-6	47	4.837e-10	4.837e-10	0.000e+00
Escaped	outlet-g4-7	48	6.498e-10	6.498e-10	0.000e+00
Escaped	outlet-g4-8	49	3.495e-10	3.495e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time:	0.000 sec
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```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model
Specific)*PushButton1(Discrete Phase)")
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 200000, escaped = 162189, trapped = 37800, incomplete = 11
report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Incomplete		11	1.366e+01	2.271e+01	1.705e+01	4.188e+00	
injection-10micron	3504 injection-10micron	128408					
Trapped	walls-g1	7	3442	4.425e-02	2.175e-01	8.142e-02	4.914e-02
injection-10mic	138146 injection-10mic	65704					
Trapped	walls-g2	8	5049	6.233e-02	2.463e+00	7.563e-02	4.856e-02
injection-10mic	74785 injection-10mic	83981					
Trapped	walls-g3	9	7130	7.092e-02	3.602e+00	8.753e-02	9.132e-02
injection-10mic	50212 injection-10mic	185846					
Trapped	walls-g4	10	17286	7.678e-02	1.682e+01	1.058e-01	2.040e-01
injection-10mic	149031 injection-10mic	97370					
Trapped	walls-g5	11	3715	8.517e-02	1.221e+01	1.247e-01	2.360e-01
injection-10mic	147576 injection-10mic	43962					
Trapped	walls-g6	12	1178	1.038e-01	5.355e+00	2.189e-01	2.473e-01
injection-10mic	179501 injection-10mic	193446					
Escaped	outlet-g6-1	14	3980	1.386e-01	6.670e+00	3.144e-01	2.966e-01
injection-10mic	47560 injection-10mic	11318					
Escaped	outlet-g6-2	15	2077	1.388e-01	2.439e+00	3.116e-01	1.362e-01
injection-10mic	123160 injection-10mic	8637					
Escaped	outlet-g6-3	16	2011	1.374e-01	3.664e+00	2.082e-01	1.431e-01
injection-10mic	50231 injection-10mic	20125					

Escaped	outlet-g6-4	17	2239	1.417e-01	4.029e+00	2.126e-01	1.885e-01
injection-10mic	123174	injection-10mic	7736				
Escaped	outlet-g6-5	18	3851	1.307e-01	2.747e+00	1.982e-01	1.299e-01
injection-10mic	42544	injection-10mic	10393				
Escaped	outlet-g6-6	19	1693	1.282e-01	4.870e+00	1.660e-01	1.302e-01
injection-10mic	58984	injection-10mic	9312				
Escaped	outlet-g6-7	20	1719	1.209e-01	2.351e+00	1.489e-01	8.041e-02
injection-10mic	26031	injection-10mic	6369				
Escaped	outlet-g6-8	21	2973	1.243e-01	6.720e+00	1.692e-01	1.539e-01
injection-10mic	70472	injection-10mic	7000				
Escaped	outlet-g6-9	22	2247	1.214e-01	2.278e-01	1.443e-01	2.416e-02
injection-10mic	85144	injection-10mic	89717				
Escaped	outlet-g6-10	23	1919	1.132e-01	3.631e-01	1.389e-01	3.554e-02
injection-10mic	80474	injection-10mic	52759				
Escaped	outlet-g6-11	24	2703	1.217e-01	8.828e-01	1.363e-01	2.716e-02
injection-10mic	42498	injection-10mic	5943				
Escaped	outlet-g6-12	25	1476	1.230e-01	4.678e-01	1.491e-01	1.628e-02
injection-10mic	31853	injection-10mic	49018				
Escaped	outlet-g6-13	26	1469	1.222e-01	4.895e-01	1.501e-01	2.200e-02
injection-10mic	173523	injection-10mic	171244				
Escaped	outlet-g6-14	27	2726	1.205e-01	3.940e-01	1.369e-01	2.106e-02
injection-10mic	170323	injection-10mic	147555				
Escaped	outlet-g6-15	28	1884	1.142e-01	4.162e-01	1.368e-01	3.239e-02
injection-10mic	122605	injection-10mic	113705				
Escaped	outlet-g6-16	29	1996	1.201e-01	2.289e-01	1.445e-01	2.435e-02
injection-10mic	134022	injection-10mic	153729				
Escaped	outlet-g6-17	30	2922	1.233e-01	3.342e+00	1.682e-01	
1.074e-01	injection-10mic	108550	injection-10mic	197022			
Escaped	outlet-g6-18	31	1768	1.210e-01	5.638e+00	1.607e-01	
2.143e-01	injection-10mic	129075	injection-10mic	194727			
Escaped	outlet-g6-19	32	1556	1.290e-01	7.154e+00	1.671e-01	
1.900e-01	injection-10mic	129865	injection-10mic	195703			
Escaped	outlet-g6-20	33	4057	1.324e-01	6.281e+00	2.020e-01	
1.648e-01	injection-10mic	133925	injection-10mic	196452			
Escaped	outlet-g6-21	34	2081	1.397e-01	2.791e+00	2.083e-01	
1.726e-01	injection-10mic	145702	injection-10mic	197160			
Escaped	outlet-g6-22	35	2140	1.386e-01	4.670e+00	2.116e-01	
1.622e-01	injection-10mic	126280	injection-10mic	179602			
Escaped	outlet-g6-23	36	2092	1.388e-01	2.262e+00	3.126e-01	
1.332e-01	injection-10mic	172658	injection-10mic	185946			

Escaped outlet-g6-24 37 4110 1.363e-01 1.001e+01 3.197e-01
 3.927e-01 injection-10mic 174422 injection-10mic 196579
 Escaped outlet-g5-1 38 5485 1.057e-01 1.701e+01 1.631e-01 4.729e-01
 injection-10mic 64783 injection-10mic 22414
 Escaped outlet-g5-2 39 4466 1.089e-01 1.137e+01 2.160e-01 4.100e-01
 injection-10mic 80538 injection-10mic 142728
 Escaped outlet-g5-3 40 4245 1.084e-01 1.677e+01 2.302e-01 5.456e-01
 injection-10mic 163318 injection-10mic 123667
 Escaped outlet-g5-4 41 5526 1.043e-01 9.990e+00 1.518e-01 3.311e-01
 injection-10mic 127984 injection-10mic 70631
 Escaped outlet-g4-1 42 7034 8.897e-02 7.755e+00 1.405e-01 2.202e-01
 injection-10mic 66602 injection-10mic 2937
 Escaped outlet-g4-2 43 12889 8.481e-02 3.418e+00 9.953e-02
 6.881e-02 injection-10mic 30698 injection-10mic 77858
 Escaped outlet-g4-3 44 9738 8.345e-02 4.852e-01 9.505e-02 1.314e-02
 injection-10mic 73139 injection-10mic 6083
 Escaped outlet-g4-4 45 12655 8.274e-02 6.045e+00 1.009e-01
 1.213e-01 injection-10mic 87496 injection-10mic 67117
 Escaped outlet-g4-5 46 12570 8.324e-02 1.146e+01 1.027e-01
 2.072e-01 injection-10mic 163101 injection-10mic 73360
 Escaped outlet-g4-6 47 9865 8.144e-02 4.218e-01 9.524e-02 1.319e-02
 injection-10mic 173203 injection-10mic 110303
 Escaped outlet-g4-7 48 13111 8.454e-02 7.110e+00 1.005e-01 9.900e-02
 injection-10mic 117966 injection-10mic 194284
 Escaped outlet-g4-8 49 6916 9.017e-02 8.519e+00 1.389e-01 1.999e-01
 injection-10mic 129997 injection-10mic 173477

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Incomplete			5.500e-13	5.500e-13	0.000e+00
Trapped	walls-g1	7	1.721e-10	1.721e-10	0.000e+00
Trapped	walls-g2	8	2.525e-10	2.525e-10	0.000e+00
Trapped	walls-g3	9	3.565e-10	3.565e-10	0.000e+00
Trapped	walls-g4	10	8.643e-10	8.643e-10	0.000e+00
Trapped	walls-g5	11	1.858e-10	1.858e-10	0.000e+00
Trapped	walls-g6	12	5.890e-11	5.890e-11	0.000e+00
Escaped	outlet-g6-1	14	1.990e-10	1.990e-10	0.000e+00

Escaped	outlet-g6-2	15	1.039e-10	1.039e-10	0.000e+00
Escaped	outlet-g6-3	16	1.006e-10	1.006e-10	0.000e+00
Escaped	outlet-g6-4	17	1.120e-10	1.120e-10	0.000e+00
Escaped	outlet-g6-5	18	1.925e-10	1.925e-10	0.000e+00
Escaped	outlet-g6-6	19	8.465e-11	8.465e-11	0.000e+00
Escaped	outlet-g6-7	20	8.595e-11	8.595e-11	0.000e+00
Escaped	outlet-g6-8	21	1.486e-10	1.486e-10	0.000e+00
Escaped	outlet-g6-9	22	1.124e-10	1.124e-10	0.000e+00
Escaped	outlet-g6-10	23	9.595e-11	9.595e-11	0.000e+00
Escaped	outlet-g6-11	24	1.352e-10	1.352e-10	0.000e+00
Escaped	outlet-g6-12	25	7.380e-11	7.380e-11	0.000e+00
Escaped	outlet-g6-13	26	7.345e-11	7.345e-11	0.000e+00
Escaped	outlet-g6-14	27	1.363e-10	1.363e-10	0.000e+00
Escaped	outlet-g6-15	28	9.420e-11	9.420e-11	0.000e+00
Escaped	outlet-g6-16	29	9.980e-11	9.980e-11	0.000e+00
Escaped	outlet-g6-17	30	1.461e-10	1.461e-10	0.000e+00
Escaped	outlet-g6-18	31	8.840e-11	8.840e-11	0.000e+00
Escaped	outlet-g6-19	32	7.780e-11	7.780e-11	0.000e+00
Escaped	outlet-g6-20	33	2.028e-10	2.028e-10	0.000e+00
Escaped	outlet-g6-21	34	1.041e-10	1.041e-10	0.000e+00
Escaped	outlet-g6-22	35	1.070e-10	1.070e-10	0.000e+00
Escaped	outlet-g6-23	36	1.046e-10	1.046e-10	0.000e+00
Escaped	outlet-g6-24	37	2.055e-10	2.055e-10	0.000e+00
Escaped	outlet-g5-1	38	2.742e-10	2.742e-10	0.000e+00
Escaped	outlet-g5-2	39	2.233e-10	2.233e-10	0.000e+00
Escaped	outlet-g5-3	40	2.122e-10	2.122e-10	0.000e+00
Escaped	outlet-g5-4	41	2.763e-10	2.763e-10	0.000e+00
Escaped	outlet-g4-1	42	3.517e-10	3.517e-10	0.000e+00
Escaped	outlet-g4-2	43	6.445e-10	6.445e-10	0.000e+00
Escaped	outlet-g4-3	44	4.869e-10	4.869e-10	0.000e+00
Escaped	outlet-g4-4	45	6.328e-10	6.328e-10	0.000e+00
Escaped	outlet-g4-5	46	6.285e-10	6.285e-10	0.000e+00
Escaped	outlet-g4-6	47	4.933e-10	4.933e-10	0.000e+00
Escaped	outlet-g4-7	48	6.556e-10	6.556e-10	0.000e+00
Escaped	outlet-g4-8	49	3.458e-10	3.458e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration: 0.000 sec
 Global reductions per iteration: 0 ops
 Global reductions time per iteration: 0.000 sec (0.0%)
 Message count per iteration: 0 messages
 Data transfer per iteration: 0.000 MB
 LE solves per iteration: 0 solves
 LE wall-clock time per iteration: 0.000 sec (0.0%)
 LE global solves per iteration: 0 solves
 LE global wall-clock time per iteration: 0.000 sec (0.0%)
 LE global matrix maximum size: 0
 AMG cycles per iteration: 0.000 cycles
 Relaxation sweeps per iteration: 0 sweeps
 Relaxation exchanges per iteration: 0 exchanges
 LE early protections (stall) per iteration: 0.000 times
 LE early protections (divergence) per iteration: 0.000 times
 Total SVARS touched: 388

Total wall-clock time: 0.000 sec

—

> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 500000, escaped = 405546, trapped = 94423, incomplete = 31

report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Incomplete		31	1.364e+01	2.248e+01	1.474e+01	2.537e+00	
injection-10micron	10130	injection-10micron	32589				
Trapped	walls-g1	7	8953	4.473e-02	2.185e-01	8.286e-02	4.990e-02
injection-10mic	95345	injection-10mic	142674				

Trapped	walls-g2	8	12408	6.204e-02	3.440e+00	7.539e-02	3.671e-02
injection-10mic	226124	injection-10mic	100895				
Trapped	walls-g3	9	17734	7.064e-02	6.302e+00	8.738e-02	8.896e-02
injection-10mic	208773	injection-10mic	15227				
Trapped	walls-g4	10	43295	7.619e-02	1.243e+01	1.052e-01	1.811e-01
injection-10mic	270928	injection-10mic	366244				
Trapped	walls-g5	11	9197	8.511e-02	5.239e+00	1.208e-01	1.058e-01
injection-10mic	284331	injection-10mic	209230				
Trapped	walls-g6	12	2836	1.030e-01	1.906e+00	2.080e-01	1.187e-01
injection-10mic	153489	injection-10mic	488929				
Escaped	outlet-g6-1	14	10205	1.411e-01	8.449e+00	3.187e-01	
3.400e-01	injection-10mic	58109	injection-10mic	61598			
Escaped	outlet-g6-2	15	5180	1.392e-01	4.711e+00	3.102e-01	1.489e-01
injection-10mic	72345	injection-10mic	16443				
Escaped	outlet-g6-3	16	4838	1.358e-01	3.837e+00	2.092e-01	1.409e-01
injection-10mic	140714	injection-10mic	57863				
Escaped	outlet-g6-4	17	5681	1.381e-01	2.233e+01	2.122e-01	3.384e-01
injection-10mic	170118	injection-10mic	62301				
Escaped	outlet-g6-5	18	9601	1.252e-01	1.054e+01	1.966e-01	1.602e-01
injection-10mic	185994	injection-10mic	9071				
Escaped	outlet-g6-6	19	4234	1.266e-01	6.501e+00	1.668e-01	1.586e-01
injection-10mic	105984	injection-10mic	11165				
Escaped	outlet-g6-7	20	4275	1.197e-01	5.844e+00	1.500e-01	1.165e-01
injection-10mic	143929	injection-10mic	88469				
Escaped	outlet-g6-8	21	7282	1.229e-01	5.478e+00	1.704e-01	1.279e-01
injection-10mic	231750	injection-10mic	19866				
Escaped	outlet-g6-9	22	5241	1.204e-01	2.273e-01	1.438e-01	2.345e-02
injection-10mic	163215	injection-10mic	346893				
Escaped	outlet-g6-10	23	5050	1.132e-01	3.787e-01	1.369e-01	3.299e-02
injection-10mic	163101	injection-10mic	198003				
Escaped	outlet-g6-11	24	6870	1.203e-01	5.084e-01	1.364e-01	2.123e-02
injection-10mic	281590	injection-10mic	214937				
Escaped	outlet-g6-12	25	3726	1.240e-01	4.404e-01	1.496e-01	1.926e-02
injection-10mic	300133	injection-10mic	220446				
Escaped	outlet-g6-13	26	3747	1.226e-01	5.547e-01	1.500e-01	2.259e-02
injection-10mic	335909	injection-10mic	258118				
Escaped	outlet-g6-14	27	6813	1.192e-01	6.336e-01	1.371e-01	2.563e-02
injection-10mic	347316	injection-10mic	436947				
Escaped	outlet-g6-15	28	4770	1.138e-01	3.541e-01	1.377e-01	3.290e-02
injection-10mic	358557	injection-10mic	411616				

Escaped	outlet-g6-16	29	4978	1.205e-01	2.309e-01	1.451e-01	2.450e-02
injection-10mic 410235 injection-10mic 260650							
Escaped	outlet-g6-17	30	7362	1.198e-01	3.083e+00	1.671e-01	
9.731e-02 injection-10mic 372147 injection-10mic 428646							
Escaped	outlet-g6-18	31	4170	1.216e-01	5.743e+00	1.540e-01	
1.420e-01 injection-10mic 345677 injection-10mic 491889							
Escaped	outlet-g6-19	32	3978	1.277e-01	7.952e+00	1.686e-01	
1.892e-01 injection-10mic 315041 injection-10mic 489962							
Escaped	outlet-g6-20	33	9927	1.215e-01	2.527e+00	1.971e-01	
1.047e-01 injection-10mic 321294 injection-10mic 475248							
Escaped	outlet-g6-21	34	5325	1.376e-01	1.355e+01	2.095e-01	
2.576e-01 injection-10mic 303090 injection-10mic 483443							
Escaped	outlet-g6-22	35	5302	1.362e-01	7.623e+00	2.050e-01	
1.450e-01 injection-10mic 234291 injection-10mic 489178							
Escaped	outlet-g6-23	36	5273	1.323e-01	5.756e+00	3.119e-01	
1.725e-01 injection-10mic 397777 injection-10mic 488815							
Escaped	outlet-g6-24	37	10437	1.425e-01	1.548e+01	3.260e-01	
4.250e-01 injection-10mic 412692 injection-10mic 375502							
Escaped	outlet-g5-1	38	13843	1.053e-01	1.495e+01	1.667e-01	
4.315e-01 injection-10mic 238981 injection-10mic 197919							
Escaped	outlet-g5-2	39	11236	1.076e-01	1.594e+01	2.250e-01	
4.450e-01 injection-10mic 206505 injection-10mic 312039							
Escaped	outlet-g5-3	40	10614	1.060e-01	1.808e+01	2.236e-01	
4.634e-01 injection-10mic 206809 injection-10mic 359588							
Escaped	outlet-g5-4	41	14071	1.048e-01	1.198e+01	1.695e-01	
4.715e-01 injection-10mic 230919 injection-10mic 371146							
Escaped	outlet-g4-1	42	17578	8.889e-02	8.589e+00	1.366e-01	
2.044e-01 injection-10mic 158461 injection-10mic 27221							
Escaped	outlet-g4-2	43	32132	8.395e-02	6.820e+00	9.996e-02	
8.084e-02 injection-10mic 59632 injection-10mic 27067							
Escaped	outlet-g4-3	44	23827	8.253e-02	1.831e+00	9.506e-02	
1.732e-02 injection-10mic 142646 injection-10mic 7474							
Escaped	outlet-g4-4	45	31867	8.242e-02	1.093e+01	1.025e-01	
1.791e-01 injection-10mic 354157 injection-10mic 273565							
Escaped	outlet-g4-5	46	31843	8.276e-02	1.411e+01	1.006e-01	
1.502e-01 injection-10mic 158332 injection-10mic 345843							
Escaped	outlet-g4-6	47	24271	8.308e-02	1.496e+00	9.512e-02	
1.651e-02 injection-10mic 298740 injection-10mic 375464							
Escaped	outlet-g4-7	48	32505	8.488e-02	5.783e+00	1.000e-01	
8.209e-02 injection-10mic 333565 injection-10mic 486102							

Escaped outlet-g4-8 49 17494 8.921e-02 1.037e+01 1.378e-01
 2.079e-01 injection-10mic 427939 injection-10mic 484615

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Initial	Final	Change
Incomplete			6.200e-13	6.200e-13	0.000e+00
Trapped	walls-g1	7	1.791e-10	1.791e-10	0.000e+00
Trapped	walls-g2	8	2.482e-10	2.482e-10	0.000e+00
Trapped	walls-g3	9	3.547e-10	3.547e-10	0.000e+00
Trapped	walls-g4	10	8.659e-10	8.659e-10	0.000e+00
Trapped	walls-g5	11	1.839e-10	1.839e-10	0.000e+00
Trapped	walls-g6	12	5.672e-11	5.672e-11	0.000e+00
Escaped	outlet-g6-1	14	2.041e-10	2.041e-10	0.000e+00
Escaped	outlet-g6-2	15	1.036e-10	1.036e-10	0.000e+00
Escaped	outlet-g6-3	16	9.676e-11	9.676e-11	0.000e+00
Escaped	outlet-g6-4	17	1.136e-10	1.136e-10	0.000e+00
Escaped	outlet-g6-5	18	1.920e-10	1.920e-10	0.000e+00
Escaped	outlet-g6-6	19	8.468e-11	8.468e-11	0.000e+00
Escaped	outlet-g6-7	20	8.550e-11	8.550e-11	0.000e+00
Escaped	outlet-g6-8	21	1.456e-10	1.456e-10	0.000e+00
Escaped	outlet-g6-9	22	1.048e-10	1.048e-10	0.000e+00
Escaped	outlet-g6-10	23	1.010e-10	1.010e-10	0.000e+00
Escaped	outlet-g6-11	24	1.374e-10	1.374e-10	0.000e+00
Escaped	outlet-g6-12	25	7.452e-11	7.452e-11	0.000e+00
Escaped	outlet-g6-13	26	7.494e-11	7.494e-11	0.000e+00
Escaped	outlet-g6-14	27	1.363e-10	1.363e-10	0.000e+00
Escaped	outlet-g6-15	28	9.540e-11	9.540e-11	0.000e+00
Escaped	outlet-g6-16	29	9.956e-11	9.956e-11	0.000e+00
Escaped	outlet-g6-17	30	1.472e-10	1.472e-10	0.000e+00
Escaped	outlet-g6-18	31	8.340e-11	8.340e-11	0.000e+00
Escaped	outlet-g6-19	32	7.956e-11	7.956e-11	0.000e+00
Escaped	outlet-g6-20	33	1.985e-10	1.985e-10	0.000e+00
Escaped	outlet-g6-21	34	1.065e-10	1.065e-10	0.000e+00
Escaped	outlet-g6-22	35	1.060e-10	1.060e-10	0.000e+00
Escaped	outlet-g6-23	36	1.055e-10	1.055e-10	0.000e+00
Escaped	outlet-g6-24	37	2.087e-10	2.087e-10	0.000e+00
Escaped	outlet-g5-1	38	2.769e-10	2.769e-10	0.000e+00

Escaped	outlet-g5-2	39	2.247e-10	2.247e-10	0.000e+00
Escaped	outlet-g5-3	40	2.123e-10	2.123e-10	0.000e+00
Escaped	outlet-g5-4	41	2.814e-10	2.814e-10	0.000e+00
Escaped	outlet-g4-1	42	3.516e-10	3.516e-10	0.000e+00
Escaped	outlet-g4-2	43	6.426e-10	6.426e-10	0.000e+00
Escaped	outlet-g4-3	44	4.765e-10	4.765e-10	0.000e+00
Escaped	outlet-g4-4	45	6.373e-10	6.373e-10	0.000e+00
Escaped	outlet-g4-5	46	6.369e-10	6.369e-10	0.000e+00
Escaped	outlet-g4-6	47	4.854e-10	4.854e-10	0.000e+00
Escaped	outlet-g4-7	48	6.501e-10	6.501e-10	0.000e+00
Escaped	outlet-g4-8	49	3.499e-10	3.499e-10	0.000e+00

Net			1.000e-08	1.000e-08	0.000e+00

> /file/stop-transcript

Performance Timer for 0 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.000 sec
Global reductions per iteration:	0 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	0 messages
Data transfer per iteration:	0.000 MB
LE solves per iteration:	0 solves
LE wall-clock time per iteration:	0.000 sec (0.0%)
LE global solves per iteration:	0 solves
LE global wall-clock time per iteration:	0.000 sec (0.0%)
LE global matrix maximum size:	0
AMG cycles per iteration:	0.000 cycles
Relaxation sweeps per iteration:	0 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.000 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	388

Total wall-clock time:	0.000 sec
------------------------	-----------

7.2.1 Associated Python Code

parcel-and-grid-independence

3/16/26, 6:23 PM

```
In [50]: import re
import pandas as pd

def parse_fluent_dpm_file(filepath):

    with open(filepath) as f:
        lines = f.readlines()

    tracked = escaped = trapped = None
    particle_rows = []
    mass_rows = []

    reading_particle_table = False
    reading_mass_table = False

    for line in lines:

        line = line.strip()

        # Global statistics
        if "number tracked" in line:
            m = re.search(r'number tracked = (\d+), escaped = (\d+), t
            tracked = int(m.group(1))
            escaped = int(m.group(2))
            trapped = int(m.group(3))

        # Detect particle fate table
        if line.startswith("Fate") and "Elapsed Time" in line:
            reading_particle_table = True
            continue

        # Detect mass transfer table
        if "Mass Transfer Summary" in line:
            reading_particle_table = False
            reading_mass_table = True
            continue

        # Particle table rows
        if reading_particle_table and (line.startswith("Trapped") or l

            parts = line.split()

            fate = parts[0]
            zone = parts[1]
            zone_id = int(parts[2])
            number = int(parts[3])

            particle_rows.append({
                "fate": fate,
                "zone": zone,
```

file:///Users/kaanborekci/Desktop/CFD-FinalProject/ANALYSIS/parcel-and-grid-independence.html

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```

        "zone_id": zone_id,
        "count": number
    })

    # Mass transfer rows
    if reading_mass_table and (line.startswith("Trapped") or line.

        parts = line.split()

        fate = parts[0]
        zone = parts[1]
        zone_id = int(parts[2])
        initial = float(parts[3])
        final = float(parts[4])

        mass_rows.append({
            "fate": fate,
            "zone": zone,
            "zone_id": zone_id,
            "initial_mass": initial,
            "final_mass": final
        })

    particle_df = pd.DataFrame(particle_rows)
    mass_df = pd.DataFrame(mass_rows)

    summary = {
        "tracked": tracked,
        "escaped": escaped,
        "trapped": trapped
    }

    return summary, particle_df, mass_df

```

```
In [52]: summary, particle_df, mass_df = parse_fluent_dpm_file("/Users/kaanbore
summary
```

```
Out[52]: {'tracked': 15000, 'escaped': 12155, 'trapped': 2845}
```

```
In [53]: particle_df
```

```
Out[53]:
```

	fate	zone	zone_id	count
0	Trapped	walls-g1	7	302
1	Trapped	walls-g2	8	382
2	Trapped	walls-g3	9	518
3	Trapped	walls-g4	10	1221
4	Trapped	walls-g5	11	295

5	Trapped	walls-g6	12	127
6	Escaped	outlet-g6-1	14	296
7	Escaped	outlet-g6-2	15	249
8	Escaped	outlet-g6-3	16	68
9	Escaped	outlet-g6-4	17	177
10	Escaped	outlet-g6-5	18	291
11	Escaped	outlet-g6-6	19	132
12	Escaped	outlet-g6-7	20	138
13	Escaped	outlet-g6-8	21	233
14	Escaped	outlet-g6-9	22	150
15	Escaped	outlet-g6-10	23	136
16	Escaped	outlet-g6-11	24	210
17	Escaped	outlet-g6-12	25	109
18	Escaped	outlet-g6-13	26	97
19	Escaped	outlet-g6-14	27	192
20	Escaped	outlet-g6-15	28	115
21	Escaped	outlet-g6-16	29	148
22	Escaped	outlet-g6-17	30	223
23	Escaped	outlet-g6-18	31	142
24	Escaped	outlet-g6-19	32	112
25	Escaped	outlet-g6-20	33	287
26	Escaped	outlet-g6-21	34	144
27	Escaped	outlet-g6-22	35	113
28	Escaped	outlet-g6-23	36	249
29	Escaped	outlet-g6-24	37	282
30	Escaped	outlet-g5-1	38	427
31	Escaped	outlet-g5-2	39	335
32	Escaped	outlet-g5-3	40	338
33	Escaped	outlet-g5-4	41	407
34	Escaped	outlet-g4-1	42	529
35	Escaped	outlet-g4-2	43	922

36	Escaped	outlet-g4-3	44	739
37	Escaped	outlet-g4-4	45	950
38	Escaped	outlet-g4-5	46	956
39	Escaped	outlet-g4-6	47	761
40	Escaped	outlet-g4-7	48	978
41	Escaped	outlet-g4-8	49	520

In [54]: mass_df

Out[54]:

	fate	zone	zone_id	initial_mass	final_mass
0	Trapped	walls-g1	7	2.013000e-10	2.013000e-10
1	Trapped	walls-g2	8	2.547000e-10	2.547000e-10
2	Trapped	walls-g3	9	3.453000e-10	3.453000e-10
3	Trapped	walls-g4	10	8.140000e-10	8.140000e-10
4	Trapped	walls-g5	11	1.967000e-10	1.967000e-10
5	Trapped	walls-g6	12	8.467000e-11	8.467000e-11
6	Escaped	outlet-g6-1	14	1.973000e-10	1.973000e-10
7	Escaped	outlet-g6-2	15	1.660000e-10	1.660000e-10
8	Escaped	outlet-g6-3	16	4.533000e-11	4.533000e-11
9	Escaped	outlet-g6-4	17	1.180000e-10	1.180000e-10
10	Escaped	outlet-g6-5	18	1.940000e-10	1.940000e-10
11	Escaped	outlet-g6-6	19	8.800000e-11	8.800000e-11
12	Escaped	outlet-g6-7	20	9.200000e-11	9.200000e-11
13	Escaped	outlet-g6-8	21	1.553000e-10	1.553000e-10
14	Escaped	outlet-g6-9	22	1.000000e-10	1.000000e-10
15	Escaped	outlet-g6-10	23	9.067000e-11	9.067000e-11
16	Escaped	outlet-g6-11	24	1.400000e-10	1.400000e-10
17	Escaped	outlet-g6-12	25	7.267000e-11	7.267000e-11
18	Escaped	outlet-g6-13	26	6.467000e-11	6.467000e-11
19	Escaped	outlet-g6-14	27	1.280000e-10	1.280000e-10
20	Escaped	outlet-g6-15	28	7.667000e-11	7.667000e-11
21	Escaped	outlet-g6-16	29	9.867000e-11	9.867000e-11

22	Escaped	outlet-g6-17	30	1.487000e-10	1.487000e-10
23	Escaped	outlet-g6-18	31	9.467000e-11	9.467000e-11
24	Escaped	outlet-g6-19	32	7.467000e-11	7.467000e-11
25	Escaped	outlet-g6-20	33	1.913000e-10	1.913000e-10
26	Escaped	outlet-g6-21	34	9.600000e-11	9.600000e-11
27	Escaped	outlet-g6-22	35	7.533000e-11	7.533000e-11
28	Escaped	outlet-g6-23	36	1.660000e-10	1.660000e-10
29	Escaped	outlet-g6-24	37	1.880000e-10	1.880000e-10
30	Escaped	outlet-g5-1	38	2.847000e-10	2.847000e-10
31	Escaped	outlet-g5-2	39	2.233000e-10	2.233000e-10
32	Escaped	outlet-g5-3	40	2.253000e-10	2.253000e-10
33	Escaped	outlet-g5-4	41	2.713000e-10	2.713000e-10
34	Escaped	outlet-g4-1	42	3.527000e-10	3.527000e-10
35	Escaped	outlet-g4-2	43	6.147000e-10	6.147000e-10
36	Escaped	outlet-g4-3	44	4.927000e-10	4.927000e-10
37	Escaped	outlet-g4-4	45	6.333000e-10	6.333000e-10
38	Escaped	outlet-g4-5	46	6.373000e-10	6.373000e-10
39	Escaped	outlet-g4-6	47	5.073000e-10	5.073000e-10
40	Escaped	outlet-g4-7	48	6.520000e-10	6.520000e-10
41	Escaped	outlet-g4-8	49	3.467000e-10	3.467000e-10

```
In [55]: import os
import re

def parse_all_runs(folder):
    all_results = []
    for file in os.listdir(folder):
        if not file.endswith(".txt"):
            continue

        filepath = os.path.join(folder, file)

        # Extract parcel number
        parcel_number = int(re.findall(r'\d+', file)[0])
```



```

summary, particle_df, mass_df = parse_fluent_dpm_file(filepath

tracked = summary["tracked"]

# Total deposition
total_fraction = summary["trapped"] / tracked

# Generation deposition
trapped_rows = particle_df[particle_df["fate"] == "Trapped"].c
trapped_rows["generation"] = trapped_rows["zone"].str.extract(

for _, row in trapped_rows.iterrows():

    all_results.append({
        "file": file,
        "parcels": parcel_number,
        "generation": row["generation"],
        "count": row["count"],
        "fraction": row["count"] / tracked,
        "total_fraction": total_fraction
    })

return pd.DataFrame(all_results)

```

```

In [56]: folder = "/Users/kaanborekci/Desktop/CFD-FinalProject/DATA/PARCEL"
df = parse_all_runs(folder)

df

```

Out [56]:

	file	parcels	generation	count	fraction	total_fraction
0	streams-150.txt	150	1	302	0.020133	0.189667
1	streams-150.txt	150	2	382	0.025467	0.189667
2	streams-150.txt	150	3	518	0.034533	0.189667
3	streams-150.txt	150	4	1221	0.081400	0.189667
4	streams-150.txt	150	5	295	0.019667	0.189667
...	...	...	...	...	...	...
67	streams-1000.txt	1000	2	2389	0.023890	0.188110
68	streams-1000.txt	1000	3	3594	0.035940	0.188110
69	streams-1000.txt	1000	4	8581	0.085810	0.188110
70	streams-1000.txt	1000	5	1794	0.017940	0.188110
71	streams-1000.txt	1000	6	565	0.005650	0.188110

72 rows × 6 columns

```
In [57]: import matplotlib.pyplot as plt

total_df = (
    df.groupby("parcels")["total_fraction"]
      .first()
      .reset_index()
      .sort_values("parcels")
)

plt.figure()

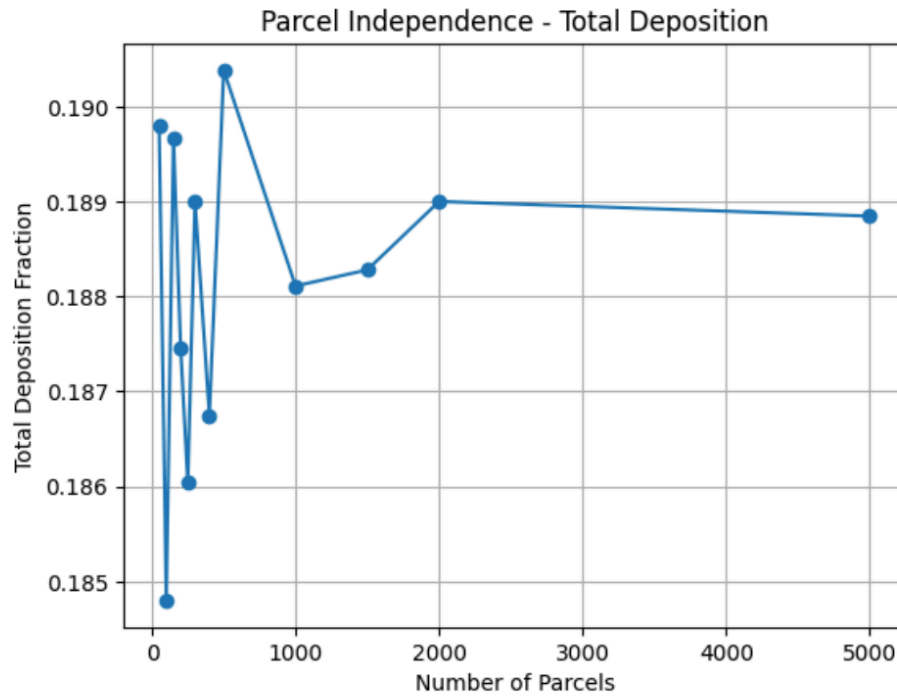
plt.plot(
    total_df["parcels"],
    total_df["total_fraction"],
    marker="o"
)

# plt.xscale("log")

plt.xlabel("Number of Parcels")
plt.ylabel("Total Deposition Fraction")
plt.title("Parcel Independence - Total Deposition")

plt.grid(True)

plt.show()
```



```
In [58]: plt.figure()

for gen in sorted(df["generation"].unique()):

    subset = df[df["generation"] == gen].sort_values("parcels")

    plt.plot(
        subset["parcels"],
        subset["fraction"],
        marker="o",
        label=f"G{gen}"
    )

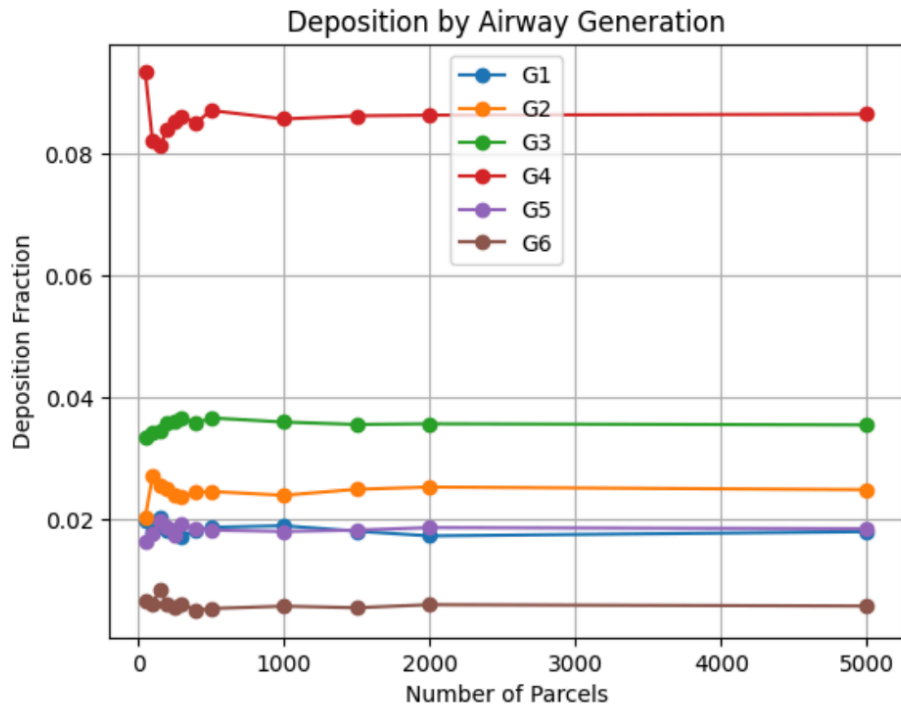
# plt.xscale("log")

plt.xlabel("Number of Parcels")
plt.ylabel("Deposition Fraction")

plt.title("Deposition by Airway Generation")

plt.legend()
plt.grid(True)

plt.show()
```



```
In [59]: pivot = df.pivot_table(
            index="parcels",
            columns="generation",
            values="fraction"
        )

pivot["total"] = df.groupby("parcels")["total_fraction"].first()
pivot.sort_index()
```

7.3 Appendix C: Grid Independence Data

Transcript Start Time: 19:28:28, 15 Mar 2026 Central Daylight Time

Current Directory:

"C:\Users\kaan\Desktop\Lung-Flow-Steady-NoP\lung-2024-final_files\dp0\FFF-2\Fluent"

Build Time: May 13 2024 11:28:30 EST

Executable Path: C:/PROGRA~1/ANSYSI~1/v242/fluvent/ntbin/win64/fluvent.exe

```
-----
ID   Hostname   Core O.S.      PID   Vendor
-----
n3   Borekci-X1  4/14 Windows-x64 21272 Intel(R) Core(TM) Ultra 7 165U
n2   Borekci-X1  3/14 Windows-x64 29520 Intel(R) Core(TM) Ultra 7 165U
n1   Borekci-X1  2/14 Windows-x64 16808 Intel(R) Core(TM) Ultra 7 165U
n0*  Borekci-X1  1/14 Windows-x64 36740 Intel(R) Core(TM) Ultra 7 165U
host Borekci-X1      Windows-x64 15436 Intel(R) Core(TM) Ultra 7 165U
```

MPI Option Selected: intel

Selected system interconnect: default

Cortex Process ID: 36560

License Server Path: 1055@rho.mcc.northwestern.edu

```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model
Specific)*PushButton1(Discrete Phase)")
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")
(cx-gui-do cx-set-list-selections "Sample Trajectories*Table1*Table1*List1(Boundaries)"
'( 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45))
(cx-gui-do cx-activate-item "Sample Trajectories*Table1*Table1*List1(Boundaries)")
(cx-gui-do cx-set-list-selections "Sample Trajectories*Table1*Table3*List1(Release from
Injections)" '( 0))
(cx-gui-do cx-activate-item "Sample Trajectories*Table1*Table3*List1(Release from
Injections)")
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 50000, escaped = 40626, trapped = 9374

(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton2(Close)")

adjoint/ mesh/ report/
close-fluent parallel/ solve/
define/ plot/ surface/
display/ preferences/ views/
file/ print-license-usage

> report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Trapped	walls-g1	7	890	4.562e-02	2.169e-01	8.380e-02	5.080e-02
injection-10mic	19299	injection-10mic	9297				
Trapped	walls-g2	8	1225	6.285e-02	5.700e-01	7.538e-02	2.151e-02
injection-10mic	34855	injection-10mic	2981				
Trapped	walls-g3	9	1779	7.240e-02	1.079e+01	9.215e-02	2.590e-01
injection-10mic	16909	injection-10mic	46047				
Trapped	walls-g4	10	4271	7.624e-02	1.608e+01	1.087e-01	3.323e-01
injection-10mic	35252	injection-10mic	41854				
Trapped	walls-g5	11	963	8.667e-02	1.776e+00	1.203e-01	9.593e-02
injection-10mic	10483	injection-10mic	7727				
Trapped	walls-g6	12	246	1.078e-01	1.022e+00	1.965e-01	8.576e-02
injection-10mic	20592	injection-10mic	36796				
Escaped	outlet-g6-1	14	1048	1.512e-01	3.120e+00	3.272e-01	2.916e-01
injection-10mic	9100	injection-10mic	4679				
Escaped	outlet-g6-2	15	695	1.405e-01	1.690e+00	3.262e-01	1.086e-01
injection-10mic	13210	injection-10mic	1924				
Escaped	outlet-g6-3	16	494	1.388e-01	1.441e+00	2.064e-01	1.172e-01
injection-10mic	4978	injection-10mic	2792				
Escaped	outlet-g6-4	17	476	1.410e-01	4.128e+00	2.228e-01	2.424e-01
injection-10mic	20345	injection-10mic	4125				
Escaped	outlet-g6-5	18	965	1.328e-01	7.244e+00	1.992e-01	2.417e-01
injection-10mic	20283	injection-10mic	926				

Escaped	outlet-g6-6	19	415	1.300e-01	3.669e+00	1.690e-01	1.828e-01
injection-10mic	23013	injection-10mic	866				
Escaped	outlet-g6-7	20	395	1.226e-01	1.572e+00	1.530e-01	1.101e-01
injection-10mic	11629	injection-10mic	2313				
Escaped	outlet-g6-8	21	734	1.242e-01	1.794e+00	1.642e-01	7.278e-02
injection-10mic	14370	injection-10mic	3452				
Escaped	outlet-g6-9	22	510	1.212e-01	2.245e-01	1.458e-01	2.468e-02
injection-10mic	16494	injection-10mic	27128				
Escaped	outlet-g6-10	23	509	1.150e-01	3.357e-01	1.359e-01	3.275e-02
injection-10mic	31637	injection-10mic	10741				
Escaped	outlet-g6-11	24	690	1.227e-01	3.717e-01	1.357e-01	1.894e-02
injection-10mic	18449	injection-10mic	19165				
Escaped	outlet-g6-12	25	362	1.292e-01	4.507e-01	1.515e-01	2.844e-02
injection-10mic	8621	injection-10mic	5382				
Escaped	outlet-g6-13	26	373	1.272e-01	3.067e-01	1.489e-01	1.369e-02
injection-10mic	42059	injection-10mic	22158				
Escaped	outlet-g6-14	27	706	1.226e-01	3.677e-01	1.374e-01	2.167e-02
injection-10mic	45063	injection-10mic	36223				
Escaped	outlet-g6-15	28	496	1.147e-01	2.887e-01	1.366e-01	3.094e-02
injection-10mic	39071	injection-10mic	45042				
Escaped	outlet-g6-16	29	517	1.213e-01	2.269e-01	1.432e-01	2.372e-02
injection-10mic	37384	injection-10mic	31831				
Escaped	outlet-g6-17	30	715	1.244e-01	1.465e+00	1.667e-01	7.801e-02
injection-10mic	40874	injection-10mic	47711				
Escaped	outlet-g6-18	31	395	1.229e-01	1.105e+00	1.478e-01	5.458e-02
injection-10mic	39950	injection-10mic	47945				
Escaped	outlet-g6-19	32	399	1.309e-01	8.873e-01	1.613e-01	4.553e-02
injection-10mic	39895	injection-10mic	45031				
Escaped	outlet-g6-20	33	971	1.349e-01	1.274e+00	2.039e-01	1.001e-01
injection-10mic	37484	injection-10mic	47656				
Escaped	outlet-g6-21	34	456	1.428e-01	3.038e+00	2.159e-01	2.396e-01
injection-10mic	25951	injection-10mic	43187				
Escaped	outlet-g6-22	35	522	1.391e-01	3.255e+00	2.083e-01	1.668e-01
injection-10mic	28083	injection-10mic	49093				
Escaped	outlet-g6-23	36	692	1.452e-01	3.038e+00	3.361e-01	1.631e-01
injection-10mic	32200	injection-10mic	41786				
Escaped	outlet-g6-24	37	1047	1.469e-01	7.539e+00	3.145e-01	
3.201e-01	injection-10mic	31532	injection-10mic	48653			
Escaped	outlet-g5-1	38	1446	1.074e-01	1.038e+01	1.679e-01	4.753e-01
injection-10mic	13118	injection-10mic	6799				

Escaped	outlet-g5-2	39	1093	1.088e-01	5.471e+00	2.346e-01	3.831e-01
injection-10mic	9221	injection-10mic	14163				
Escaped	outlet-g5-3	40	1066	1.083e-01	4.784e+00	2.060e-01	2.320e-01
injection-10mic	21271	injection-10mic	16022				
Escaped	outlet-g5-4	41	1352	1.069e-01	8.678e+00	1.608e-01	4.013e-01
injection-10mic	45166	injection-10mic	31380				
Escaped	outlet-g4-1	42	1757	8.771e-02	2.795e+00	1.412e-01	1.817e-01
injection-10mic	13984	injection-10mic	978				
Escaped	outlet-g4-2	43	3242	8.487e-02	4.168e+00	1.014e-01	1.051e-01
injection-10mic	26219	injection-10mic	2862				
Escaped	outlet-g4-3	44	2386	8.354e-02	1.969e-01	9.489e-02	1.214e-02
injection-10mic	36750	injection-10mic	15534				
Escaped	outlet-g4-4	45	3194	8.353e-02	6.874e+00	9.971e-02	1.289e-01
injection-10mic	11224	injection-10mic	8527				
Escaped	outlet-g4-5	46	3160	8.343e-02	1.076e+01	1.040e-01	2.640e-01
injection-10mic	39204	injection-10mic	30996				
Escaped	outlet-g4-6	47	2404	8.510e-02	1.973e-01	9.536e-02	1.362e-02
injection-10mic	18961	injection-10mic	40398				
Escaped	outlet-g4-7	48	3216	8.578e-02	2.266e+00	1.002e-01	6.639e-02
injection-10mic	24358	injection-10mic	48312				
Escaped	outlet-g4-8	49	1728	9.079e-02	1.686e+00	1.313e-01	1.135e-01
injection-10mic	35207	injection-10mic	47931				

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Initial	Final	Mass Flow (kg/s) Change
Trapped	walls-g1	7	1.780e-10	1.780e-10	0.000e+00
Trapped	walls-g2	8	2.450e-10	2.450e-10	0.000e+00
Trapped	walls-g3	9	3.558e-10	3.558e-10	0.000e+00
Trapped	walls-g4	10	8.542e-10	8.542e-10	0.000e+00
Trapped	walls-g5	11	1.926e-10	1.926e-10	0.000e+00
Trapped	walls-g6	12	4.920e-11	4.920e-11	0.000e+00
Escaped	outlet-g6-1	14	2.096e-10	2.096e-10	0.000e+00
Escaped	outlet-g6-2	15	1.390e-10	1.390e-10	0.000e+00
Escaped	outlet-g6-3	16	9.880e-11	9.880e-11	0.000e+00
Escaped	outlet-g6-4	17	9.520e-11	9.520e-11	0.000e+00
Escaped	outlet-g6-5	18	1.930e-10	1.930e-10	0.000e+00
Escaped	outlet-g6-6	19	8.300e-11	8.300e-11	0.000e+00

Escaped	outlet-g6-7	20	7.900e-11	7.900e-11	0.000e+00
Escaped	outlet-g6-8	21	1.468e-10	1.468e-10	0.000e+00
Escaped	outlet-g6-9	22	1.020e-10	1.020e-10	0.000e+00
Escaped	outlet-g6-10	23	1.018e-10	1.018e-10	0.000e+00
Escaped	outlet-g6-11	24	1.380e-10	1.380e-10	0.000e+00
Escaped	outlet-g6-12	25	7.240e-11	7.240e-11	0.000e+00
Escaped	outlet-g6-13	26	7.460e-11	7.460e-11	0.000e+00
Escaped	outlet-g6-14	27	1.412e-10	1.412e-10	0.000e+00
Escaped	outlet-g6-15	28	9.920e-11	9.920e-11	0.000e+00
Escaped	outlet-g6-16	29	1.034e-10	1.034e-10	0.000e+00
Escaped	outlet-g6-17	30	1.430e-10	1.430e-10	0.000e+00
Escaped	outlet-g6-18	31	7.900e-11	7.900e-11	0.000e+00
Escaped	outlet-g6-19	32	7.980e-11	7.980e-11	0.000e+00
Escaped	outlet-g6-20	33	1.942e-10	1.942e-10	0.000e+00
Escaped	outlet-g6-21	34	9.120e-11	9.120e-11	0.000e+00
Escaped	outlet-g6-22	35	1.044e-10	1.044e-10	0.000e+00
Escaped	outlet-g6-23	36	1.384e-10	1.384e-10	0.000e+00
Escaped	outlet-g6-24	37	2.094e-10	2.094e-10	0.000e+00
Escaped	outlet-g5-1	38	2.892e-10	2.892e-10	0.000e+00
Escaped	outlet-g5-2	39	2.186e-10	2.186e-10	0.000e+00
Escaped	outlet-g5-3	40	2.132e-10	2.132e-10	0.000e+00
Escaped	outlet-g5-4	41	2.704e-10	2.704e-10	0.000e+00
Escaped	outlet-g4-1	42	3.514e-10	3.514e-10	0.000e+00
Escaped	outlet-g4-2	43	6.484e-10	6.484e-10	0.000e+00
Escaped	outlet-g4-3	44	4.772e-10	4.772e-10	0.000e+00
Escaped	outlet-g4-4	45	6.388e-10	6.388e-10	0.000e+00
Escaped	outlet-g4-5	46	6.320e-10	6.320e-10	0.000e+00
Escaped	outlet-g4-6	47	4.808e-10	4.808e-10	0.000e+00
Escaped	outlet-g4-7	48	6.432e-10	6.432e-10	0.000e+00
Escaped	outlet-g4-8	49	3.456e-10	3.456e-10	0.000e+00

-----	-----	-----	-----	-----	-----
Net		1.000e-08	1.000e-08	0.000e+00	

```

> (cx-gui-do cx-activate-item
"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface
Integrals)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table1*DropDownList1(Report Type)"
'( 11))
(cx-gui-do cx-activate-item "Surface Integrals*Table1*DropDownList1(Report Type)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*DropDownList2" '( 4))

```

```
(cx-gui-do cx-activate-item "Surface Integrals*Table2*DropDownList2")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

inlet	4.0496854

```
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '())
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 3 4
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

outlet-g4-1	1.0983247
outlet-g4-2	1.681357
outlet-g4-3	2.179203
outlet-g4-4	2.2644579
outlet-g4-5	2.2734119
outlet-g4-6	2.1798247
outlet-g4-7	1.6759643
outlet-g4-8	1.1074701
outlet-g5-1	1.2984729
outlet-g5-2	1.2179441
outlet-g5-3	1.2238523
outlet-g5-4	1.3078351
outlet-g6-1	0.43068655
outlet-g6-10	2.2465125
outlet-g6-11	2.1735406
outlet-g6-12	2.1587982
outlet-g6-13	2.1625391
outlet-g6-14	2.1788768
outlet-g6-15	2.244111
outlet-g6-16	2.2402204

outlet-g6-17	1.7759756
outlet-g6-18	1.780712
outlet-g6-19	1.6669428
outlet-g6-2	0.43822971
outlet-g6-20	1.6604873
outlet-g6-21	0.4957943
outlet-g6-22	0.50508526
outlet-g6-23	0.450892
outlet-g6-24	0.44378983
outlet-g6-3	0.49252962
outlet-g6-4	0.4833784
outlet-g6-5	1.6489656
outlet-g6-6	1.6557514
outlet-g6-7	1.7749393
outlet-g6-8	1.7708013
outlet-g6-9	2.2433339

Net 1.6118758

(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton2(Cancel)")
/file/stop-transcript

Performance Timer for 127 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.262 sec
Global reductions per iteration:	374 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	7922 messages
Data transfer per iteration:	0.734 MB
LE solves per iteration:	3 solves
LE wall-clock time per iteration:	0.213 sec (81.2%)
LE global solves per iteration:	9 solves
LE global wall-clock time per iteration:	0.001 sec (0.5%)
LE global matrix maximum size:	71
AMG cycles per iteration:	9.016 cycles
Relaxation sweeps per iteration:	4417 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.024 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	397
DPM updates per iteration:	0.0157 updates
DPM wall-clock time per iteration:	0.707 sec (269.3%)

Total wall-clock time: 33.335 sec
Total dpm solve time: 89.780 sec
Total dpm i/o time: 0.000 sec

Transcript Stop Time: 19:31:07, 15 Mar 2026 Central Daylight Time
Total Transcript Time: 2 Minute 39 Second.

—
Transcript Start Time: 19:08:22, 15 Mar 2026 Central Daylight Time
Current Directory:
"C:\Users\kaan\Desktop\Lung-Flow-Steady-NoP\lung-2024-final_files\dp0\FFF-3\Fluent"

Build Time: May 13 2024 11:28:30 EST
Executable Path: C:/PROGRA~1/ANSYSI~1/v242/fluvent/ntbin/win64/fluvent.exe

ID Hostname Core O.S. PID Vendor

n3 Borekci-X1 4/14 Windows-x64 36116 Intel(R) Core(TM) Ultra 7 165U
n2 Borekci-X1 3/14 Windows-x64 35384 Intel(R) Core(TM) Ultra 7 165U
n1 Borekci-X1 2/14 Windows-x64 34616 Intel(R) Core(TM) Ultra 7 165U
n0* Borekci-X1 1/14 Windows-x64 38404 Intel(R) Core(TM) Ultra 7 165U
host Borekci-X1 Windows-x64 23988 Intel(R) Core(TM) Ultra 7 165U

MPI Option Selected: intel
Selected system interconnect: default

Cortex Process ID: 36288
License Server Path: 1055@rho.mcc.northwestern.edu

```
> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model  
Specific)*PushButton1(Discrete Phase)")  
(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")  
(cx-gui-do cx-set-list-selections "Sample Trajectories*Table1*Table3*List1(Release from  
Injections)" '( 0))  
(cx-gui-do cx-activate-item "Sample Trajectories*Table1*Table3*List1(Release from  
Injections)")
```

```
(cx-gui-do cx-set-list-selections "Sample Trajectories*Table1*Table1*List1(Boundaries)"
'( 3))
(cx-gui-do cx-activate-item "Sample Trajectories*Table1*Table1*List1(Boundaries)")
(cx-gui-do cx-set-list-selections "Sample Trajectories*Table1*Table1*List1(Boundaries)"
'( 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45))
(cx-gui-do cx-activate-item "Sample Trajectories*Table1*Table1*List1(Boundaries)")
(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")
```

DPM Iteration

number tracked = 50000, escaped = 38748, trapped = 11252

```
adjoint/      mesh/      report/
close-fluent  parallel/  solve/
define/       plot/      surface/
display/      preferences/ views/
file/         print-license-usage
```

> report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,	
Index	Injection, Index	Name	Id	Min	Max	Avg	Std Dev	Min
Max								

Trapped	walls-g1	7	1045	4.523e-02	2.996e-01	9.267e-02	6.541e-02	
injection-10mic	24137	injection-10mic	22235					
Trapped	walls-g2	8	1426	6.044e-02	1.303e-01	7.365e-02	1.457e-02	
injection-10mic	21050	injection-10mic	15978					
Trapped	walls-g3	9	2028	6.948e-02	4.245e-01	7.886e-02	1.007e-02	
injection-10mic	20547	injection-10mic	7126					
Trapped	walls-g4	10	5204	7.430e-02	5.234e-01	9.121e-02	3.442e-02	
injection-10mic	30442	injection-10mic	42676					
Trapped	walls-g5	11	1216	8.332e-02	4.294e-01	1.002e-01	3.657e-02	
injection-10mic	9620	injection-10mic	20917					
Trapped	walls-g6	12	333	9.757e-02	4.679e-01	1.892e-01	1.012e-01	
injection-10mic	28638	injection-10mic	372					

Escaped	outlet-g6-1	14	521	1.454e-01	2.012e+00	3.312e-01	2.318e-01
injection-10mic 8569	injection-10mic 11054						
Escaped	outlet-g6-2	15	440	1.364e-01	5.573e-01	3.811e-01	1.265e-01
injection-10mic 22373	injection-10mic 92						
Escaped	outlet-g6-3	16	582	1.354e-01	3.234e-01	2.094e-01	2.994e-02
injection-10mic 17970	injection-10mic 4237						
Escaped	outlet-g6-4	17	627	1.405e-01	3.252e-01	1.851e-01	2.296e-02
injection-10mic 21178	injection-10mic 6321						
Escaped	outlet-g6-5	18	718	1.253e-01	4.631e-01	1.732e-01	4.584e-02
injection-10mic 11205	injection-10mic 8410						
Escaped	outlet-g6-6	19	507	1.261e-01	2.831e-01	1.454e-01	1.684e-02
injection-10mic 18058	injection-10mic 5052						
Escaped	outlet-g6-7	20	381	1.149e-01	2.079e-01	1.327e-01	1.815e-02
injection-10mic 9342	injection-10mic 15736						
Escaped	outlet-g6-8	21	599	1.185e-01	2.799e-01	1.462e-01	2.946e-02
injection-10mic 19761	injection-10mic 3638						
Escaped	outlet-g6-9	22	489	1.164e-01	2.203e-01	1.400e-01	2.336e-02
injection-10mic 13087	injection-10mic 20078						
Escaped	outlet-g6-10	23	399	1.085e-01	2.890e-01	1.296e-01	2.727e-02
injection-10mic 14704	injection-10mic 36419						
Escaped	outlet-g6-11	24	561	1.191e-01	4.873e-01	1.340e-01	2.232e-02
injection-10mic 13938	injection-10mic 26686						
Escaped	outlet-g6-12	25	362	1.301e-01	3.832e-01	1.451e-01	1.955e-02
injection-10mic 3648	injection-10mic 35186						
Escaped	outlet-g6-13	26	326	1.214e-01	3.723e-01	1.443e-01	1.771e-02
injection-10mic 35022	injection-10mic 25616						
Escaped	outlet-g6-14	27	542	1.192e-01	3.924e-01	1.339e-01	2.186e-02
injection-10mic 34605	injection-10mic 34340						
Escaped	outlet-g6-15	28	491	1.102e-01	2.792e-01	1.334e-01	3.152e-02
injection-10mic 40208	injection-10mic 29308						
Escaped	outlet-g6-16	29	509	1.162e-01	2.207e-01	1.397e-01	2.269e-02
injection-10mic 29253	injection-10mic 38126						
Escaped	outlet-g6-17	30	624	1.181e-01	2.772e-01	1.435e-01	2.764e-02
injection-10mic 30266	injection-10mic 42055						
Escaped	outlet-g6-18	31	373	1.149e-01	2.478e-01	1.311e-01	1.647e-02
injection-10mic 43201	injection-10mic 33708						
Escaped	outlet-g6-19	32	496	1.255e-01	2.287e-01	1.446e-01	1.428e-02
injection-10mic 46383	injection-10mic 32938						
Escaped	outlet-g6-20	33	744	1.270e-01	4.792e-01	1.706e-01	4.630e-02
injection-10mic 38419	injection-10mic 37943						

Escaped	outlet-g6-21	34	555	1.425e-01	5.634e-01	1.877e-01	2.813e-02
injection-10mic	29337	injection-10mic	45166				
Escaped	outlet-g6-22	35	547	1.382e-01	2.885e-01	2.053e-01	2.313e-02
injection-10mic	26138	injection-10mic	48668				
Escaped	outlet-g6-23	36	519	1.394e-01	1.779e+00	3.875e-01	1.613e-01
injection-10mic	37944	injection-10mic	34928				
Escaped	outlet-g6-24	37	470	1.391e-01	1.200e+00	3.011e-01	1.814e-01
injection-10mic	18678	injection-10mic	48890				
Escaped	outlet-g5-1	38	1466	1.009e-01	3.975e-01	1.208e-01	1.375e-02
injection-10mic	15676	injection-10mic	30394				
Escaped	outlet-g5-2	39	1001	1.090e-01	1.552e+00	1.801e-01	9.489e-02
injection-10mic	29165	injection-10mic	15245				
Escaped	outlet-g5-3	40	1094	1.078e-01	1.483e+00	1.867e-01	1.212e-01
injection-10mic	40224	injection-10mic	39202				
Escaped	outlet-g5-4	41	1416	1.030e-01	2.574e-01	1.207e-01	1.230e-02
injection-10mic	46405	injection-10mic	7651				
Escaped	outlet-g4-1	42	3162	8.930e-02	4.845e-01	1.090e-01	1.864e-02
injection-10mic	11380	injection-10mic	11224				
Escaped	outlet-g4-2	43	2574	8.149e-02	1.779e-01	9.369e-02	1.002e-02
injection-10mic	15201	injection-10mic	11847				
Escaped	outlet-g4-3	44	2161	8.250e-02	1.813e-01	9.238e-02	1.044e-02
injection-10mic	16863	injection-10mic	17226				
Escaped	outlet-g4-4	45	2768	8.040e-02	3.437e-01	9.225e-02	1.452e-02
injection-10mic	22220	injection-10mic	26050				
Escaped	outlet-g4-5	46	2903	8.029e-02	3.916e-01	9.230e-02	1.432e-02
injection-10mic	32049	injection-10mic	5101				
Escaped	outlet-g4-6	47	2106	7.938e-02	1.969e-01	9.229e-02	1.075e-02
injection-10mic	34970	injection-10mic	30500				
Escaped	outlet-g4-7	48	2598	8.281e-02	6.308e-01	9.374e-02	1.451e-02
injection-10mic	42424	injection-10mic	36168				
Escaped	outlet-g4-8	49	3117	8.646e-02	4.362e-01	1.087e-01	2.037e-02
injection-10mic	33820	injection-10mic	40361				

(*)- Mass Transfer Summary -(*)

Fate	Zone Name	Zone Id	Mass Flow (kg/s)		
			Initial	Final	Change
Trapped	walls-g1	7	2.090e-10	2.090e-10	0.000e+00
Trapped	walls-g2	8	2.852e-10	2.852e-10	0.000e+00

Trapped	walls-g3	9	4.056e-10	4.056e-10	0.000e+00
Trapped	walls-g4	10	1.041e-09	1.041e-09	0.000e+00
Trapped	walls-g5	11	2.432e-10	2.432e-10	0.000e+00
Trapped	walls-g6	12	6.660e-11	6.660e-11	0.000e+00
Escaped	outlet-g6-1	14	1.042e-10	1.042e-10	0.000e+00
Escaped	outlet-g6-2	15	8.800e-11	8.800e-11	0.000e+00
Escaped	outlet-g6-3	16	1.164e-10	1.164e-10	0.000e+00
Escaped	outlet-g6-4	17	1.254e-10	1.254e-10	0.000e+00
Escaped	outlet-g6-5	18	1.436e-10	1.436e-10	0.000e+00
Escaped	outlet-g6-6	19	1.014e-10	1.014e-10	0.000e+00
Escaped	outlet-g6-7	20	7.620e-11	7.620e-11	0.000e+00
Escaped	outlet-g6-8	21	1.198e-10	1.198e-10	0.000e+00
Escaped	outlet-g6-9	22	9.780e-11	9.780e-11	0.000e+00
Escaped	outlet-g6-10	23	7.980e-11	7.980e-11	0.000e+00
Escaped	outlet-g6-11	24	1.122e-10	1.122e-10	0.000e+00
Escaped	outlet-g6-12	25	7.240e-11	7.240e-11	0.000e+00
Escaped	outlet-g6-13	26	6.520e-11	6.520e-11	0.000e+00
Escaped	outlet-g6-14	27	1.084e-10	1.084e-10	0.000e+00
Escaped	outlet-g6-15	28	9.820e-11	9.820e-11	0.000e+00
Escaped	outlet-g6-16	29	1.018e-10	1.018e-10	0.000e+00
Escaped	outlet-g6-17	30	1.248e-10	1.248e-10	0.000e+00
Escaped	outlet-g6-18	31	7.460e-11	7.460e-11	0.000e+00
Escaped	outlet-g6-19	32	9.920e-11	9.920e-11	0.000e+00
Escaped	outlet-g6-20	33	1.488e-10	1.488e-10	0.000e+00
Escaped	outlet-g6-21	34	1.110e-10	1.110e-10	0.000e+00
Escaped	outlet-g6-22	35	1.094e-10	1.094e-10	0.000e+00
Escaped	outlet-g6-23	36	1.038e-10	1.038e-10	0.000e+00
Escaped	outlet-g6-24	37	9.400e-11	9.400e-11	0.000e+00
Escaped	outlet-g5-1	38	2.932e-10	2.932e-10	0.000e+00
Escaped	outlet-g5-2	39	2.002e-10	2.002e-10	0.000e+00
Escaped	outlet-g5-3	40	2.188e-10	2.188e-10	0.000e+00
Escaped	outlet-g5-4	41	2.832e-10	2.832e-10	0.000e+00
Escaped	outlet-g4-1	42	6.324e-10	6.324e-10	0.000e+00
Escaped	outlet-g4-2	43	5.148e-10	5.148e-10	0.000e+00
Escaped	outlet-g4-3	44	4.322e-10	4.322e-10	0.000e+00
Escaped	outlet-g4-4	45	5.536e-10	5.536e-10	0.000e+00
Escaped	outlet-g4-5	46	5.806e-10	5.806e-10	0.000e+00
Escaped	outlet-g4-6	47	4.212e-10	4.212e-10	0.000e+00
Escaped	outlet-g4-7	48	5.196e-10	5.196e-10	0.000e+00
Escaped	outlet-g4-8	49	6.234e-10	6.234e-10	0.000e+00

Net	1.000e-08	1.000e-08	0.000e+00
-----	-----------	-----------	-----------

```
> (cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton2(Close)")
(cx-gui-do cx-activate-item
"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface
Integrals)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table1*DropDownList1(Report Type)"
'( 11))
(cx-gui-do cx-activate-item "Surface Integrals*Table1*DropDownList1(Report Type)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*DropDownList2" '( 4))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*DropDownList2")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

inlet	4.2198657
-------	-----------

```
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 1 3
4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33
34 35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 3 4
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

outlet-g4-1	1.1471375
outlet-g4-2	1.8047741
outlet-g4-3	2.0411496
outlet-g4-4	2.171926
outlet-g4-5	2.1841765
outlet-g4-6	2.0501187
outlet-g4-7	1.8107353

outlet-g4-8	1.1590329
outlet-g5-1	1.0110078
outlet-g5-2	0.92010889
outlet-g5-3	0.93663186
outlet-g5-4	1.026717
outlet-g6-1	0.45543376
outlet-g6-10	2.0344926
outlet-g6-11	1.9937113
outlet-g6-12	1.980348
outlet-g6-13	1.9856373
outlet-g6-14	1.9996107
outlet-g6-15	2.042714
outlet-g6-16	2.0458399
outlet-g6-17	1.806262
outlet-g6-18	1.8041101
outlet-g6-19	1.7241025
outlet-g6-2	0.46097571
outlet-g6-20	1.7181414
outlet-g6-21	0.50909606
outlet-g6-22	0.51584777
outlet-g6-23	0.4690341
outlet-g6-24	0.46371453
outlet-g6-3	0.50727756
outlet-g6-4	0.50069644
outlet-g6-5	1.7185625
outlet-g6-6	1.7244023
outlet-g6-7	1.8081651
outlet-g6-8	1.8110535
outlet-g6-9	2.0373913

 Net 1.5542082

(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton2(Cancel)")
 /file/stop-transcript

Performance Timer for 250 iterations on 4 compute nodes

Average wall-clock time per iteration:	0.601 sec
Global reductions per iteration:	376 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	9786 messages
Data transfer per iteration:	1.126 MB

LE solves per iteration: 3 solves
 LE wall-clock time per iteration: 0.477 sec (79.3%)
 LE global solves per iteration: 8 solves
 LE global wall-clock time per iteration: 0.002 sec (0.3%)
 LE global matrix maximum size: 67
 AMG cycles per iteration: 8.296 cycles
 Relaxation sweeps per iteration: 5602 sweeps
 Relaxation exchanges per iteration: 0 exchanges
 LE early protections (stall) per iteration: 0.040 times
 LE early protections (divergence) per iteration: 0.000 times
 Total SVARS touched: 396
 DPM updates per iteration: 0.0120 updates
 DPM wall-clock time per iteration: 0.469 sec (78.1%)

Total wall-clock time: 150.215 sec
 Total dpm solve time: 117.350 sec
 Total dpm i/o time: 0.000 sec

Transcript Stop Time: 19:11:32, 15 Mar 2026 Central Daylight Time
 Total Transcript Time: 3 Minute 10 Second.

—
 Transcript Start Time: 18:27:11, 15 Mar 2026 Central Daylight Time
 Current Directory:
 "C:\Users\kaan\Desktop\Lung-Flow-Steady-NoP\lung-2024-final_files\dp0\FFF-4\Fluent"

Build Time: May 13 2024 11:28:30 EST
 Executable Path: C:/PROGRA~1/ANSYSI~1/v242/fluvent/ntbin/win64/fluvent.exe

ID	Hostname	Core	O.S.	PID	Vendor
n3	Borekci-X1	4/14	Windows-x64	15564	Intel(R) Core(TM) Ultra 7 165U
n2	Borekci-X1	3/14	Windows-x64	16216	Intel(R) Core(TM) Ultra 7 165U
n1	Borekci-X1	2/14	Windows-x64	15860	Intel(R) Core(TM) Ultra 7 165U
n0*	Borekci-X1	1/14	Windows-x64	20636	Intel(R) Core(TM) Ultra 7 165U
host	Borekci-X1		Windows-x64	10356	Intel(R) Core(TM) Ultra 7 165U

MPI Option Selected: intel
 Selected system interconnect: default

Cortex Process ID: 32784

License Server Path: 1055@rho.mcc.northwestern.edu

> (cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model Specific)*PushButton1(Discrete Phase)")

(cx-gui-do cx-activate-item "MenuBar*PopupMenuDiscretePhase*Sample...")

(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton1(Apply)")

DPM Iteration

number tracked = 50000, escaped = 39655, trapped = 10345

(cx-gui-do cx-activate-item "Sample Trajectories*PanelButtons*PushButton2(Close)")

(cx-gui-do cx-activate-item

"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface Integrals)")

(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton2(Cancel)")

(cx-gui-do cx-activate-item "Ribbon*Frame1*Frame6(Results)*Table1*Table3(Model Specific)*PushButton1(Discrete Phase)")

adjoint/	mesh/	report/
close-fluent	parallel/	solve/
define/	plot/	surface/
display/	preferences/	views/
file/	print-license-usage	

> report/dpm-summary

Fate	Zone	Zone Number	Elapsed Time (s)				Injection,
Index	Injection, Index						
	Name	Id	Min	Max	Avg	Std Dev	Min
Max							

Trapped walls-g1 7 781 4.476e-02 7.354e-01 6.358e-02 3.446e-02
injection-10mic 23999 injection-10mic 48923

Trapped walls-g2 8 1451 6.190e-02 1.220e-01 7.172e-02 1.144e-02
injection-10mic 47230 injection-10mic 34484

Trapped	walls-g3	9	1902	6.913e-02	4.282e-01	7.958e-02	1.461e-02
injection-10mic	24078	injection-10mic	8164				
Trapped	walls-g4	10	4647	7.460e-02	6.493e-01	9.369e-02	3.866e-02
injection-10mic	15639	injection-10mic	11211				
Trapped	walls-g5	11	1266	8.281e-02	4.424e-01	1.064e-01	4.783e-02
injection-10mic	41867	injection-10mic	19252				
Trapped	walls-g6	12	298	9.779e-02	8.655e-01	2.033e-01	1.231e-01
injection-10mic	12243	injection-10mic	44991				
Escaped	outlet-g6-1	14	1101	1.455e-01	1.852e+00	4.501e-01	3.932e-01
injection-10mic	4655	injection-10mic	8728				
Escaped	outlet-g6-2	15	412	1.386e-01	3.986e-01	2.914e-01	7.998e-02
injection-10mic	8454	injection-10mic	196				
Escaped	outlet-g6-3	16	538	1.397e-01	4.363e-01	2.084e-01	3.067e-02
injection-10mic	36173	injection-10mic	3261				
Escaped	outlet-g6-4	17	715	1.396e-01	6.238e-01	1.905e-01	3.321e-02
injection-10mic	11049	injection-10mic	13146				
Escaped	outlet-g6-5	18	712	1.249e-01	4.044e-01	1.741e-01	4.722e-02
injection-10mic	23163	injection-10mic	3912				
Escaped	outlet-g6-6	19	433	1.263e-01	2.987e-01	1.491e-01	1.857e-02
injection-10mic	9616	injection-10mic	8054				
Escaped	outlet-g6-7	20	306	1.172e-01	3.343e-01	1.392e-01	2.537e-02
injection-10mic	8811	injection-10mic	2373				
Escaped	outlet-g6-8	21	616	1.213e-01	3.881e-01	1.504e-01	3.331e-02
injection-10mic	10256	injection-10mic	3720				
Escaped	outlet-g6-9	22	524	1.146e-01	2.240e-01	1.391e-01	2.290e-02
injection-10mic	18627	injection-10mic	14089				
Escaped	outlet-g6-10	23	505	1.093e-01	2.466e-01	1.309e-01	2.425e-02
injection-10mic	20149	injection-10mic	12768				
Escaped	outlet-g6-11	24	600	1.177e-01	4.897e-01	1.344e-01	2.598e-02
injection-10mic	13992	injection-10mic	14250				
Escaped	outlet-g6-12	25	398	1.255e-01	7.356e-01	1.471e-01	3.194e-02
injection-10mic	5600	injection-10mic	6383				
Escaped	outlet-g6-13	26	363	1.285e-01	2.934e-01	1.463e-01	1.777e-02
injection-10mic	41575	injection-10mic	21070				
Escaped	outlet-g6-14	27	561	1.179e-01	2.695e-01	1.338e-01	1.926e-02
injection-10mic	35886	injection-10mic	36861				
Escaped	outlet-g6-15	28	437	1.086e-01	4.038e-01	1.322e-01	2.889e-02
injection-10mic	27289	injection-10mic	36563				
Escaped	outlet-g6-16	29	531	1.175e-01	2.241e-01	1.422e-01	2.423e-02
injection-10mic	19333	injection-10mic	25191				

Escaped	outlet-g6-17	30	606	1.196e-01	3.569e-01	1.526e-01	3.196e-02
injection-10mic	45895	injection-10mic	30981				
Escaped	outlet-g6-18	31	315	1.158e-01	2.054e-01	1.363e-01	1.817e-02
injection-10mic	37731	injection-10mic	35613				
Escaped	outlet-g6-19	32	445	1.232e-01	7.425e-01	1.541e-01	3.828e-02
injection-10mic	31678	injection-10mic	46463				
Escaped	outlet-g6-20	33	715	1.271e-01	4.146e-01	1.769e-01	4.960e-02
injection-10mic	32806	injection-10mic	48077				
Escaped	outlet-g6-21	34	704	1.369e-01	4.194e-01	1.881e-01	2.650e-02
injection-10mic	24357	injection-10mic	43422				
Escaped	outlet-g6-22	35	554	1.391e-01	5.779e-01	2.076e-01	3.588e-02
injection-10mic	21202	injection-10mic	43447				
Escaped	outlet-g6-23	36	440	1.413e-01	5.393e-01	2.862e-01	8.256e-02
injection-10mic	35094	injection-10mic	42264				
Escaped	outlet-g6-24	37	1074	1.416e-01	1.535e+00	4.452e-01	
3.933e-01	injection-10mic	45525	injection-10mic	49969			
Escaped	outlet-g5-1	38	1565	1.040e-01	5.333e-01	1.238e-01	1.690e-02
injection-10mic	10540	injection-10mic	16485				
Escaped	outlet-g5-2	39	1346	1.068e-01	1.478e+00	1.960e-01	1.252e-01
injection-10mic	5905	injection-10mic	9771				
Escaped	outlet-g5-3	40	1288	1.085e-01	1.487e+00	1.874e-01	1.124e-01
injection-10mic	36913	injection-10mic	34586				
Escaped	outlet-g5-4	41	1545	1.020e-01	3.798e-01	1.237e-01	1.751e-02
injection-10mic	21673	injection-10mic	26545				
Escaped	outlet-g4-1	42	2092	8.793e-02	7.064e-01	1.137e-01	3.066e-02
injection-10mic	7167	injection-10mic	9181				
Escaped	outlet-g4-2	43	3054	8.276e-02	2.570e-01	9.384e-02	1.061e-02
injection-10mic	12391	injection-10mic	7255				
Escaped	outlet-g4-3	44	2409	8.021e-02	2.013e-01	9.212e-02	1.139e-02
injection-10mic	11078	injection-10mic	42548				
Escaped	outlet-g4-4	45	2659	8.054e-02	3.780e-01	9.164e-02	1.506e-02
injection-10mic	5985	injection-10mic	30837				
Escaped	outlet-g4-5	46	2660	8.028e-02	5.069e-01	9.224e-02	1.658e-02
injection-10mic	43757	injection-10mic	29402				
Escaped	outlet-g4-6	47	2329	8.158e-02	2.042e-01	9.265e-02	1.260e-02
injection-10mic	36536	injection-10mic	33131				
Escaped	outlet-g4-7	48	2983	8.252e-02	3.709e-01	9.366e-02	1.060e-02
injection-10mic	38512	injection-10mic	30574				
Escaped	outlet-g4-8	49	2120	8.788e-02	4.851e-01	1.141e-01	2.553e-02
injection-10mic	26641	injection-10mic	48549				

(*)- Mass Transfer Summary -(*)

Fate	Zone	Zone	Mass Flow (kg/s)		
	Name	Id	Initial	Final	Change

Trapped	walls-g1	7	1.562e-10	1.562e-10	0.000e+00
Trapped	walls-g2	8	2.902e-10	2.902e-10	0.000e+00
Trapped	walls-g3	9	3.804e-10	3.804e-10	0.000e+00
Trapped	walls-g4	10	9.294e-10	9.294e-10	0.000e+00
Trapped	walls-g5	11	2.532e-10	2.532e-10	0.000e+00
Trapped	walls-g6	12	5.960e-11	5.960e-11	0.000e+00
Escaped	outlet-g6-1	14	2.202e-10	2.202e-10	0.000e+00
Escaped	outlet-g6-2	15	8.240e-11	8.240e-11	0.000e+00
Escaped	outlet-g6-3	16	1.076e-10	1.076e-10	0.000e+00
Escaped	outlet-g6-4	17	1.430e-10	1.430e-10	0.000e+00
Escaped	outlet-g6-5	18	1.424e-10	1.424e-10	0.000e+00
Escaped	outlet-g6-6	19	8.660e-11	8.660e-11	0.000e+00
Escaped	outlet-g6-7	20	6.120e-11	6.120e-11	0.000e+00
Escaped	outlet-g6-8	21	1.232e-10	1.232e-10	0.000e+00
Escaped	outlet-g6-9	22	1.048e-10	1.048e-10	0.000e+00
Escaped	outlet-g6-10	23	1.010e-10	1.010e-10	0.000e+00
Escaped	outlet-g6-11	24	1.200e-10	1.200e-10	0.000e+00
Escaped	outlet-g6-12	25	7.960e-11	7.960e-11	0.000e+00
Escaped	outlet-g6-13	26	7.260e-11	7.260e-11	0.000e+00
Escaped	outlet-g6-14	27	1.122e-10	1.122e-10	0.000e+00
Escaped	outlet-g6-15	28	8.740e-11	8.740e-11	0.000e+00
Escaped	outlet-g6-16	29	1.062e-10	1.062e-10	0.000e+00
Escaped	outlet-g6-17	30	1.212e-10	1.212e-10	0.000e+00
Escaped	outlet-g6-18	31	6.300e-11	6.300e-11	0.000e+00
Escaped	outlet-g6-19	32	8.900e-11	8.900e-11	0.000e+00
Escaped	outlet-g6-20	33	1.430e-10	1.430e-10	0.000e+00
Escaped	outlet-g6-21	34	1.408e-10	1.408e-10	0.000e+00
Escaped	outlet-g6-22	35	1.108e-10	1.108e-10	0.000e+00
Escaped	outlet-g6-23	36	8.800e-11	8.800e-11	0.000e+00
Escaped	outlet-g6-24	37	2.148e-10	2.148e-10	0.000e+00
Escaped	outlet-g5-1	38	3.130e-10	3.130e-10	0.000e+00
Escaped	outlet-g5-2	39	2.692e-10	2.692e-10	0.000e+00
Escaped	outlet-g5-3	40	2.576e-10	2.576e-10	0.000e+00
Escaped	outlet-g5-4	41	3.090e-10	3.090e-10	0.000e+00

Escaped	outlet-g4-1	42	4.184e-10	4.184e-10	0.000e+00
Escaped	outlet-g4-2	43	6.108e-10	6.108e-10	0.000e+00
Escaped	outlet-g4-3	44	4.818e-10	4.818e-10	0.000e+00
Escaped	outlet-g4-4	45	5.318e-10	5.318e-10	0.000e+00
Escaped	outlet-g4-5	46	5.320e-10	5.320e-10	0.000e+00
Escaped	outlet-g4-6	47	4.658e-10	4.658e-10	0.000e+00
Escaped	outlet-g4-7	48	5.966e-10	5.966e-10	0.000e+00
Escaped	outlet-g4-8	49	4.240e-10	4.240e-10	0.000e+00

-----		-----			
Net		1.000e-08	1.000e-08	0.000e+00	

```
> (cx-gui-do cx-activate-item
"Ribbon*Frame1*Frame6(Results)*Table1*Table3(Reports)*PushButton4(Surface
Integrals)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

inlet	4.2615679

```
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '())
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-set-list-selections "Surface Integrals*Table2*Table4*List1(Surfaces)" '( 3 4
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
35 36 37 38))
(cx-gui-do cx-activate-item "Surface Integrals*Table2*Table4*List1(Surfaces)")
(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton1(OK)")
```

Mass-Weighted Average	
Total Pressure	[Pa]

outlet-g4-1	1.1612432
outlet-g4-2	1.7586683
outlet-g4-3	2.1326607
outlet-g4-4	2.3090998
outlet-g4-5	2.3109999
outlet-g4-6	2.1379088
outlet-g4-7	1.7591435
outlet-g4-8	1.160557
outlet-g5-1	1.186605

outlet-g5-2	1.1178511
outlet-g5-3	1.1183198
outlet-g5-4	1.186365
outlet-g6-1	0.49648957
outlet-g6-10	2.1134732
outlet-g6-11	2.0753322
outlet-g6-12	2.0604963
outlet-g6-13	2.0608151
outlet-g6-14	2.0757107
outlet-g6-15	2.1141049
outlet-g6-16	2.1143213
outlet-g6-17	1.7865428
outlet-g6-18	1.7855513
outlet-g6-19	1.7047645
outlet-g6-2	0.50153705
outlet-g6-20	1.699661
outlet-g6-21	0.54012823
outlet-g6-22	0.54654743
outlet-g6-23	0.5003054
outlet-g6-24	0.4953133
outlet-g6-3	0.54753172
outlet-g6-4	0.54123598
outlet-g6-5	1.700937
outlet-g6-6	1.7061205
outlet-g6-7	1.7867454
outlet-g6-8	1.787809
outlet-g6-9	2.1130588

 Net 1.6097149

(cx-gui-do cx-activate-item "Surface Integrals*PanelButtons*PushButton2(Cancel)")
 /file/stop-transcript

Performance Timer for 225 iterations on 4 compute nodes

Average wall-clock time per iteration:	3.055 sec
Global reductions per iteration:	378 ops
Global reductions time per iteration:	0.000 sec (0.0%)
Message count per iteration:	12133 messages
Data transfer per iteration:	1.789 MB
LE solves per iteration:	3 solves
LE wall-clock time per iteration:	2.562 sec (83.8%)

LE global solves per iteration:	8 solves
LE global wall-clock time per iteration:	0.003 sec (0.1%)
LE global matrix maximum size:	96
AMG cycles per iteration:	8.591 cycles
Relaxation sweeps per iteration:	3930 sweeps
Relaxation exchanges per iteration:	0 exchanges
LE early protections (stall) per iteration:	0.004 times
LE early protections (divergence) per iteration:	0.000 times
Total SVARS touched:	401
DPM updates per iteration:	0.0267 updates
DPM wall-clock time per iteration:	1.945 sec (63.7%)

Total wall-clock time:	687.391 sec
Total dpm solve time:	437.650 sec
Total dpm i/o time:	0.000 sec

Transcript Stop Time: 18:30:11, 15 Mar 2026 Central Daylight Time
Total Transcript Time: 3 Minute 0 Second.

7.3.1 Associated Python Code

Grid:

```
In [61]: import re
import pandas as pd
import numpy as np

# PARSER – to handle both old (static pressure) and new (total pressure)

def parse_fluent_grid_file(filepath):
    with open(filepath, encoding='utf-8', errors='replace') as f:
        lines = f.readlines()

        tracked = escaped = trapped = None
        particle_rows = []
        mass_rows = []
        static_pressure_rows = []
        total_pressure_rows = []

        reading_particle_table = False
        reading_mass_table = False
        reading_static_pressure = False
        reading_total_pressure = False

        for raw_line in lines:
            line = raw_line.strip()
```

file:///Users/kaanborekci/Desktop/CFD-FinalProject/ANALYSIS/parcel-and-grid-independence.html

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```

# — DPM summary line —
if "number tracked" in line:
    m = re.search(r'number tracked = (\d+), escaped = (\d+), t
    if m:
        tracked = int(m.group(1))
        escaped = int(m.group(2))
        trapped = int(m.group(3))

# — Section header detection —
if line.startswith("Fate") and "Elapsed Time" in line:
    reading_particle_table = True
    reading_mass_table = False
    reading_static_pressure = False
    reading_total_pressure = False
    continue

if "Mass Transfer Summary" in line:
    reading_particle_table = False
    reading_mass_table = True
    reading_static_pressure = False
    reading_total_pressure = False
    continue

if "Area-Weighted Average" in line:
    reading_particle_table = False
    reading_mass_table = False
    reading_total_pressure = False
    continue

if "Mass-Weighted Average" in line:
    reading_particle_table = False
    reading_mass_table = False
    reading_static_pressure = False
    continue

if "Static Pressure" in line and "[Pa]" in line:
    reading_static_pressure = True
    reading_particle_table = False
    reading_mass_table = False
    reading_total_pressure = False
    continue

if "Total Pressure" in line and "[Pa]" in line:
    reading_total_pressure = True
    reading_static_pressure = False
    reading_particle_table = False
    reading_mass_table = False
    continue

# — Stop on Fluent GUI commands or transcript end —
if line.startswith("(cx-gui)" or "/file/stop-transcript" in li

```

```

        reading_particle_table = False
        reading_mass_table = False
        reading_static_pressure = False
        reading_total_pressure = False
        continue

# — Particle fate table —
if reading_particle_table and (line.startswith("Trapped") or line.
    parts = line.split()
    if len(parts) >= 4:
        particle_rows.append({
            "fate": parts[0],
            "zone": parts[1],
            "zone_id": int(parts[2]),
            "count": int(parts[3]),
        })

# — Mass transfer table —
if reading_mass_table and (line.startswith("Trapped") or line.
    parts = line.split()
    if len(parts) >= 5:
        mass_rows.append({
            "fate": parts[0],
            "zone": parts[1],
            "zone_id": int(parts[2]),
            "initial_mass": float(parts[3]),
            "final_mass": float(parts[4]),
        })

# — Static pressure table (area-weighted) —
if reading_static_pressure:
    if line.startswith("----") or line == "":
        continue
    parts = line.split()
    if len(parts) >= 2:
        try:
            value = float(parts[-1])
            static_pressure_rows.append({
                "zone": parts[0],
                "pressure_pa": value
            })
        except ValueError:
            pass

# — Total pressure table (mass-weighted) —
if reading_total_pressure:
    if line.startswith("----") or line == "":
        continue
    parts = line.split()
    if len(parts) >= 2:
        try:
            value = float(parts[-1])

```

```

        total_pressure_rows.append({
            "zone": parts[0],
            "total_pressure_pa": value
        })
    except ValueError:
        pass

summary = {
    "tracked": tracked,
    "escaped": escaped,
    "trapped": trapped
}

particle_df = pd.DataFrame(particle_rows)
mass_df = pd.DataFrame(mass_rows)
static_pressure_df = pd.DataFrame(static_pressure_rows)
total_pressure_df = pd.DataFrame(total_pressure_rows)

return summary, particle_df, mass_df, static_pressure_df, total_pressure_df

def extract_grid_scalars(filepath):
    (summary, particle_df, mass_df,
     static_pressure_df, total_pressure_df) = parse_fluent_grid_file(filepath)

    tracked = summary["tracked"]
    total_dep = summary["trapped"] / tracked

    # Per-generation deposition
    trapped_df = particle_df[particle_df["fate"] == "Trapped"].copy()
    trapped_df["generation"] = trapped_df["zone"].str.extract(r"g(\d+)")
    local_dep = {
        f"G{int(row['generation'])}": row["count"] / tracked
        for _, row in trapped_df.iterrows()
    }

    # — Total pressure (mass-weighted) —
    tp_map = dict(zip(total_pressure_df["zone"], total_pressure_df["total_pressure_pa"]))
    inlet_total = tp_map.get("inlet")
    # "Net" in the outlets-only report = mass-flow-weighted outlet average
    outlet_avg_total = tp_map.get("Net")
    total_pressure_drop = (inlet_total - outlet_avg_total
                           if inlet_total is not None and outlet_avg_total is not None
                           else None)

    # — Static pressure (area-weighted, may be empty for new files) —
    sp_map = dict(zip(static_pressure_df["zone"], static_pressure_df["static_pressure_pa"]))
    inlet_static = sp_map.get("inlet")

    scalars = {
        "tracked": tracked,
        "escaped": summary["escaped"],
        "trapped": summary["trapped"],
        "total_pressure_drop": total_pressure_drop,
        "inlet_total": inlet_total,
        "outlet_avg_total": outlet_avg_total,
        "inlet_static": inlet_static,
        "local_dep": local_dep
    }

```

```

        "trapped": summary["trapped"],
        "total_deposition_fraction": total_dep,
        # Total pressure (primary)
        "inlet_total_pressure_pa": inlet_total,
        "outlet_avg_total_pressure_pa": outlet_avg_total,
        "total_pressure_drop_pa": total_pressure_drop,
        # Static pressure (legacy, may be None)
        "inlet_static_pressure_pa": inlet_static,
        "net_static_pressure_pa": sp_map.get("Net"),
    }

    for k, v in local_dep.items():
        scalars[f"dep_{k}"] = v

    return scalars

# GCI FUNCTIONS (based on Çelik et al. 2008)

def representative_mesh_size_2d(domain_area, n_elements):
    return np.sqrt(domain_area / n_elements)

def refinement_ratio(h_coarser, h_finer):
    return h_coarser / h_finer

def solve_p(f1, f2, f3, r21, r32, tol=1e-8, max_iter=100):
    e21 = f2 - f1
    e32 = f3 - f2

    if abs(e21) < 1e-14 or abs(e32) < 1e-14:
        return None, "indeterminate"

    s = 1.0 * np.sign(e32 / e21)

    if (e32 / e21) > 0:
        convergence = "monotonic"
    else:
        convergence = "oscillatory"

    # Fixed-point iteration per Çelik Eq. 3a-3c
    p = abs(np.log(abs(e32 / e21))) / np.log(r21)

    for _ in range(max_iter):
        p_old = p
        qp = np.log(abs((r21**p - s) / (r32**p - s)))
        p = abs(np.log(abs(e32 / e21)) + qp) / np.log(r21)
        if abs(p - p_old) < tol:
            break

    return p, convergence

```

```

def richardson_extrapolated_value(f1, f2, r21, p):
    return f1 + (f1 - f2) / (r21**p - 1.0)

def gci(f_fine, f_coarse, r, p, Fs=1.25):
    return Fs * abs((f_coarse - f_fine) / f_fine) / (r**p - 1.0)

def asymptotic_ratio(GCI21, GCI32, r21, p):
    return GCI32 / (GCI21 * r21**p)

def run_scalar_gci(name, f1, f2, f3, N1, N2, N3, A_domain):
    h1 = representative_mesh_size_2d(A_domain, N1)
    h2 = representative_mesh_size_2d(A_domain, N2)
    h3 = representative_mesh_size_2d(A_domain, N3)

    r21 = refinement_ratio(h2, h1)
    r32 = refinement_ratio(h3, h2)

    result = {
        "name": name,
        "f1": f1, "f2": f2, "f3": f3,
        "h1": h1, "h2": h2, "h3": h3,
        "r21": r21, "r32": r32,
    }

    if r21 <= 1.3 or r32 <= 1.3:
        result["status"] = "invalid_refinement_ratio"
        return result

    p, convergence = solve_p(f1, f2, f3, r21, r32)

    if p is None:
        result["status"] = "indeterminate"
        return result

    f_ext = richardson_extrapolated_value(f1, f2, r21, p)
    GCI21 = gci(f1, f2, r21, p)
    GCI32 = gci(f2, f3, r32, p)
    AR = asymptotic_ratio(GCI21, GCI32, r21, p)

    result.update({
        "convergence_type": convergence,
        "p": p,
        "f_ext": f_ext,
        "GCI21": GCI21,
        "GCI32": GCI32,
        "asymptotic_ratio": AR,
        "fine_mesh_rel_error_percent": abs((f_ext - f1) / f_ext) * 100
    })
    return result

```

In [62]: `# RUN GCI ANALYSIS`


```

N1 = 688209 # ultra-fine
N2 = 228640 # fine
N3 = 109206 # medium
A_domain = 0.0066366965 # m^2

ultra = extract_grid_scalars("/Users/kaanborekci/Desktop/CFD-FinalProj
fine = extract_grid_scalars("/Users/kaanborekci/Desktop/CFD-FinalProj
med = extract_grid_scalars("/Users/kaanborekci/Desktop/CFD-FinalProj

# Print raw values
print("=" * 70)
print("RAW SCALAR VALUES")
print("=" * 70)
for label, d in [("Ultra-fine", ultra), ("Fine", fine), ("Medium", med)]:
    print(f"\n--- {label} ---")
    for k, v in d.items():
        if v is None:
            print(f" {k}: None")
        elif isinstance(v, float):
            print(f" {k}: {v:.6f}")
        else:
            print(f" {k}: {v}")

# Build GCI results
results = []

results.append(run_scalar_gci(
    "Overall deposition fraction",
    ultra["total_deposition_fraction"],
    fine["total_deposition_fraction"],
    med["total_deposition_fraction"],
    N1, N2, N3, A_domain
))

results.append(run_scalar_gci(
    "Inlet total pressure [Pa]",
    ultra["inlet_total_pressure_pa"],
    fine["inlet_total_pressure_pa"],
    med["inlet_total_pressure_pa"],
    N1, N2, N3, A_domain
))

results.append(run_scalar_gci(
    "Total pressure drop [Pa]",
    ultra["total_pressure_drop_pa"],
    fine["total_pressure_drop_pa"],
    med["total_pressure_drop_pa"],
    N1, N2, N3, A_domain
))

for gen in ["G1", "G2", "G3", "G4", "G5", "G6"]:
    key = f"dep_{gen}"

```

```

        results.append(run_scalar_gci(
            f"Local deposition fraction {gen}",
            ultra[key],
            fine[key],
            med[key],
            N1, N2, N3, A_domain
        ))

results_df = pd.DataFrame(results)
print("\n" + "=" * 70)
print("GCI RESULTS")
print("=" * 70)
cols = ["name", "f1", "f2", "f3", "convergence_type", "r21", "r32",
        "p", "f_ext", "GCI21", "GCI32", "asymptotic_ratio",
        "fine_mesh_rel_error_percent"]
print(results_df[cols].to_string(index=False))

```

```

=====
RAW SCALAR VALUES
=====

```

```

--- Ultra-fine ---

```

```

tracked: 50000
escaped: 39655
trapped: 10345
total_deposition_fraction: 0.206900
inlet_total_pressure_pa: 4.261568
outlet_avg_total_pressure_pa: 1.609715
total_pressure_drop_pa: 2.651853
inlet_static_pressure_pa: None
net_static_pressure_pa: None
dep_G1: 0.015620
dep_G2: 0.029020
dep_G3: 0.038040
dep_G4: 0.092940
dep_G5: 0.025320
dep_G6: 0.005960

```

```

--- Fine ---

```

```

tracked: 50000
escaped: 38748
trapped: 11252
total_deposition_fraction: 0.225040
inlet_total_pressure_pa: 4.219866
outlet_avg_total_pressure_pa: 1.554208
total_pressure_drop_pa: 2.665658
inlet_static_pressure_pa: None
net_static_pressure_pa: None
dep_G1: 0.020900
dep_G2: 0.028520
dep_G3: 0.040560
dep_G4: 0.104080
dep_G5: 0.024320

```

dep_G6: 0.006660

--- Medium ---

tracked: 50000
 escaped: 40626
 trapped: 9374
 total_deposition_fraction: 0.187480
 inlet_total_pressure_pa: 4.049685
 outlet_avg_total_pressure_pa: 1.611876
 total_pressure_drop_pa: 2.437810
 inlet_static_pressure_pa: None
 net_static_pressure_pa: None
 dep_G1: 0.017800
 dep_G2: 0.024500
 dep_G3: 0.035580
 dep_G4: 0.085420
 dep_G5: 0.019260
 dep_G6: 0.004920

GCI RESULTS

	name	f1	f2	f3	convergence_typ		
e	r21	r32	p	f_ext	GCI21	GCI32	asymptotic_rat
io	fine_mesh_rel_error_percent						
	Overall deposition fraction	0.206900	0.225040	0.187480			oscillator
y	1.734938	1.446948	1.707102	0.195282	0.070188	0.237369	1.3203
14		5.949110					
	Inlet total pressure [Pa]	4.261568	4.219866	4.049685			monotoni
c	1.734938	1.446948	4.172108	4.266221	0.001365	0.013731	1.0098
82		0.109079					
	Total pressure drop [Pa]	2.651853	2.665658	2.437810			oscillator
y	1.734938	1.446948	7.466204	2.651624	0.000108	0.007231	1.0931
48		0.008652					
	Local deposition fraction G1	0.015620	0.020900	0.017800			oscillator
y	1.734938	1.446948	0.808790	0.006216	0.752567	0.532385	0.4530
54		151.290131					
	Local deposition fraction G2	0.029020	0.028520	0.024500			monotoni
c	1.734938	1.446948	5.862238	0.029041	0.000887	0.022817	1.0175
32		0.070918					
	Local deposition fraction G3	0.038040	0.040560	0.035580			oscillator
y	1.734938	1.446948	1.589786	0.036241	0.059102	0.192031	1.3531
97		4.962792					
	Local deposition fraction G4	0.092940	0.104080	0.085420			oscillator
y	1.734938	1.446948	1.182775	0.080815	0.163077	0.408930	1.3068
83		15.003579					
	Local deposition fraction G5	0.025320	0.024320	0.019260			monotoni
c	1.734938	1.446948	4.701646	0.025401	0.004002	0.055564	1.0411
18		0.319132					
	Local deposition fraction G6	0.005960	0.006660	0.004920			oscillator
y	1.734938	1.446948	2.176718	0.005658	0.063340	0.264453	1.2583
87		5.337700					

In [63]: results_df

Out [63]:	name	f1	f2	f3	h1	h2	h3
0	Overall deposition fraction	0.206900	0.225040	0.187480	0.000098	0.00017	0.000247 1.73
1	Inlet total pressure [Pa]	4.261568	4.219866	4.049685	0.000098	0.00017	0.000247 1.73
2	Total pressure drop [Pa]	2.651853	2.665658	2.437810	0.000098	0.00017	0.000247 1.73
3	Local deposition fraction G1	0.015620	0.020900	0.017800	0.000098	0.00017	0.000247 1.73
4	Local deposition fraction G2	0.029020	0.028520	0.024500	0.000098	0.00017	0.000247 1.73
5	Local deposition fraction G3	0.038040	0.040560	0.035580	0.000098	0.00017	0.000247 1.73
6	Local deposition fraction G4	0.092940	0.104080	0.085420	0.000098	0.00017	0.000247 1.73
7	Local deposition fraction G5	0.025320	0.024320	0.019260	0.000098	0.00017	0.000247 1.73
8	Local deposition fraction G6	0.005960	0.006660	0.004920	0.000098	0.00017	0.000247 1.73